Who Buys the Longshot?
The Favourite–Longshot Bias and Retail Flow in Prediction Markets

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Abstract

We study the favourite–longshot bias on event-contract exchanges with three stratified random samples: quotes on 19,007 settled non-financial Kalshi markets from 2021 to 2026, 2.7 million Kalshi trade prints, and 2.6 million fills by 237,569 identified Polymarket wallets. On Kalshi, contracts priced at 10 cents or less return −55% per dollar at the mid-quote and −75% after the spread and fee, while favourites roughly break even. The calibration slope of outcomes on prices is 1.13 ($t = 6.5$ against one). The bias appears in every category and does not fade as resolution approaches. It is no weaker in liquid markets and has persisted while the venue grew a thousand-fold. It is steepest in multi-outcome and deadline contracts. Small orders hold the longshots: 39% of contracts in taker orders of $10 or less are longshots, against 1% in orders above $1,000. On Polymarket, retail wallets account for 86% of net longshot buying. The one-cent tick, the spread and the fee bound the bias, so taking the other side as a taker does not pay. But they do not explain it: it survives away from the tick floor, at one-hour horizons and in the most liquid markets. Insider trading and belief dispersion fit poorly, while one mild probability-weighting function fits every category. The evidence points to lottery demand by small traders, met at prices that just cover the cost of supplying it.

1 Introduction

Bettors overpay for unlikely outcomes. This favourite–longshot bias was first documented at the racetrack [23]. It has since been found in most betting markets, and it is one of the most robust anomalies in the economics of risk [37, 43, 49]. Event-contract exchanges now give the old question a new setting. In the United States, Kalshi settles hundreds of thousands of contracts a month on sports, weather, politics, entertainment and what speakers will say. A large retail brokerage routes its customers' orders to it [40]. Polymarket runs a comparable venue outside the United States and records every trade on a public blockchain. These markets are often read as probability forecasts [52], so whether their prices are biased, and for whom, matters beyond the traders themselves.

We ask three questions.

1. How large is the favourite–longshot bias on a regulated event-contract exchange, and how does it vary across event categories, horizons, contract structures, liquidity and time?

2. Who buys the longshots, and who bears the losses: takers or makers, small or large orders, new or experienced accounts?

3. How much of the bias is mechanical, the result of the tick, the fees and the cost of capital, and how much is preference, information or belief dispersion?

We answer them with three data sets, each a stratified random sample with design weights (Section 3):

- bid and ask quotes at fixed leads before resolution for 19,007 settled Kalshi markets in 4,770 events, from 2021 to 2026;
- 2.69 million public Kalshi trade prints with size, price and aggressor side, covering 15,329 markets in 18 strata;
- 2.57 million Polymarket fills by 237,569 identified wallets in 2,765 markets, and matched Kalshi and Polymarket prices for 441 games.

The bias is large per dollar, small per contract, and everywhere. On Kalshi the calibration slope of outcomes on prices is 1.13 ($t = 6.5$ against one). Longshots, sides priced at 10 ¢ or less, return −55% per dollar at the mid-quote and −75% after the spread and the taker fee. Favourites priced at 90 ¢ or more return +1.1% at the mid. The bias:

- appears in every event category;
- does not fade as resolution approaches (the slope is 1.18 one hour ahead);
- is no weaker in the most liquid markets;
- has held in every quarter since 2025, while the venue grew by about three orders of magnitude.

It is steepest in multi-outcome events and "by date" deadline contracts. Pre-game sports prices with a known start time are close to calibrated.

Small traders buy the longshots, and the price of immediacy falls on them. In Kalshi's trade prints:

- 39% of the contracts bought in taker orders of at most $10 are longshots, against 1.2% in orders above $1,000. This holds in all 18 strata.
- In the median stratum, longshot positions lose for both liquidity roles. Takers lose more, in 13 of 18 strata, because they pay a half-spread and fee that together come to 14% of a longshot's stake.
- Averaged over all prices, takers and makers earn about the same.
- The per-dollar longshot loss is similar at every order size, so small traders bear most of it because they hold most longshots, not because they trade at worse prices.

On Polymarket, where wallets are identified, retail wallets are the only significant net buyers of longshots and account for 86% of net longshot buying. No class of professional sellers takes the other side: the flow is absorbed by many counterparties. Profits are extremely concentrated, with the top 1% of wallets taking 85% of all gains, but wallet skill does not persist from one half of the sample to the next.

Mechanics bound the bias; demand keeps prices at the bound. Mechanics explain part of the bias.

- About half of the loss in the cheapest price bucket is the one-cent tick floor: contracts worth a tenth of a cent cannot be quoted lower.
- Averaged across markets, the longshot overpricing lies inside the no-arbitrage band set by the spread and the fee, so an arbitrageur taking the other side as a taker earns nothing.

But mechanics do not explain why prices sit near the edge of that band. The slope is unchanged without the tick-floor buckets, and the bias does not shrink at short horizons, where capital costs vanish. In the most heavily traded markets it exceeds the band. An insider explanation fits poorly: Shin's insider share is small, and the field of longshots in multi-outcome events stays overpriced after the Shin adjustment. Proxies for belief dispersion, such as the number of outcomes, turnover and the presence of an external forecast, do not predict where the bias is steeper. A single mild inverse-S probability-weighting function, with the same curvature ($\gamma \approx 0.89$) in every category, describes prices well. Fitted on the first half of the sample, it removes 61% of the longshot overpricing in the second. The evidence points to demand for lottery-like payoffs by small traders, met by liquidity suppliers at prices that just cover their costs.

Contributions. We contribute to three literatures.

- *The favourite–longshot bias.* We measure it on an order-book exchange across categories, horizons and four contract structures. We also separate the part that the tick and fees impose mechanically from the part that does not. Racetrack data rarely allow this [17, 47].
- *Prediction markets.* We extend Bürgi et al. [14], who document the bias in Kalshi trade prices and its stronger form for takers, to a volume-blind sample of all non-financial categories. We add complete trade histories classified by size and timing, and identified wallets on a second venue [1, 41].
- *Retail lottery demand.* The longshot is the purest lottery ticket. The concentration of longshot buying in small accounts mirrors the evidence on lottery stocks and retail options trading [3, 13, 31]. Here the claim has a fixed payoff date and no cash-flow risk, so its mispricing can be measured exactly.

The rest of the paper is organised as follows. Section 2 reviews related work and Section 3 describes the venues and data. Section 4 sets out the measures, the no-arbitrage band and the hypotheses. Section 5 describes the methods and Section 6 reports the results. Section 7 discusses their interpretation and limits, and Section 8 concludes.

2 Related literature

The favourite–longshot bias in betting markets. The tendency of bettors to overpay for unlikely outcomes is one of the oldest anomalies in economics.

- Griffith [23] found it in American horse racing, and Ali [2] and Hausch et al. [25] documented it on large samples of parimutuel races.
- Thaler and Ziemba [49] and Sauer [43] survey the early evidence.

- The bias is not universal. Woodland and Woodland [54] find a reverse bias in baseball money-line betting, and its size differs between parimutuel and fixed-odds settings.
- Whelan [51] shows that fixed-odds markets generate the bias even with competitive bookmakers once bettors are risk-averse and heterogeneous.
- Moskowitz [36] uses sports betting contracts as a laboratory for asset pricing and finds momentum consistent with delayed overreaction.

Explanations. Ottaviani and Sørensen [37] organise the explanations into four groups.

- *Risk preference.* Ali [2] infers risk-loving bettors; Golec and Tamarkin [20] argue that bettors love skewness rather than variance.
- *Probability weighting.* Jullien and Salanié [26] find that cumulative prospect theory [27, 50] fits racetrack odds better than expected utility. Snowberg and Wolfers [47] use exotic bets to separate the two and conclude that misperception of probabilities explains the bias better than risk-love.
- *Information.* Shin [44, 45] model bookmakers who protect themselves against insiders by shading longshot odds, yielding an estimable insider share.
- *Belief heterogeneity.* Ali [2] and Gandhi and Serrano-Padial [17] show that dispersed beliefs among budget-constrained bettors produce the bias even without preference distortions. Gandhi and Serrano-Padial [17] find that a minority of noise traders with dispersed beliefs fits US racetrack data better than preference-based models.

Our setting can tell these explanations apart in ways racetracks cannot, because we observe who trades, in which role, at what size and when.

Prediction markets. Prices in prediction markets are routinely read as probabilities [52, 53], though Manski [35] shows that, with heterogeneous beliefs, the price need not equal the mean belief.

- Page and Clemen [38] find that Intrade prices were well calibrated for events resolving soon but biased for events far in the future. They show in a model that time to expiration itself pushes prices toward the bias.
- Rothschild and Sethi [41] classify trader strategies with account-level Intrade data. They document a single trader who accounted for over a third of bets on one candidate and kept that candidate's price above a rival exchange's.
- On Kalshi, Bürgi et al. [14] document a favourite–longshot bias that is stronger in prices accepted by takers than by makers. They explain it with a modest overestimation of small probabilities.

We extend this line of work in three directions:

- across the full range of event categories and contract structures;
- with all public trades classified by role, size and timing;
- with identified on-chain wallets and matched markets on a second venue.

Lottery demand and retail trading in financial markets. The bias has close analogues in securities markets.

- *Lottery-like stocks.* Individual investors overweight stocks with lottery-like payoffs [31]. Such stocks earn low subsequent returns [3], as probability weighting predicts [10].
- *Options.* Lottery-like equity options earn large negative returns [12]. Retail options traders, now a majority of volume, concentrate in cheap, short-dated contracts and lose money on average [13].
- *Attention.* Attention drives individual investors' buying [5]. Attention-induced herding by app-based retail traders predicts negative returns [9].

A prediction-market longshot is the purest lottery ticket: a cheap claim that pays a large multiple with small probability, and that is often bought through the same retail apps. Our results speak to whether lottery demand, and the retail flow behind it, prices such claims in markets with a known payoff date and no fundamental cash-flow risk.

3 Institutional setting and data

Kalshi. Kalshi is a US exchange for event contracts, regulated by the Commodity Futures Trading Commission as a designated contract market.

- *Contracts.* Each binary contract pays \$1 if its event occurs and nothing otherwise, and is fully collateralised. A trader who buys YES at p posts p; a trader who buys NO posts $1 - p$.
- *Trading.* Trading takes place in a central limit order book. A bid for NO at q is an offer of YES at $1 - q$, so each market has a single book. Most markets are quoted on a one-cent grid from 1 to 99 ¢.
- *Fees.* A taker pays $0.07 \times m \times C\,P(1 - P)$ for C contracts at price P, where m is a series multiplier equal to one for almost all series. Some series also charge makers, with coefficient 0.0175 [28]. Per contract, the fee is largest at $P = 50$ ¢ and vanishes at the edges of the price range. Per dollar staked it is $0.07\,m(1 - P)$, which is largest for longshots.
- *Access.* Retail customers trade through Kalshi's own app and website. Since March 2025 they can also trade through a prediction-market hub inside Robinhood's brokerage app, which routes orders to Kalshi [40]. Order flow on Kalshi therefore includes flow from a large retail brokerage.

Contracts are organised as *series* (a recurring question), *events* (one instance of the question) and *markets* (one binary leg of an event). We classify every market into one of four contract structures.

- *Single binary:* an event with one or two legs, such as a game-winner market.
- *Multi-outcome:* an event with at least three mutually exclusive legs, of which exactly one pays, such as the winner of an award or a tournament.
- *Strike ladder:* legs on thresholds or brackets of one quantity, such as a city's daily high temperature, a point spread or a game total.
- *Deadline:* a contract that pays if an event happens by a stated date ("by date T").

Multi-outcome events and ladders produce most longshots, because all but one or two of their legs are unlikely.

Table 1: Data sets

	Kalshi quote panel	Kalshi trades	Polymarket wallets
Period (resolution)	Jul 2021–Oct 2026	15 Aug–31 Oct 2026	Jan 2024–Oct 2026
Frame (markets)	3,100,088	1,128,256	636,520
Sampling unit	event	event	market
Strata	group × month × structure	group or league	group × volume × month
Sample (markets)	23,475	15,329	2,765
Observations	63,753 quotes	2.69 million prints	2.57 million fills
Identities	none	aggressor side	wallet, role
Used in	Sections 6.1 and 6.4	Section 6.2	Section 6.3

Non-financial contracts only. Kalshi trade prints are those made before the pre-resolution cut of Section 3.2. Polymarket fills are all fills in sampled markets; analyses of returns use fills before each market's information time (Section 5.3).

Polymarket. Polymarket's international exchange matches orders off chain and settles them on chain, with outcome shares collateralised in a dollar stablecoin. Every fill therefore records the wallet on each side, which lets us follow identified traders. The venue is not open to US persons. Its markets are binary, and multi-outcome questions are listed as groups of binary markets, as on Kalshi. Trading was fee-free on almost all markets until early 2026. Taker fees of the form $\theta\, q(1-q)$ were then introduced on almost every market by May 2026 (Appendix C). We use Polymarket only as a source of data and place no orders.

Scope. We study *non-financial* contracts: sports, weather, politics (including elections and world affairs), entertainment, mentions (whether a speaker says a phrase during an event), science and technology, and a small residual group. We exclude contracts on crypto prices, equity indices, interest rates, economic releases and company metrics. Their prices are tied to traded underlyings with their own dynamics and arbitrage relations. In the events we study, the event contract is the only market on the event, so its price is the market's probability assessment.

Three data sets. Each research question uses the data best suited to it. Table 1 summarises them.

- *Kalshi quote panel* (Section 3.1): bid and ask quotes at fixed leads before the scheduled resolution of a stratified random sample of settled markets. It measures the bias across categories, horizons and contract structures, and is the basis of the structural analysis.
- *Kalshi trades* (Section 3.2): complete public trade histories of a second stratified sample, with each print's size, price and aggressor side. It measures who pays the bias by liquidity role, trade size and timing.
- *Polymarket wallets* (Section 3.3): complete on-chain trade histories of a stratified sample of Polymarket markets, with wallet identities, plus Kalshi and Polymarket prices for matched games. It measures which traders buy longshots, how concentrated profits are, and how the two venues compare.

All three samples are drawn at random within strata, and design weights make pooled estimates representative of their frames. The two Kalshi samples draw events without regard to volume. The Polymarket sample stratifies by volume tier and excludes the largest markets (Section 3.3). All data are public; we use no account-level information from either venue.

3.1 Kalshi markets and prices

Frame. The frame contains every settled single-leg Kalshi market in a non-financial series, 3,100,088 markets in total. Markets settled before Kalshi's historical cutoff of 11 September 2026 come from the public historical endpoint, read series by series back to the first listings in July 2021. Later markets, to 31 October 2026, come from the settled-market table of Section 3. Sports series are read back to July 2024, because Kalshi listed almost no sports contracts before 2025. Each sports series is capped at its newest 60,000 markets; three MLB player-prop series reach the cap and are covered from June–July 2026. Multivariate (parlay) collections are not single-leg markets and are excluded, as are financial contracts (Section 3).

Scheduled time. Leads are measured from a scheduled time T_m that is known when the market lists:

- *start*: the scheduled start written in the ticker of a sports event;
- *date*: the event date written in the ticker. This is midnight US Eastern for most groups; for daily weather, the start of the local measurement day; and for sports without a start time, midnight at UTC+14, which is before any local start on that date;
- *deadline*: the date stated in the rules of a "by date" contract;
- *close*: otherwise, the scheduled close, equal to the expected expiration minus the series' modal gap between expiration and close.

Ticker dates take precedence when they fall before the scheduled close, because many markets stay open well after their outcome is known. The 2024 presidential market, for example, traded until the January 2025 inauguration. Close times of markets that closed early, settlement times and the post-outcome `last_price` are never used.

Sample and panel. A market is eligible if its scheduled time is known and falls by 31 October 2026, it opened at least an hour before that time, and its event traded at least 10 contracts. That leaves 1,880,593 markets. We draw a stratified random sample of events, keeping all their legs, from cells defined by group, month of T_m and structure. Each group has a market budget spread evenly over its months, and quota that one structure cannot use in a cell passes to the others in that month. Events with more than 30 candidate legs keep a random 30, and ladder events keep whole ladders up to 60 legs. The sample has 23,475 markets in 5,640 events, with hourly candles from the public candlestick endpoints. Of the 80,581 market–lead observations, 79% have a usable quote. The analysis uses 61,992 pre-event quotes on 19,007 markets in 4,770 events and 1,670 series, plus 1,761 sports in-play quotes (Table 24). The panel is shared with the structural analysis (Section 5.4).

3.2 Kalshi trades

The trade-level analysis uses Kalshi's public trade prints. Each print records the time to the microsecond, the number of contracts, the YES price and the taker's side. We fetch complete histories from the public `/historical/trades` and `/markets/trades` endpoints, at no more than three requests per second, and keep the raw responses unchanged.

Frame and sample. The frame is every settled, non-financial market that closed between 15 August and 31 October 2026 and traded at least once: 1,128,256 markets in 210,854 events. Sports account for 98.6% of the frame's volume. To cover the other groups, we draw a stratified random sample.

- There are 18 strata: five non-sports groups and 13 sports leagues (Table 31).
- Events are drawn uniformly within each stratum, independently of volume.
- Every traded leg of a drawn event is kept, up to 20 legs drawn at random.

The sample has 2,300 events and 15,329 markets. Design weights ω (inverse inclusion probabilities) make pooled estimates representative of the frame. The histories are complete: for the median market in every group, the contracts printed equal the reported volume.

Pre-resolution filter. We drop prints made after the outcome could be known from the clock:

- all prints in the last 30 minutes before a market's close (or its scheduled expiration, if earlier);
- for non-sports markets that closed early because the event occurred, all prints in the last four hours before the close.

The cut-offs follow the data (Table 33). In sports, the share of volume at $\leq 3\,¢$ or $\geq 97\,¢$ is 9–10% between 30 and 60 minutes before the close and 65–75% in the last five minutes. In-play sports prints before the cut are kept and studied. A *strict* variant keeps only pre-game sports prints and other prints more than 24 hours before the close. The filter removes 34% of the 4.04 million fetched prints. That leaves 2,685,117 prints in 1,643,534 taker orders, 14,164 markets and 2,151 events, worth 277 million contracts and $132 million of taker notional (Table 32).

Replication sample. Earlier work on this project left 9.8 million prints in 18,333 other settled markets. These are mainly pre-game windows of sports game-winners, capped histories of high-volume markets and the final months of "by date" deadline contracts. The sample is selected on volume, so we use it only for replication (Table 15).

3.3 Polymarket wallets and matched markets

Polymarket's international exchange settles on chain, so every fill names the wallet on each side. We read only its public APIs (Gamma for markets, the Data API for trades, account statistics and prices), cache raw responses unchanged, stay at or below three requests per second, and place no orders (Appendix C.1).

- *Frame.* Every resolved binary market with at least $1,000 of volume whose event ended between January 2024 and October 2026: 636,520 markets in 164,311 events after financial tags are excluded. Sports make up 73% of the markets and 68% of the volume.
- *Sample.* A seeded stratified random draw by group × volume tier × resolution month (six markets per stratum below $100k, two at $100k–1M, one per group and half-year at $1M–6M), with design weights N_h/n_h. The 1,340 markets above $6M (42% of frame volume) cost over 150 requests each and are not sampled.

- *Trades.* The complete trade history of each sampled market, every wallet, with maker and taker rows and each fill's liquidity role: 2,765 markets, 2,574,475 fills, 237,569 wallets. 72% of fills precede the information time. Account creation dates come from a pseudo-random subset of wallets selected by address (5,144 dated wallets).
- *Venues.* 882 Kalshi team markets in 441 NBA, NHL, NFL, college-football and WNBA games (August–October 2026) matched to the same Polymarket outcome, with Kalshi hourly bid–ask candles and Polymarket's hourly price history. Polymarket US's data host disallows Anthropic's agents in its `robots.txt`, so of that venue we use only its published fee schedule.

4 Conceptual framework

This section defines the objects we measure and the explanations we test. Throughout, a contract pays \$1 if its event occurs. We work with the *side price* $q \in \{0.01, \ldots, 0.99\}$: the price paid for whichever side, YES or NO, a trader buys. Let $w \in \{0, 1\}$ indicate that the side paid out and let π be its true probability.

4.1 Measuring the bias

Definition 1 (Calibration and the favourite–longshot bias)**.** The calibration function is $c(q) = \mathbb{E}[w \mid q]$. Prices exhibit a *favourite–longshot bias* (FLB) if $c(q) < q$ for small q and $c(q) > q$ for large q: longshots are overpriced and favourites underpriced.

We summarise the bias in three ways.

1. *Returns by price bucket.* The return per dollar of buying a side at q with fee f per contract is

$$r = \frac{w - q - f}{q + f}. \tag{1}$$

   We report gross ($f = 0$) and net-of-fee returns.

2. *Logit slope.* In the regression

$$\operatorname{logit} \mathbb{P}(w = 1 \mid q) = \alpha + \beta \operatorname{logit} q, \tag{2}$$

   $\beta > 1$ indicates an FLB, $\beta < 1$ a reverse bias, and $\beta = 1, \alpha = 0$ calibration.

3. *Longshot–favourite spread.* The difference in mean return per dollar between longshots ($q \leq 10\,¢$, LS) and favourites ($q \geq 90\,¢$, FAV).

A binary contract has two sides priced q and $1 - q$ at the same moment. If each observation enters with both sides, the calibration function is symmetric by construction: $c(1 - q) = 1 - c(q)$, so $\alpha = 0$ and the bias is carried entirely by β. If only the YES side enters, a preference for buying YES ("YES bias") appears as $\alpha < 0$. We therefore report both the side-symmetric and the YES-only versions.

4.2 Mechanical limits: tick, fees and capital

Some of the bias cannot be traded away even by a risk-neutral, fully informed arbitrageur. Three frictions matter.

- *Tick floor.* Prices live on a one-cent grid. A contract with $\pi < 0.01$ cannot trade below 1 ¢, so buyers at the floor lose at least $0.01 - \pi$ per contract. The loss in the lowest bucket is partly mechanical.
- *Fees.* A Kalshi taker pays $f(q) = \theta\, q(1-q)$ per contract, with $\theta = 0.07$ times a series multiplier. Makers pay $\theta_m = 0.0175$ on some series and nothing on others. The fee per contract is the same for both sides of a trade, but per dollar it is $\theta(1-q)$, about 7% for a longshot and much less for a favourite.
- *Capital.* Exploiting overpriced longshots means buying the opposite side at $1-q$ and holding it to resolution, a horizon of T years. This ties up $1-q$ per contract at an opportunity cost ρ per year.

Proposition 1 (No-arbitrage band)**.** *Ignore the bid–ask spread. A risk-neutral taker buys the NO side of a longshot priced q only if $q - \pi > \theta q(1-q) + \rho T(1-q)$, and buys the YES side only if $\pi - q > \theta q(1-q) + \rho T q$. Prices are therefore consistent with no arbitrage whenever*

$$1 - \theta(1-q) - \rho T\,\frac{1-q}{q} \;\le\; \frac{\pi}{q} \;\le\; 1 + \theta(1-q) + \rho T. \tag{3}$$

Proof. Buying NO at $1-q$ pays $1-\pi$ in expectation and costs $(1-q) + \theta q(1-q)$ plus $\rho T(1-q)$ of capital. Rearranging the condition for a positive expected profit, and dividing by q, gives the lower bound. The upper bound follows in the same way for buying YES at q, which ties up q of capital. □

The lower bound has a feature that matters for longshots. Its capital term scales with $(1-q)/q$, so the band widens sharply as the price falls and as the horizon lengthens.

- At $q = 2$ ¢, $T = 30$ days and $\rho = 5\%$, the capital term alone allows the true probability to sit about 20% below the price.
- At $q = 50$ ¢, the same capital term allows less than 1%.

Mechanical frictions therefore predict a bias concentrated in cheap, long-dated longshots, which grows with the horizon [38]. Bid–ask spreads widen the band further.

4.3 Behavioural and informational explanations

Three classes of explanation predict a bias beyond the band. They make different predictions about *who* buys longshots.

Preferences: risk-love and probability weighting. A representative bettor with convex utility [2] or skewness preference [20] pays a premium for lottery-like payoffs. Under rank-dependent or cumulative prospect theory [26, 27, 50], prices reflect decision weights $w(\pi)$ rather than probabilities. With the linear-in-log-odds weighting function $\operatorname{logit} w(\pi) = \delta + \gamma \operatorname{logit} \pi$, inverting $q = w(\pi)$ gives regression (2) with $\beta = 1/\gamma$ and $\alpha = -\delta/\gamma$. A logit slope above one is thus equivalent to inverse-S weighting ($\gamma < 1$), and β measures its strength. Snowberg and Wolfers [47] argue that misperception of probabilities, rather than risk-love, best explains racetrack prices.

Information: insiders. In Shin [44, 45], a market maker who faces a share z of perfectly informed traders protects himself by shading prices. The shading is proportionally largest on longshots, because an insider's bet on a longshot is the most costly to the market maker. For an event with n mutually exclusive outcomes priced $q_1, \ldots, q_n$ with sum $S = \sum_j q_j > 1$, the implied probabilities are

$$\pi_i = \frac{\sqrt{z^2 + 4(1-z)\, q_i^2 / S} - z}{2(1-z)}, \tag{4}$$

with z chosen so that $\sum_i \pi_i = 1$. The insider share z is a per-event measure of adverse selection, and it predicts that informed flow, not noise, is on the longshot.

Heterogeneous beliefs. When risk-neutral traders with dispersed beliefs face budget constraints [2, 17], the most optimistic traders set the price of each outcome. Since optimism about a longshot moves its price proportionally more, longshots trade above their mean-belief probability. The bias then rises with belief dispersion and with the share of noise traders.

4.4 Testable implications

The explanations are not mutually exclusive, but they make distinct predictions about which traders lose on longshots and when.

Hypothesis 1 (Mechanical)**.** *If frictions are the whole story, the bias lies inside the band* (3)*. It is concentrated at the tick floor and grows with horizon, and it shows up in gross but not net returns.*

Hypothesis 2 (Preference and retail demand)**.** *If the bias reflects demand for lottery-like payoffs by unsophisticated traders [10, 31], then:*

- *longshot losses fall on takers, small trades and new or inexperienced accounts;*
- *the bias widens when retail attention is high [5, 9];*
- *makers and experienced traders earn the other side.*

Hypothesis 3 (Information)**.** *If longshot prices are shaded against insiders, then:*

- *longshot order flow predicts outcomes, at least for large or early trades;*
- *the estimated z is higher where private information is plausible, as in politics, announcements and corporate events, than where outcomes are publicly observed, as in weather and game scores.*

Hypothesis 4 (Belief dispersion)**.** *If belief heterogeneity drives the bias, it is larger in events with many outcomes, high turnover and retail-heavy participation, and smaller where a public forecast anchors beliefs.*

Section 6.1 measures the bias against these benchmarks. Sections 6.2 and 6.3 identify who buys longshots. Section 6.4 compares the explanations directly.

5 Empirical methodology

All workstreams share the definitions of Section 4 and a common set of conventions.

- *Side price and outcome.* Every observation is a side bought at price q, with outcome w. Buying YES at YES price p gives $q = p$; buying NO gives $q = 1 - p$.
- *Price buckets.* We use twelve buckets of q: 1–2, 3–5, 6–10, 11–20, 21–35, 36–50, 51–65, 66–80, 81–90, 91–95, 96–98 and 99 ¢. Longshots (LS) are $q \leq 10$ ¢ and favourites (FAV) $q \geq 90$ ¢.
- *Returns.* Returns per contract are $w - q - f$, and returns per dollar follow (1).
- *Fees.* Unless stated otherwise, fees are Kalshi's exact schedule: 0.07 times the series multiplier times $q(1-q)$ for takers, and $0.0175\,q(1-q)$ for makers on series that charge makers. Kalshi computes fees per fill to \$0.000001 and rebates rounding over-payment [29], so exact fees are the fees paid.
- *Categories.* Kalshi series are grouped into Sports, Weather, Politics (including elections and world affairs), Entertainment, Mentions, Science & Tech, and Other. Polymarket markets are mapped to the same groups from their tags. Financial contracts (crypto prices, equity indices, interest rates, economic releases and company metrics) are outside the scope of this study and are excluded throughout.
- *Timing.* Prices used for calibration are observed before the outcome is known:
  - at fixed leads before the scheduled resolution;
  - before the start for sports, except where in-play prices are studied explicitly;
  - and never after settlement.

  Markets that closed early because the event occurred are timed by their scheduled, not realised, close.
- *Inference.* Outcomes within an event, such as the legs of a ladder or the teams in a game, are dependent. Standard errors are therefore clustered by event [15], and we report the number of clusters.
- *Robustness.* Every main result is shown both market-weighted and volume-weighted where the two differ, and separately for the two halves of the sample.

5.1 Calibration and the cross-section of the bias

Observations. For market m with scheduled time T_m (defined in Section 3.1) and lead L, the observation time is $t = T_m - L$. The pre-event leads are $L \in \{1, 6, 24, 72, 168, 720\}$ hours. Sports events whose ticker carries the start time also have in-play observations at $t = T_m + 0.5$, +1 and +2 hours. The quote at t is the YES bid b and ask a at the close of the last hourly candle that ended at or before t. Kalshi omits many hours without activity, and the last close is taken as the standing quote. Its age is recorded and used in a robustness check. The mid is $p = (a + b)/2$. An observation enters the analysis when the market is open at t, stays open at least 15 minutes after t, and its quote is *usable*: $a > b$ and $a - b \leq 10$ ¢. A one-sided book at the tick floor ($b = 0$, $a = 1$ ¢) is usable and has mid 0.5 ¢.

Sides, prices and returns. Each quote yields two side observations: YES, with side price $q = p$ and payoff $w = y$, and NO, with $q = 1 - p$ and $w = 1 - y$, where $y = 1$ if the market settled YES. The executable taker price is $q^a = a$ for YES and $1 - b$ for NO; a resting (maker) order buys at $q^b = b$ or $1 - a$. Returns per dollar are $r = (w - q - f)/(q + f)$ as in Section 4, with the taker fee $f(q) = \theta\, q(1 - q)$, where $\theta = 0.07$ times the series fee multiplier (one for almost all series). We report three versions: gross at the mid ($q = p$ or $1 - p$, $f = 0$), gross at the ask ($q = q^a$, $f = 0$), and net of the exact taker fee at the ask ($q = q^a$, $f = f(q^a)$). The difference between the first two is the half-spread; the difference between the last two is the fee. Kalshi computes fees per fill to \$0.000001 and rebates rounding over-payment [29], so we use exact fees and show per-order rounding to the cent only for comparison. Fees follow the schedule and series multipliers in force in November 2026 throughout; Kalshi's schedule differed in parts of the earlier sample, so net returns before 2025 are indicative. A side is a *longshot* (LS) if $q \le 10\,¢$ and a *favourite* (FAV) if $q \ge 90\,¢$. Price buckets are those of the shared definitions: 1–2, 3–5, 6–10, 11–20, 21–35, 36–50, 51–65, 66–80, 81–90, 91–95, 96–98 ¢ and 99 ¢.

Weights. A market has up to six pre-event observations. *Market-weighted* statistics give each observation weight $1/n_m$, where n_m is the number of observations of market m in the sample being summarised, so that every market counts once. *Volume-weighted* statistics multiply this by the number of contracts traded in market m in the seven days up to t. That count is known at t. We do not weight by lifetime volume, because it includes trading after t, and a longshot that later wins attracts heavy trading on its way up.

Three measures of the bias.

1. *Returns by price bucket*: the weighted mean of r within each bucket of the side price (the mid for gross returns, the taker price for net returns).

2. *Calibration slope*: the logit regression of Section 4, estimated with weights on YES sides,

$$\mathbb{P}(y = 1 \mid p) = \Lambda(\alpha + \beta \operatorname{logit} p), \tag{5}$$

   with p clipped to $[0.5\,¢, 99.5\,¢]$. Calibrated prices have $\alpha = 0$ and $\beta = 1$. With $\beta > 1$, true odds are more extreme than quoted odds: longshots are overpriced and favourites underpriced, which is the favourite–longshot bias. Fitting on NO sides gives the same β and the opposite α, so β summarises both tails and α is the YES-only intercept (a preference for buying YES appears as $\alpha < 0$). We report $t_{\beta=1} = (\hat{\beta} - 1)/\mathrm{se}(\hat{\beta})$.

3. *Longshot–favourite spread*: the difference between the mean return per dollar of LS sides and of FAV sides. We estimate it as the coefficient on a longshot indicator in a weighted regression of r on a constant and that indicator, over LS and FAV sides.

Standard errors are clustered by event [15], because the legs of an event and the leads of a market share one outcome. Each table reports the number of observations, the number of clusters and t-statistics.

Cross-sections.

- *Category*: the coarse groups of the shared definitions.
- *Horizon*: each lead L separately, which gives one observation per market and lead.

- *Contract structure*, from the event and leg design:
  - a *ladder leg* belongs to a set of at least three legs of one event that differ only by a numeric strike (temperature buckets, run totals, point spreads, player-prop thresholds);
  - a *deadline* contract pays if an occurrence happens before a date T stated in the rules, classified from the rules text with the hazard study's parser;
  - a *multi-outcome* leg belongs to an event with at least three non-ladder legs that Kalshi flags as mutually exclusive;
  - everything else is a *binary*: single-market events, two-leg head-to-head events and independent legs that share an event, such as the words of a mentions event.
- *Liquidity*: within-group terciles of the 7-day volume, the quoted spread and the open interest at t.
- *Calendar time*: quarters of T_m.

Multi-outcome events and the Shin model. For a mutually exclusive event with K legs, all quoted at t and exactly one settling YES, the overround is $\sum_k a_k - 1$ at the asks and $\sum_k p_k - 1$ at the mids. Following Shin [44, 45] and Section 4, quoted prices π_k with $\Pi = \sum_k \pi_k > 1$ map to probabilities

$$\hat{p}_k(z) = \frac{\sqrt{z^2 + 4(1-z)\pi_k^2/\Pi} - z}{2(1-z)}, \tag{6}$$

where z is the share of insider trading and solves $\sum_k \hat{p}_k(z) = 1$. We compute z per event and lead from the asks. We then compare the realised frequency of each leg with three probability forecasts: the raw mid, the mid normalised to sum to one, and $\hat{p}_k$ [48]. Legs are grouped by their price rank within the event. We also measure the *field*: the summed price of all legs at or below 10 ¢ against the number of them that win.

Limits to arbitrage.

- *Fees.* We show the taker fee per dollar staked, $f(q)/q = \theta(1-q)$, with and without per-order rounding.
- *Taking the other side.* For every quote with a longshot side, we compute the net return of buying the opposite (favourite) side as a taker at q^a. We also compute it for resting a bid at q^b as a maker, paying the maker fee $\theta_m q(1-q)$, with $\theta_m = 0.0175$ times the multiplier, only on series that charge it. The maker figure assumes a fill at the quote without adverse selection, so it is an upper bound; who actually fills is the subject of Section 5.2. Exploiting overpriced longshots this way is the arbitrage whose limits Shleifer and Vishny [46] describe.
- *No-arbitrage band.* Per bucket, we compare the mispricing per contract, $\bar{q} - \bar{w}$ at the mid, with the cost of trading against it as a taker: the half-spread plus the taker fee on the opposite side. The capital term of the band in Section 4 is computed at a 5% annual cost of capital over the lead.
- *Tick floor.* Prices move on a 1 ¢ grid; some markets use 0.1 ¢ steps. On the 1 ¢ grid, a contract whose fair value is below 0.5 ¢ cannot have a lower mid, and a taker cannot pay less than 1 ¢. In the 1–2 ¢ bucket we split the loss per contract, $\bar{q} - \bar{w}$, into a mechanical part and a

remainder. The mechanical part is the loss of the quotes at the floor $q_{\min}$ (a 0.5 ¢ mid with no bid, or a 1 ¢ ask for takers), $s \max(q_{\min} - \bar{w}_f, 0)$, where s is their weight share and $\bar{w}_f$ their win rate. The remainder comes from prices above the floor. Markets on the 0.1 ¢ grid serve as a comparison.

Robustness. We repeat the main measures on several variants of the sample:

- the first and second halves of the sample, split at the median observation time;
- without quotes whose mid is at or below 2 ¢ or at or above 98 ¢ (the 1–2 ¢ and 99 ¢ buckets and their mirror images);
- with quotes at most 24 hours old, and with spreads of at most 5 ¢;
- by the source of the scheduled time;
- with design weights, the inverse of each event's sampling probability;
- with standard errors clustered by market, series or month instead of by event.

Many cells are reported. We emphasise patterns that hold across groups and in both halves rather than single cells.

5.2 Trade-level flow: who pays the bias

This subsection asks who pays the bias measured in Section 5.1. It uses every public print in the trade sample of Section 3.2. Kalshi's public trade feed has no account identities, but it does name the side that crossed the spread. That is enough to separate the two parties to each trade by *liquidity role*.

Takers, makers and positions. A print i records a number of contracts n_i, a YES price p_i and the taker's side.

- A taker who bought YES holds YES at $q_i = p_i$. A taker who bought NO holds NO at $q_i = 1 - p_i$.
- The resting order on the other side (the *maker*) holds the opposite side at $1 - q_i$.
- Gross of fees, the two positions are mirror images. The taker earns $w_i - q_i$ per contract and the maker earns $-(w_i - q_i)$.
- Fees break the symmetry. The taker pays $f_i^T = 0.07\,\kappa\, q_i(1-q_i)$, where κ is the series multiplier. The maker pays $f_i^M = 0.0175\,\kappa\, q_i(1-q_i)$ on series that charge makers, and nothing otherwise.

Each print therefore contributes two *positions*, one per role. We sort positions by the price of the side held, q for the taker and $1 - q$ for the maker, into the buckets of Section 5.

Weights and returns. Let ω_i be the design weight of the print's market (the inverse probability that the sampling design of Section 3.2 drew it). For any cell of positions we report two averages.

$$\bar{\Delta} = \frac{\sum_i \omega_i n_i (w_i - q_i - f_i)}{\sum_i \omega_i n_i} \qquad \text{(contract-weighted, \$ per contract)}, \tag{7}$$

$$\bar{r} = \frac{\sum_i \omega_i n_i (w_i - q_i - f_i)}{\sum_i \omega_i n_i (q_i + f_i)} \qquad \text{(dollar-weighted, per dollar)}. \tag{8}$$

Here q_i, w_i and f_i refer to the position's own side. Equation (8) is the dollar-weighted mean of the per-dollar return (1), that is, aggregate profit over aggregate stake. Unless a table says otherwise, returns are net of the exact fee. Gross returns set $f_i = 0$. As a retail-perspective variant we also report a print-weighted mean, in which each fill counts once regardless of its size.

Taker orders and trade size. Kalshi prints one fill for each resting order that a taker order executes against. A taker order that sweeps several resting orders therefore appears as several prints. These prints share the market, the microsecond timestamp and the taker side. We group such prints into one *taker order* o, with dollar notional $N_o = \sum_{i \in o} n_i q_i$. Trade size is N_o, bucketed as $\leq$ \$10, \$10–100, \$100–1k and $>$ \$1k. Every print inherits the size of its order. Small trade size is the standard proxy for retail flow when identities are hidden [7, 34]. It is only a proxy: a professional can split orders, and a retail trader can send a large one. Makers' order sizes are not observed. A print shows only the part of the resting order that was filled.

Timing and attention. Each print is classified by the following variables. Each is known at the time of the print, except the marquee label.

- *Clock.* Hour of day and weekday or weekend, in US Eastern time.
- *In play.* A sports print is in play if it is at or after the game's scheduled start. The start is the time encoded in the ticker where there is one. Otherwise it is the occurrence time minus three hours, Kalshi's convention for game markets. A start is used only if it falls between 30 minutes and 12 hours before the close.
- *Primetime.* The game is scheduled to start between 19:00 and 23:59 ET.
- *Time to expiry.* Hours from the print to the market's scheduled expiration.
- *New listing.* The print falls in the first 24 hours after the market opened.
- *Volume surge.* The market's volume in the hour before the print, $V_{t-1h,t}$, is compared with its own average hourly volume before that hour, $\bar{V}_{t-1h}$. Both use past prints only, and the average needs at least six hours of history. A surge is $V_{t-1h,t} \geq 3\,\bar{V}_{t-1h}$, which captures attention shocks in the spirit of Barber and Odean [5].
- *Marquee.* The event's total volume is in the top decile of its stratum. This label is descriptive and uses ex post volume.

Mid-quote proxy, price impact and markouts. The public trade history contains no quotes. We therefore build a mid-quote from the prints themselves.

- A YES-taker print executes at the YES ask, and a NO-taker print executes at the YES bid.

- Let a_t be the YES price of the market's last YES-taker print at or before t, and b_t that of its last NO-taker print.
- The proxy is $m_t = (a_t + b_t)/2$. It is defined only when both prints are at most 60 minutes old and $a_t \geq b_t$.
- Impact, markout and order-flow analyses also require a tight proxy, $a_t - b_t \leq 5\,¢$, both before the order and at each later horizon. This keeps them to markets where the proxy tracks the book.

For a taker order at time t with fill price p and direction $d = +1$ (bought YES) or $d = -1$ (bought NO), define, in cents per contract,

$$s = d\,(p - m_{t^-}), \qquad I_h = d\,(m_{t+h} - m_{t^-}), \qquad M_h = d\,(m_{t+h} - p) = I_h - s, \tag{9}$$

where m_{t^-} is the proxy just before the order. These are the effective half-spread s, the price impact I_h and the taker's markout M_h at horizons h of one minute and one hour. A mark at $t + h$ is used only if $t + h$ is before the market's pre-resolution cut, so no markout reads a price set after the outcome was known. The markout to settlement, $M_\infty = d\,(Y - p)$ with Y the YES outcome, is the taker's gross profit per contract. It splits into the mispricing of the mid and the cost of immediacy, $M_\infty = d\,(Y - m_{t^-}) - s$. Persistent impact signals information, and impact that reverses signals inventory or noise [18, 24].

We validated the proxy against order books recorded from Kalshi's WebSocket feed in two three-hour windows.

- *Saturday evening ET (8 November, 00–03 UTC).* The window covers 236 markets and 193,907 taker orders. 98% of its prints are in game-winner markets, mainly college football and NBA games during the evening slate. The proxy exists for 96% of orders. It is within one cent of the book mid for 96.9% of them and within two cents for 98.9%. Its mean absolute error is 0.16 ¢.
- *Tuesday morning ET (10 November, 14–17 UTC).* The window covers 174 markets and 11,717 orders. It is mainly pre-game NBA and NHL markets, with daily temperature, mentions and political markets. The proxy exists for 86% of orders, is within one cent for 94.4% and within two cents for 97.0%. Its mean absolute error is 0.27 ¢.

In both windows the proxy's correlation with the book mid exceeds 0.998, and 99.1% of taker orders execute at or beyond the book mid on their own side. This confirms the meaning of the taker-side field. The recorded books are among the more liquid on the venue, with a median spread of one cent. In thin markets the proxy is staler and more often missing.

Order-flow regressions. For the information content of flow we aggregate taker orders to market-hours h, before each market's pre-resolution cut. Order sizes fall into three classes: small ($\leq$ \$100), medium (\$100–1k) and large ($>$ \$1k). Class c's signed imbalance is

$$\mathrm{OIB}_{c,h} = \frac{\sum_{o \in (c,h)} d_o n_o}{\sum_{o \in h} n_o} \in [-1, 1]. \tag{10}$$

It is normalised by the hour's total volume, so the three classes sum to the hour's net taker imbalance. We then estimate

$$\Delta_h^{(k)} = \alpha + \sum_c \beta_c^{(k)}\,\mathrm{OIB}_{c,h} + \gamma^{(k)}\,(m_h^{\mathrm{end}} - m_h^{\mathrm{start}}) + \varepsilon_h. \tag{11}$$

The left-hand side $\Delta_h^{(k)}$ is measured in cents and takes three forms: the same-hour mid change $m_h^{\text{end}} - m_h^{\text{start}}$ (with $\gamma = 0$), the next-hour change, and the drift to settlement, $Y - m_h^{\text{end}}$. A class whose flow moves the price and is confirmed at settlement ($\beta_c^{(\infty)} > 0$) carries information in the sense of Kyle [32]. A class whose same-hour impact reverses ($\beta_c^{(\text{next})} < 0$, $\beta_c^{(\infty)} \leq 0$) behaves as noise. The settlement drift needs no mid at the outcome, because Y is observed.

Inference. Standard errors are clustered by event [15]. For weighted means we compute them from per-event sums of $\omega_i n_i (x_i - \bar{x})$, which gives exactly the cluster-robust variance. A difference between two cells, such as takers against makers or YES-takers against NO-takers, uses the per-event influence functions of both means. This accounts for the two cells sharing events. Regression (11) uses CR1 cluster-robust errors. Every table reports t-statistics and the number of event clusters.

5.3 Wallet-level and cross-venue analysis

Polymarket records every fill on chain with the wallet on each side, so we can follow identified traders. This subsection defines the Polymarket return measures, the exact wallet accounting, the wallet classes, and the tests of concentration, persistence, retail dynamics and cross-venue pricing. Notation follows Section 5: q is the price of the side bought, w is 1 if that side pays out, f is the fee per contract and $r = (w - q - f)/(q + f)$ is the return per dollar of holding to settlement.

Information cut-off. For each market we fix an information time t^{info} that does not depend on the outcome. For a sports game it is the scheduled start (`gameStartTime`); otherwise it is the scheduled end date. If a market closed before that schedule, the event happened early, and we set t^{info} to 24 hours before the close. Unless stated otherwise, all calibration and trade-level return measures use only fills and prices strictly before t^{info}, so in-play sports trading and post-outcome sweeps at 99.9 ¢ are excluded.

Calibration from prices at fixed leads. For a lead $L \in \{1\text{ h}, 24\text{ h}, 7\text{ d}\}$, the price of market m is the last trade at or before $t_m^{\text{info}} - L$, expressed in units of outcome 0 ("YES"): a trade in outcome 1 at price p is a YES price of $1 - p$. Each market contributes both sides, YES at p and NO at $1 - p$. We report the mean of r by side-price bucket, with weights equal to the design weight of the market ("market-weighted") or the design weight times market volume ("volume-weighted"). The calibration slope β is the coefficient of the logistic regression (2) estimated on the outcome-0 side of each market (the other side duplicates it); with $\beta > 1$ prices are too close to $\frac{1}{2}$, longshots too dear and favourites too cheap.

Trade-level returns. A buy fill of outcome k at price p has $q = p$ and $w = \mathbf{1}\{k \text{ wins}\}$. Polymarket charges takers $f = \theta_m q(1 - q)$ per share on fee-enabled markets, where θ_m is the market's published rate (mostly 0.03–0.05 by category, 0 for geopolitics); makers pay nothing. We use the rate in each market's fee schedule, and $f = 0$ for maker fills and for markets without fees, which covers almost every market before March 2026 (Appendix C.2). Each table states whether returns are gross ($f = 0$) or net. Fill-level means are either "dollar-weighted" (weight = notional × design weight) or "fill-weighted" (weight = design weight).

Exact wallet accounting. We express each fill in YES-equivalent units. Buying s shares of outcome 0 at p adds $+s$ to exposure e and $-sp$ to cash c; selling reverses both signs. Buying

outcome 1 at p pays $s(1-Y)$ at settlement, so it adds $-s$ to e and $+s(1-p)$ to c. The PnL of wallet i in market m is then

$$\pi_{im} = \sum_{j\in(i,m)} c_j + Y_m \sum_{j\in(i,m)} e_j - \sum_{j\in(i,m)} f_j s_j, \tag{12}$$

where $Y_m = 1$ if outcome 0 wins. In these units a split (\$1 for one share of each outcome) or a merge is PnL-neutral, so (12) is exact from trades alone. Each fill appears once for the taker and once for every maker it matched. In a fully observed fee-free market, therefore, $\sum_i e_{im} = 0$ and $\sum_i \pi_{im} = 0$. We use this identity to check the data. Where fees apply, Polymarket takes the taker's fee in shares on buys, and the identity holds up to those shares.

Wallet classes. Following Rothschild and Sethi [41], who group Intrade accounts by their trading behaviour, we classify traders by observable conduct rather than by identity. Each fill is classified from the wallet's own history in the sample strictly before the fill. Let n be its earlier fills, μ the share of those that were maker fills, τ the share of the markets it had entered that it had already traded in both YES-equivalent directions, and $\tilde{v}$ the median notional of its last 200 fills, including the current one. Classes are assigned in this order:

1. **Market maker (MM):** $n \geq 20$, $\mu \geq 0.7$ and $\tau \geq 0.5$.
2. **Whale:** $\tilde{v} \geq \$1{,}000$.
3. **Active retail:** $\tilde{v} \leq \$100$ and $n \geq 20$.
4. **Retail:** $\tilde{v} \leq \$100$ and $n < 20$.
5. **Mid:** all other fills ($\$100 < \tilde{v} < \$1{,}000$).

Account age, which separates new from established retail, comes from the creation date (`join_date`) that Polymarket reports for each wallet. Age is measured at the fill, so it is known at trade time. Leaderboard rank and lifetime counters are measured today, after the fact, and are used only descriptively. The robustness grid moves one cut-off at a time (small size \$50 or \$250; whale size \$500 or \$5,000; active at 10 or 50 earlier fills; looser or stricter MM criteria). Table 40 reports the results.

Who buys and who sells longshots. Besides the longshot share of each class's buying, we measure net longshot flow. A fill in a token priced at most 10 ¢ counts as a longshot purchase or sale at its notional. A fill in a token priced at least 90 ¢ counts as the opposite trade in the complementary longshot, at that longshot's cost $s(1-p)$. Net flows sum to zero across all participants, so the classes that buy longshots on net are matched by the classes that sell them. For positions opened by a buy before t^{info}, the exit value added (proceeds − shares sold × w)/cost compares the PnL actually realised with that of holding every share to settlement.

Concentration and persistence. Wallet PnL sums (12) over the sampled markets. We report the share of all gains (losses) held by the top x% of winning (losing) wallets, which is the Lorenz curve read from the top, and the corresponding Gini coefficients. For persistence, the sample is split by resolution date into H1 (January 2024 to May 2025) and H2 (June 2025 to October 2026). Wallets with at least five markets and \$50 bought in each half are ranked on the shrunk H1 return

$$\tilde{R}_i^{\text{H1}} = \frac{\sum_{m\in\text{H1}} \pi_{im}}{\sum_{m\in\text{H1}} B_{im} + K}, \tag{13}$$

where B_{im} is the amount bought and K is the median amount bought per wallet-half. They are then evaluated on their raw H2 return, using the Spearman rank correlation with a wallet bootstrap and the H2 return of the top minus the bottom H1 quintile. Two controls separate skill from luck and style:

- **Edge versus entry price.** The wallet's mean buy return minus the mean return of all buys in the same price bucket and half. This removes persistence that comes only from buying favourites rather than longshots.
- **Luck placebo.** We redraw each market's outcome as $\tilde{Y}_m \sim \text{Bernoulli}(p_m^{\text{close}})$, where p_m^{close} is the last price before t^{info}, and recompute (12) with the same trades. Under the placebo, prices are calibrated and no wallet can forecast outcomes beyond p^{close}, but wallets keep the execution advantage of trading at better prices than the close.

Retail dynamics and herding. Account creation dates are looked up for a pseudo-random subset of sampled wallets: those whose address is divisible by eight, in address order. Inflow is the weekly count of creations among them. Around five salient events fixed in advance (the 2024 US presidential election, Super Bowls LIX and LX, the 2025 papal conclave and the 2025 New York City mayoral election), we compare creations in the 14 days centred on the event with a baseline four to eight weeks earlier. Herding of small wallets is measured in one-hour windows with at least five small (at most \$100) taker buys by retail or active-retail wallets, using the measure of Lakonishok et al. [33]:

$$H_{mt} = \left|\tfrac{k_{mt}}{n_{mt}} - \bar{p}_m\right| - \mathbb{E}\left|\tfrac{K}{n_{mt}} - \bar{p}_m\right|, \qquad K \sim \text{Bin}(n_{mt}, \bar{p}_m). \tag{14}$$

Here k_{mt} counts YES-side buys in the window and $\bar{p}_m$ is the market's average YES share of small buys. $H > 0$ means buys cluster on one side more than independent orders would. To test for reversal, we regress the YES-price change over the next $h \in \{1, 6, 24\}$ hours on the window's dollar imbalance $I_{mt} \in [-1, 1]$ and on the previous hour's price change, using only windows that end before t^{info}.

Cross-venue comparison. For Kalshi game markets matched one-to-one to Polymarket outcomes (Section 3.3), we compare synchronous hourly prices at leads of 1–24 hours before the scheduled start. The Kalshi price is the bid–ask midpoint of the hourly candle and the Polymarket price is the hourly price history. The gap is $g = p^{\text{PM}} - p^{\text{K}}$, grouped by the two venues' average side price. A gap exceeds fees if $|g|$ is larger than the sum of the two taker fees at the observed prices, $0.07\,p^{\text{K}}(1-p^{\text{K}})$ on Kalshi and $\theta_m p^{\text{PM}}(1-p^{\text{PM}})$ on Polymarket. The comparison is descriptive.

Inference. Standard errors are clustered by event (Polymarket event id, or game for the cross-venue tests), as in Section 5. Population estimates use the design weights of the stratified sample.

5.4 Structural decomposition

This subsection sets the four explanations of Section 4 against each other on the quote panel of Section 5.1. We keep that panel's usable quotes, side observations, leads, market and volume weights and event clustering. No trading strategy is evaluated, so no portfolio ledgers or Sharpe ratios are computed.

The mechanical band. Take a side with mid q^m, bid q^b and ask q^a, observed T years before the scheduled time. Its price minus outcome splits exactly into

$$q^m - w = \underbrace{(q^m - q^b)}_{\text{half-spread}} + \underbrace{\theta\, q^b(1-q^b)}_{\text{fee}} + \underbrace{\rho T(1-q^b)}_{\text{capital}} + R, \qquad R = q^b - \theta\, q^b(1-q^b) - \rho T(1-q^b) - w. \tag{15}$$

R is the profit per contract of the arbitrage behind Proposition 1: sell the side by buying the other side as a taker at $1-q^b$, pay the fee and hold to resolution. Its expectation is non-positive exactly when the side's true probability lies above the lower edge of the band (3) evaluated at the bid. With $q^b = q$ it is that edge. A mean residual $\bar{R}$ that is not positive therefore means that the measured mispricing lies inside the band.

- *Tick floor.* We split the half-spread into the half-spread of sides pinned at the floor, with no bid and an ask of one tick (1 ¢ on the standard grid, 0.1 ¢ on Kalshi's sub-cent grid), and the rest.
- *Parameters.* θ is 0.07 times the series fee multiplier. The cost of capital is $\rho = 5\%$ a year, with $\rho = 0$ and 10% as robustness checks.
- *Makers.* The maker variant $R^{mk} = q^a - \theta_m q^a(1-q^a) - \rho T(1-q^a) - w$ sells the side at its ask by resting a bid on the other side. It ignores the adverse selection of resting orders, so it is an upper bound.
- *Cells.* Longshot sides ($q^m \le 10$ ¢) by price bucket, lead, group and structure. For each cell we report the mean components and $\bar{R}$, with event-clustered standard errors. The share of the gross mispricing that the band can rationalise is $\min(\text{band}, \text{gross})/\text{gross}$.
- *Horizon.* Hypothesis 1 predicts that the gross mispricing grows with T. We regress the gross mispricing and R of longshot sides (cents) on the log lead with price-bucket fixed effects, by weighted least squares with event-clustered errors.

Probability weighting. We fit two weighting functions $q = w(\pi)$ to outcomes.

- *Linear in log odds (LLO).* $\operatorname{logit} w(\pi) = \delta + \gamma \operatorname{logit} \pi$ [22]. As Section 4 shows, inverting it gives regression (2) with $\beta = 1/\gamma$ and $\alpha = -\delta/\gamma$.
- *Prelec.* $w(\pi) = \exp\{-\delta(-\ln \pi)^\gamma\}$ [39], so that

$$\mathbb{P}(y = 1 \mid q) = \exp\{-[(-\ln q)/\delta]^{1/\gamma}\}. \tag{16}$$

In both, $\gamma < 1$ is inverse-S weighting: small probabilities are overweighted and large ones underweighted. Estimation is by weighted maximum likelihood on YES sides, with the mid clipped to $[0.5\,¢, 99.5\,¢]$ as in Section 5.1. Weights are market weights, with volume weights as a check. Standard errors are sandwich estimates clustered by event, using the delta method for γ and δ. We compare four models:

- calibrated prices, $\mathbb{P}(y = 1) = q$;
- a constant shift, $\operatorname{logit} \mathbb{P}(y = 1) = \alpha + \operatorname{logit} q$;
- LLO;

- Prelec.

Hypothesis 2, read as probability weighting, predicts $\gamma < 1$ with one function across categories. We test equal slopes across groups, structures and leads with a cluster-robust Wald test from one fully interacted logit. Leads of the same market are therefore handled jointly.

For out-of-sample evidence, we split the months of the scheduled time at the month that divides the quotes most evenly. The first half runs to 2025-07. We fit each model on one half and compute its mean log score on the other, in both directions. Per-quote differences against calibrated prices are averaged with event-clustered t-statistics, a clustered version of the test of Diebold and Mariano [16] with a proper scoring rule [19].

Shin's insider share. Shin's model needs every outcome of an event. We use two kinds of partition:

- events that Kalshi flags as mutually exclusive, with two or more legs;
- strike ladders whose legs are one lower tail, one upper tail and buckets between them, read from the markets' strike types.

An event enters at a lead only if every leg has a usable quote there and exactly one leg settles YES. That is, the number of quoted legs must equal the event's number of settled legs.

We solve (4) for z from the asks and from the mids, setting $z = 0$ when the prices sum to at most one. On an exchange, the overround of the asks is mostly the bid–ask spread. We therefore also fit z to outcomes. For a given z, we choose the scale S so that the probabilities of (4) sum to one; $z = 0$ then gives the normalisation $q_i / \sum_j q_j$. The fitted value is

$$\hat{z} = \arg \max_{z \in [0,\, 0.6]} \sum_e \omega_e \log \pi_{e, i_e^*}(z), \tag{17}$$

where i_e^* is the winning leg of event-lead e and ω_e weights each event once across its leads. We search a grid with step 0.005 and form 95% percentile intervals from 400 event-bootstrap draws.

If Shin's mechanism drives the bias, $\hat{z}$ should match the z implied by the overround. It should also be higher where private information is plausible, in Politics, Entertainment, Mentions and Science & Tech, than where outcomes are publicly observed, in Sports and Weather. We compare raw, normalised and Shin probabilities for each leg [48], using the log score and the price minus outcome of longshot legs. We also regress z from the asks on the legs' mean half-spread, the log number of legs and group dummies.

Belief dispersion. Hypothesis 4 concerns where the bias is steeper. We let the calibration slope vary with characteristics Z_k of the quote:

$$\operatorname{logit} \mathbb{P}(y = 1) = a_g + \Big(b + \sum_k b_k Z_k\Big) \operatorname{logit} p + \sum_k c_k Z_k, \tag{18}$$

where a_g are group intercepts. A coefficient $b_k > 0$ means a stronger bias where Z_k is high. The characteristics are:

- *Number of outcomes* (log). This is the number of legs of mutually exclusive events, ladders and multi-outcome events, and two otherwise.
- *Turnover*, $\log\{(1 + V_{24})/(1 + OI)\}$: contracts traded in the 24 hours before the quote relative to open interest at the quote.

- *Quoted spread.*
- *Horizon*, the log lead.
- *Forecast anchor*, a 0/1 indicator for Sports and Weather. Sportsbook lines anchor sports prices and National Weather Service forecasts anchor weather prices.

Continuous characteristics are standardised. All are known at the quote time; the number of legs is part of the event's design.

We estimate three versions:

- a pooled version with one base slope, which identifies the anchor across groups;
- a within-group version with group-specific slopes b_g, which absorbs the anchor;
- separate fits by group, with a sign test across groups.

A least-squares regression of longshot overpricing in cents on the same characteristics, with price-bucket fixed effects, checks that the result does not depend on the logit form.

Synthesis. We reduce each hypothesis to the predictions listed in Table 51 and test each at the 5% level. We also ask how much of the second-half longshot loss each explanation removes when its map from prices to probabilities replaces the price:

- the band, which covers at most the gross mispricing;
- LLO and Prelec, fitted on the first half;
- Shin, in partition events.

These maps are alternative readings of one calibration curve, not additive components. We say in Section 6.4.5 what this design cannot separate.

6 Results

6.1 The longshot bias across categories, horizons and contract types

This subsection measures the bias on Kalshi with the three measures of Section 5.1. The benchmarks of Section 4 guide the reading: a bias confined to the tick floor and the no-arbitrage band points to mechanics, and a bias beyond them to preferences, information or beliefs. Unless stated otherwise, statistics are market-weighted, pool the pre-event leads from 1 hour to 30 days, and use t-statistics clustered by event.

6.1.1 Size of the bias

The bias is large per dollar and small per contract. Table 2 and Figures 1 and 2 give the headline.

- The calibration slope is $\hat{\beta} = 1.13$ ($t_{\beta=1} = 6.5$): quoted odds are too moderate. This is the favourite–longshot bias familiar from racetracks [23, 47, 49].
- Longshot sides return -55.1% per dollar at the mid ($t = -10.0$) and -74.7% net of spread and fees ($t = -25.3$).

- Favourite sides return 1.1% per dollar at the mid ($t = 9.8$), but -0.7% at the ask after the fee ($t = -2.4$).
- Volume weighting steepens the slope to $\hat{\beta} = 1.48$ ($t = 3.6$) and deepens the longshot return to -85.6% at the mid ($t = -17.7$). In each of the six main groups the slope is steeper under volume weights (1.29 to 1.80, against 1.07 to 1.25 market-weighted).

The bias is concentrated in the tails (Table 3). In the 1–2 ¢ bucket, the mean mid is 0.87 ¢ and the side wins 0.38% of the time, a gross return of -62.0% per dollar ($t = -7.5$). The 3–5 ¢ and 6–10 ¢ buckets return -54.1% and -29.2%. Between about 20 ¢ and 80 ¢ the realised frequency tracks the price, and no bucket differs from zero by more than a few percent per dollar. At the top, the 99 ¢ bucket earns 0.4% at the mid ($t = 5.6$) and -0.4% net of the fee. Per contract, these tails are small: the 1–2 ¢ bucket loses well under one cent. The large per-dollar losses come from small prices, not from large errors in probability. Bürgi et al. [14] report a similar pattern in Kalshi trade prices.

YES bias. The intercept of the YES-side regression is $\hat{\alpha} = 0.07$ ($t = 2.0$). A positive $\hat{\alpha}$ means YES outcomes are slightly more frequent than YES prices imply, holding the slope fixed. So within this sample, longshot losses are not driven by a preference for buying YES contracts.

Every category shows the bias. The longshot return at the mid is negative in every group, and the calibration slope is above one in every group (Table 2). The differences between groups are modest.

- *Entertainment* has the steepest slope ($\hat{\beta} = 1.25$, $t = 4.8$), followed by *Science & Tech* (1.18, $t = 2.6$) and *Weather* (1.15, $t = 2.9$).
- *Mentions* (1.13, $t = 2.0$), *Sports* before the game (1.10, $t = 2.6$) and in play (1.09, $t = 1.1$), and *Politics* (1.07, $t = 1.8$) are flatter.
- The small *Other* group (116 markets: transport and similar) has $\hat{\beta} = 1.46$.

Longshot returns rank the groups differently from the slope, because they also depend on how much mass sits at the tick floor. Mentions longshots lose -75.8% per dollar ($t = -12.1$) and Weather longshots -30.7% ($t = -1.4$). Only Entertainment stands out from the pooled slope by more than its sampling error. Sports and politics are the groups with the most liquid external references, bookmaker lines and polls, and their slopes are the lowest. That is consistent with an external anchor disciplining tail prices, but the cross-group evidence is too weak to identify it. In baseball betting the bias is absent or even reversed [54].

6.1.2 Horizon

Pooled across groups, the slope is 1.12 at 30 days ($t = 2.3$), 1.16 at 7 days, 1.23 at 24 hours ($t = 7.8$) and 1.18 at one hour ($t = 7.1$) (Table 4 and Figure 3). Prices do not become calibrated as resolution approaches; if anything, the bias steepens in the last day. The pattern is clearest in Entertainment, where the slope is 1.29 at 30 days, 1.14 at 7 days and 1.47–1.59 within the last day ($t = 5.3$ at 24 hours). Within other groups the cells are noisier and the confidence intervals overlap across leads.

Page and Clemen [38] find calibration good at short horizons and a bias in long-dated contracts, which they attribute to the cost of tying up capital. Kalshi's short horizons do not remove the bias. This finding does not support a purely capital-cost explanation. The capital term of the no-arbitrage band in Section 4 scales with the horizon, so it predicts a bias that shrinks towards

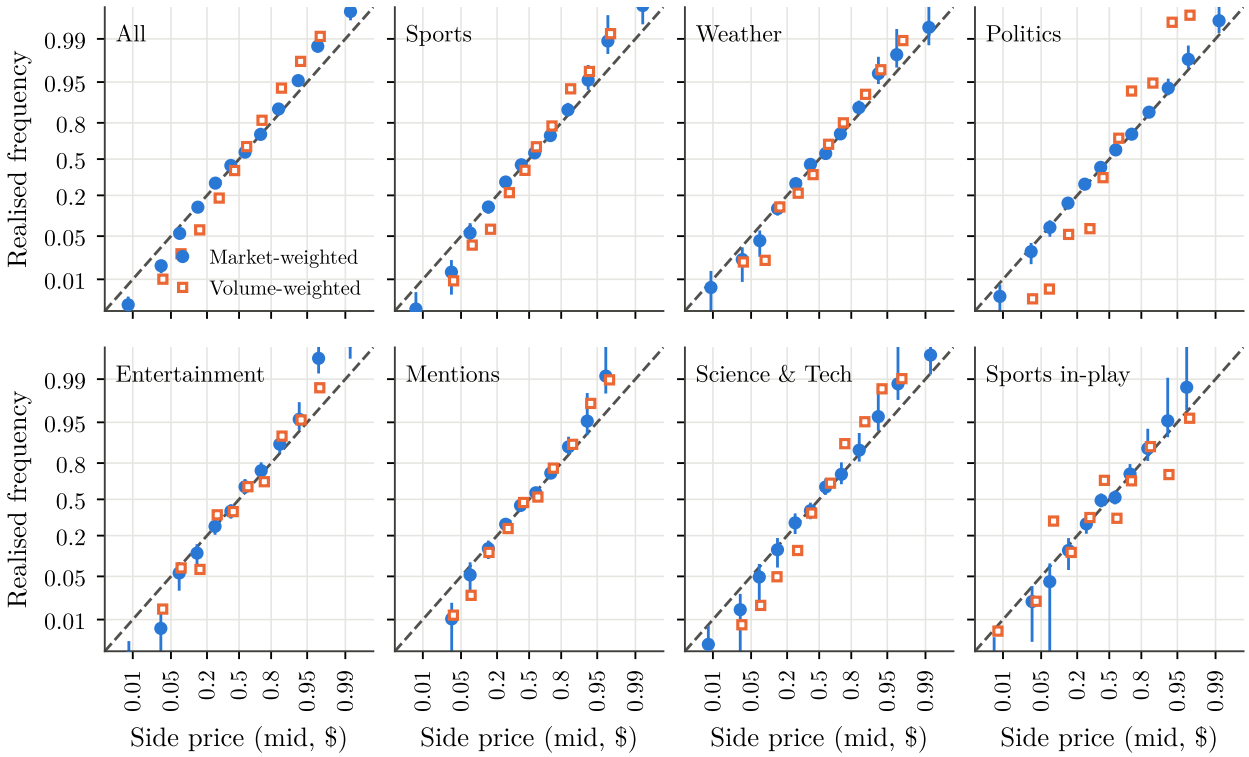


Figure 1: Calibration by group. Realised frequency of the side bought against its mid price, by price bucket, on log-odds axes. Dots: market-weighted means with 95% confidence intervals clustered by event; open squares: volume-weighted (contracts traded in the 7 days to the quote). Pre-event leads pooled; the last panel shows sports in-play quotes. The dashed line is perfect calibration.

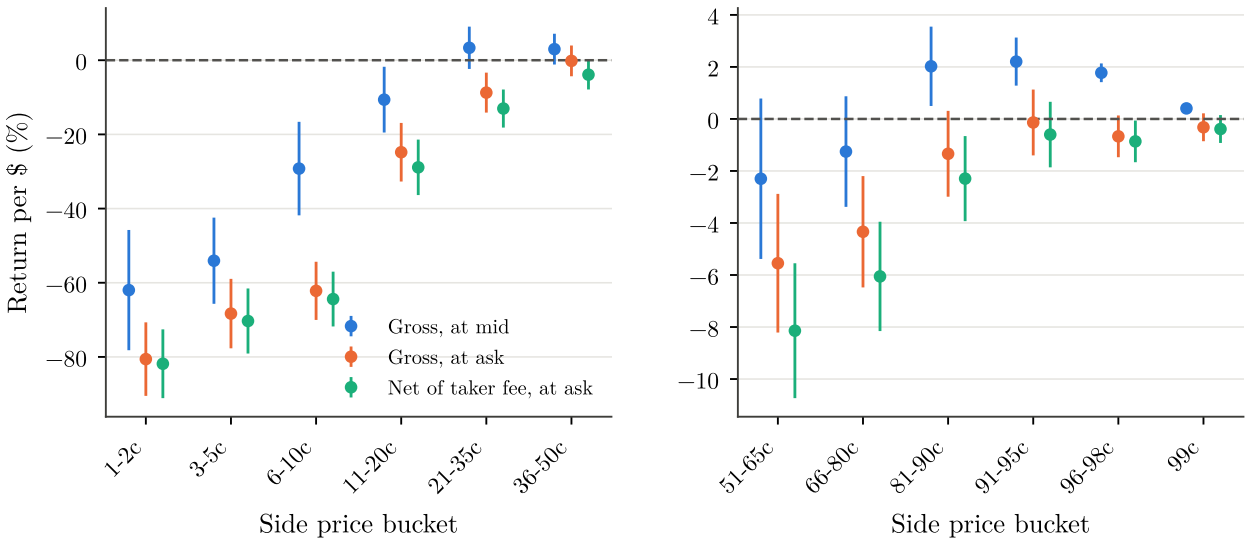


Figure 2: Return per dollar by side-price bucket, all groups, market-weighted, pre-event leads pooled: gross at the mid, gross at the ask, and net of the exact taker fee at the ask. Bars are 95% confidence intervals clustered by event. The two panels share buckets but not the vertical scale.

Table 2: The favourite–longshot bias by group

Group	Markets	Events	$\hat{\beta}$	LS	FAV	LS−FAV	LS	FAV	LS−FAV
				gross, at mid			net of taker fee, at ask		
A. Market-weighted									
All groups	19,007	4,770	1.129	−55.1	+1.1	−56.2	−74.7	−0.7	−74.0
			(6.5)	(−10.0)	(9.8)	(−10.1)	(−25.3)	(−2.4)	(−24.1)
Sports	6,284	1,521	1.099	−57.7	+1.1	−58.8	−76.8	−0.6	−76.1
			(2.6)	(−4.3)	(4.8)	(−4.3)	(−11.0)	(−0.9)	(−10.6)
Weather	2,901	1,065	1.152	−30.7	+1.5	−32.2	−61.4	−0.6	−60.7
			(2.9)	(−1.4)	(4.0)	(−1.4)	(−5.2)	(−0.9)	(−5.0)
Politics	3,906	968	1.075	−42.5	+0.5	−43.0	−67.0	−1.8	−65.2
			(1.8)	(−4.5)	(2.2)	(−4.4)	(−12.2)	(−3.1)	(−11.3)
Entertainment	2,665	530	1.254	−71.0	+1.4	−72.3	−84.6	−0.1	−84.5
			(4.8)	(−8.2)	(6.9)	(−8.3)	(−18.7)	(−0.1)	(−17.6)
Mentions	1,760	103	1.127	−75.8	+1.6	−77.4	−85.0	+0.5	−85.5
			(2.0)	(−12.1)	(3.4)	(−11.6)	(−17.0)	(0.5)	(−15.3)
Science & Tech	1,375	510	1.175	−64.0	+1.2	−65.2	−77.3	−0.4	−76.9
			(2.6)	(−6.0)	(3.7)	(−6.0)	(−11.0)	(−0.5)	(−10.1)
Other	116	73	1.463	−95.2	+2.9	−98.1	−95.7	+3.2	−98.9
			(2.4)	(−19.3)	(5.7)	(−18.2)	(−21.4)	(3.9)	(−19.0)
Sports in-play	812	242	1.086	−74.1	+1.4	−75.5	−82.3	−0.1	−82.2
			(1.1)	(−8.5)	(3.0)	(−8.3)	(−14.2)	(−0.1)	(−12.8)
B. Volume-weighted (contracts traded in the 7 days to t)									
All groups	19,007	4,770	1.477	−85.6	+1.6	−87.3	−90.5	+1.8	−92.3
			(3.6)	(−17.7)	(5.0)	(−17.6)	(−28.7)	(3.0)	(−25.8)
Sports	6,284	1,521	1.373	−73.2	+1.8	−75.1	−81.4	+2.5	−83.9
			(2.0)	(−5.6)	(3.2)	(−5.7)	(−9.7)	(1.9)	(−8.9)
Weather	2,901	1,065	1.291	−81.0	+1.3	−82.3	−87.2	+2.1	−89.3
			(2.0)	(−5.5)	(5.7)	(−5.5)	(−8.7)	(1.5)	(−7.9)
Politics	3,906	968	1.795	−96.1	+2.0	−98.1	−97.8	+1.6	−99.4
			(6.6)	(−67.0)	(2.6)	(−54.6)	(−120.7)	(6.0)	(−102.6)
Entertainment	2,665	530	1.344	−85.3	+0.9	−86.2	−90.7	+0.3	−91.0
			(3.0)	(−9.2)	(1.9)	(−8.9)	(−13.9)	(0.2)	(−11.7)
Mentions	1,760	103	1.346	−92.0	+1.4	−93.4	−93.6	+2.1	−95.7
			(4.6)	(−33.0)	(4.7)	(−31.6)	(−40.9)	(2.8)	(−33.3)
Science & Tech	1,375	510	1.564	−89.4	+1.4	−90.8	−94.1	+1.8	−95.9
			(4.0)	(−22.4)	(6.8)	(−22.1)	(−39.5)	(3.4)	(−36.3)
Other	116	73	1.431	−61.8	+0.2	−62.0	−66.7	+2.0	−68.6
			(1.5)	(−1.5)	(0.1)	(−1.4)	(−1.9)	(0.5)	(−1.8)
Sports in-play	812	242	0.810	+69.8	−9.7	+79.5	+53.1	−18.3	+71.4
			(−0.6)	(0.5)	(−0.9)	(0.6)	(0.4)	(−1.2)	(0.5)

Pre-event quotes at leads of 1 h to 30 d before the scheduled time, pooled; sports in-play: 0.5–2 h after the scheduled start. $\hat{\beta}$: slope of the logit regression (5), with $t_{\beta=1}$ in parentheses. LS, FAV: mean return per dollar (%) of sides priced at most 10 ¢ or at least 90 ¢; LS−FAV: their difference. Gross: at the mid, no fee. Net: at the ask, net of the exact taker fee, sides bucketed by the ask. Panel A gives each market weight one; panel B weights quotes by contracts traded in the 7 days to the quote. t-statistics clustered by event in parentheses.

Table 3: Calibration and returns by price bucket, all groups

Bucket	Sides	Events	$\bar{q}$ (¢)	Win (%)	Gross, mid (%)	Gross, ask (%)	Net, ask (%)
1-2c	26,184	2,118	0.87	0.38	−62.0 (−7.5)	−80.6	−81.8 (−17.3)
3-5c	9,521	2,007	3.53	1.67	−54.1 (−9.1)	−68.3	−70.3 (−15.7)
6-10c	4,847	1,707	7.53	5.52	−29.2 (−4.5)	−62.2	−64.4 (−17.1)
11-20c	6,366	1,919	15.18	13.73	−10.6 (−2.4)	−24.8	−28.9 (−7.6)
21-35c	8,230	2,146	27.63	28.52	+3.4 (1.2)	−8.7	−13.0 (−5.0)
36-50c	6,978	2,018	42.68	43.92	+3.0 (1.4)	−0.2	−3.9 (−1.9)
51-65c	6,996	2,058	57.93	56.59	−2.3 (−1.5)	−5.5	−8.1 (−6.2)
66-80c	8,254	2,150	72.95	72.02	−1.3 (−1.2)	−4.3	−6.1 (−5.6)
81-90c	6,485	1,939	85.42	87.15	+2.0 (2.6)	−1.3	−2.3 (−2.8)
91-95c	5,548	1,826	93.20	95.26	+2.2 (4.7)	−0.1	−0.6 (−0.9)
96-98c	10,721	2,041	96.96	98.68	+1.8 (9.7)	−0.7	−0.9 (−2.1)
99c	23,854	1,983	99.24	99.65	+0.4 (5.6)	−0.3	−0.4 (−1.4)

YES and NO sides of every usable pre-event quote, market-weighted. $\bar{q}$: mean mid price of the side; Win: weighted share that paid out. Returns per dollar; the ask columns bucket sides by the ask. t-statistics clustered by event in parentheses.

zero at short leads. At one hour the opportunity cost of capital is negligible, yet the slope is at least as steep as at 30 days.

Sports pre-game and in-play. For sports events with a scheduled start time in the ticker, the horizon is exact. Their pre-game prices are close to calibrated: $\hat{\beta} = 0.98$ at 24 hours, 1.05 at 6 hours and 1.09 one hour before the start ($t = 0.9$). Half an hour into the game the slope is 1.15 ($t = 1.4$), and it is 1.03 and 1.08 one and two hours in. None of these differs significantly from one. Longshots still lose in play: sides priced at most 10 ¢, such as a team trailing late, return −74.1% per dollar at the mid ($t = -8.5$). Most in-play longshot quotes sit in the 1–2 ¢ bucket, where the tick floor binds (Section 6.1.5).

Table 4: Calibration slope by lead and group

Lead	All	Sports	Weather	Politics	Entert.	Mentions	Sci. & Tech
1h	1.180	1.092	1.158	1.190	1.501	1.218	1.326
	(7.1)	(0.9)	(2.7)	(3.2)	(4.1)	(2.5)	(2.8)
6h	1.196	1.053	1.190	1.228	1.590	1.135	1.372
	(7.7)	(0.4)	(3.1)	(3.8)	(4.7)	(1.8)	(3.1)
24h	1.229	0.976	1.335	1.238	1.469	1.151	1.320
	(7.8)	(−0.1)	(1.3)	(3.9)	(5.3)	(2.2)	(2.9)
3d	1.166		1.345	1.096	1.278	1.311	1.259
	(5.9)		(1.6)	(1.9)	(3.6)	(4.1)	(2.7)
7d	1.159		1.197	1.074	1.143		1.348
	(3.8)		(1.0)	(1.1)	(1.8)		(3.1)
30d	1.123		1.316	1.200	1.288		1.085
	(2.3)		(1.3)	(2.1)	(2.6)		(0.6)
in-play 0.5h		1.146					
		(1.4)					
in-play 1h		1.026					
		(0.3)					
in-play 2h		1.083					
		(0.7)					

$\hat{\beta}$ of (5), one quote per market and lead, market-weighted; $t_{\beta=1}$ clustered by event in parentheses. The Sports column uses only events with a start time in the ticker, so that the horizon is exact; in-play rows are 0.5, 1 and 2 hours after that start. Cells with fewer than 100 quotes or 30 events are blank.

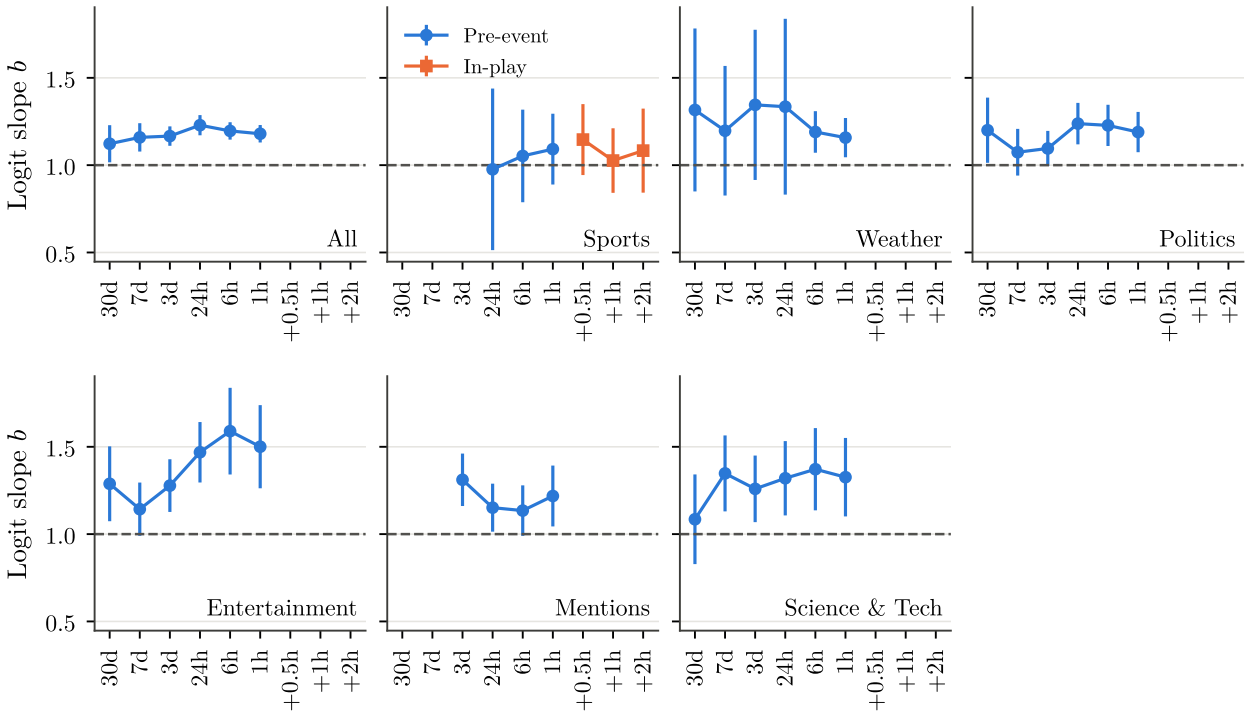


Figure 3: Term structure of the calibration slope $\hat{\beta}$ by group, from 30 days to 1 hour before the scheduled time. The sports panel uses events with a start time in the ticker, before and 0.5, 1 and 2 hours after the start (in-play). One quote per market and lead; 95% confidence intervals clustered by event. The dashed line ($\beta = 1$) is calibration.

6.1.3 Contract structure

Table 5 and Figure 4 compare the four contract structures. Multi-outcome events and deadline contracts are the most biased; ladder legs and stand-alone binaries the least.

Deadline contracts. "By date" contracts have $\hat{\beta} = 1.22$ ($t = 3.9$), and their longshots return -68.3% per dollar ($t = -10.7$). Only multi-outcome longshots lose more per dollar.

Ladder legs. Strike-ladder legs have $\hat{\beta} = 1.12$ ($t = 2.9$). Their tails return -30.9% per dollar at the mid ($t = -2.3$), and the body (35–65 ¢) 0.4%. The implied distribution has tails that are too fat relative to outcomes, the cross-strike form of the bias, but the tail loss is smaller than in the other structures.

Binaries. Stand-alone binaries have $\hat{\beta} = 1.12$ ($t = 3.7$).

Multi-outcome events. Mutually exclusive events with at least three outcomes have $\hat{\beta} = 1.28$ ($t = 4.4$). Table 6 shows where their mispricing sits.

- *Overround.* Summed asks exceed one by 9.7% ($t = 16.6$) in multi-outcome events and by 2.8% in two-leg events. That is the cost of buying the whole book, and it is mostly spread. In two-leg events the mids sum to one (-0.1%). In multi-outcome events the mids themselves exceed one by 2.3% ($t = 7.8$): even at the mid, the book is overpriced.
- *Shin's insider share.* The median z is 0.017 in multi-outcome and 0.020 in two-leg events. These are of the same order as estimates for bookmaker markets [45], but on an exchange z mostly restates the bid–ask spread. Median z is similar across groups (Table 30).
- *The field.* Per event and lead, the legs priced at or below 10 ¢ cost 9.7 ¢ together and pay out 5.4 ¢ on average: an overpricing of 4.3 ¢ ($t = 3.4$). Realised minus price is 1.46 ¢ for the favourite leg ($t = 0.8$) and -0.55 ¢ for legs ranked sixth or lower ($t = -3.8$).

Figure 21 in the appendix plots these errors by rank. Shin's adjustment moves probability from longshots to favourites, but with z near 0.017 the move is small. The field remains overpriced by 4.0 ¢ after the adjustment ($t = 3.2$), and the Brier score barely changes (0.0611 raw, 0.0614 Shin). The longshot overpricing in these events is therefore not the price protection against insiders that Shin's model implies. It is closer to many small bets spread over the field. Within multi-outcome events the slope does not rise with the number of outcomes: it is 1.26 for 3–5 legs, 1.17 for 6–10, 1.41 for 11–20 and 1.25 for more.

6.1.4 Liquidity and calendar time

Liquidity. The bias is not confined to thin markets (Figure 5, Table 26). Across within-group terciles of trailing 7-day volume, the slope is 1.13, 1.13 and 1.12 ($t = 4.1$ in the top tercile). Across open-interest terciles it is 1.12, 1.14 and 1.13, and across spread terciles 1.17 (tightest), 1.11 and 1.13 (widest). Longshot returns at the mid are, if anything, more negative in the most active tercile: -60.9%, against -53.3% in the least active. Volume-weighted estimates are steeper than all terciles (Section 6.1.1). The very largest markets, which dominate volume-weighted averages but are a small part of the top tercile, are the most biased.

Table 5: The bias by contract structure

				Gross, at mid			Net, at ask	
Structure	Markets	Events	$\hat{\beta}$	LS	35–65 ¢	FAV	LS	FAV
Binary	6,028	2,644	1.120	−64.1	+0.2	+1.7	−79.0	−0.0
			(3.7)	(−9.0)	(0.4)	(6.4)	(−17.8)	(−0.0)
Multi-outcome	3,441	474	1.283	−71.8	+0.6	+1.1	−84.4	+0.4
			(4.4)	(−8.3)	(0.6)	(7.2)	(−18.1)	(0.6)
Ladder leg	7,029	1,030	1.118	−30.9	+0.4	+0.9	−61.6	−1.4
			(2.9)	(−2.3)	(0.9)	(3.9)	(−8.8)	(−2.9)
Deadline	2,509	644	1.224	−68.3	+3.3	+0.9	−81.1	−1.1
			(3.9)	(−10.7)	(2.4)	(4.0)	(−20.0)	(−1.4)

Structures as defined in Section 5.1. Returns per dollar (%) of sides in three price regions; market-weighted; pre-event leads pooled; t-statistics clustered by event in parentheses ($t_{\beta=1}$ under $\hat{\beta}$).

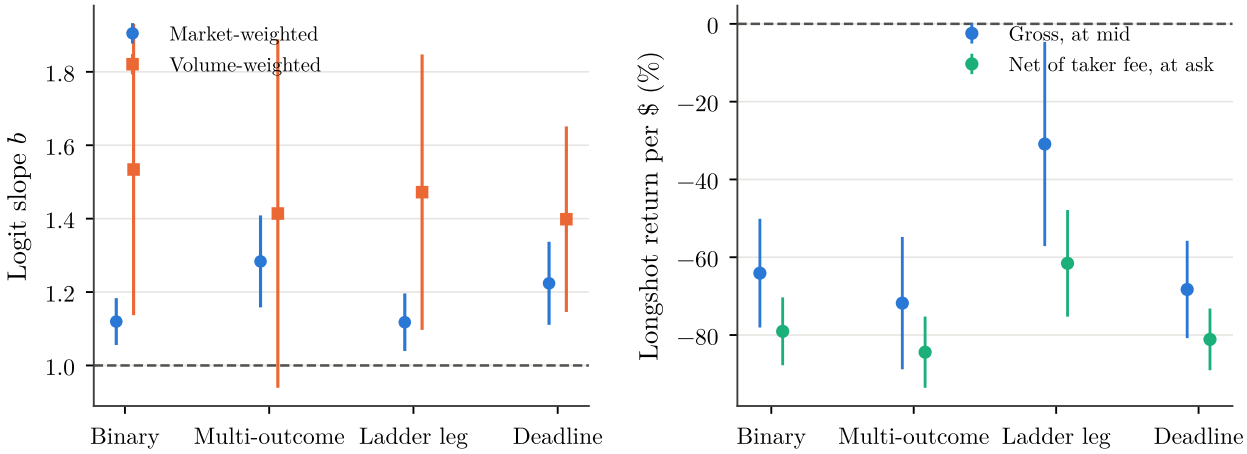


Figure 4: Contract structure. Left: calibration slope, market- and volume-weighted. Right: return per dollar on longshot sides ($q \leq 10$ ¢), gross at the mid and net of the taker fee at the ask. 95% confidence intervals clustered by event.

Table 6: Mutually exclusive events: overround, Shin z and where the mispricing sits

	Events	Event-leads	Legs	$\sum a - 1$ (%)	$\sum p - 1$ (%)	$\sum b - 1$ (%)	Mean z	Median z
Two-leg events	740	2,009	2	+2.8 (41.3)	−0.1 (−3.2)	−3.0	0.030	0.020
Multi-outcome events (≥ 3 legs)	381	1,289	4	+9.7 (16.6)	+2.3 (7.8)	−5.2	0.025	0.017

Leg rank	Legs	Mid (¢)	Win (%)	Win − mid (¢)	Win − normalised (¢)	Win − Shin (¢)
1 (favourite)	1,289	68.4	69.8	+1.46 (0.8)	+3.08 (1.6)	+3.33 (1.7)
2	1,289	18.2	16.4	−1.80 (−1.1)	−1.55 (−0.9)	−1.65 (−1.0)
3-5	2,565	6.8	6.5	−0.33 (−0.4)	−0.24 (−0.3)	−0.34 (−0.4)
6+	2,933	0.9	0.4	−0.55 (−3.8)	−0.47 (−3.3)	−0.44 (−3.1)
Field (≤ 10 ¢ legs)	762	9.7	5.4	−4.34 (−3.4)	−3.77 (−3.0)	−3.97 (−3.2)

Top: events flagged mutually exclusive by Kalshi with every leg quoted at the lead and exactly one leg settling YES; one observation per event and lead. a, p, b: YES ask, mid and bid; z from the asks via (6). Bottom: legs of events with at least three legs, ranked by mid within the event and lead. Normalised: mid divided by the event's sum of mids. Shin: $\hat{p}_k(z)$. Field: per event and lead, all legs priced at or below 10 ¢ together (their summed price against the number that win). t-statistics clustered by event in parentheses.

Calendar time. The slope is above one in 17 of the 22 quarters from 2021Q3 to 2026Q4 (Figure 5 and Table 27).

- Before 2025 the venue was small (about 1,239 in-scope markets a quarter in 2022), and the quarterly slopes are noisy, ranging from 0.84 to 1.40.
- From 2025, as sports listings took the venue to 1,209,781 settled markets in 2026Q3, the slope settles between 1.08 and 1.19 in all 8 quarters.

Growth by three orders of magnitude neither removed the bias nor visibly reduced it. Whatever capital entered on the other side was not enough to remove it.

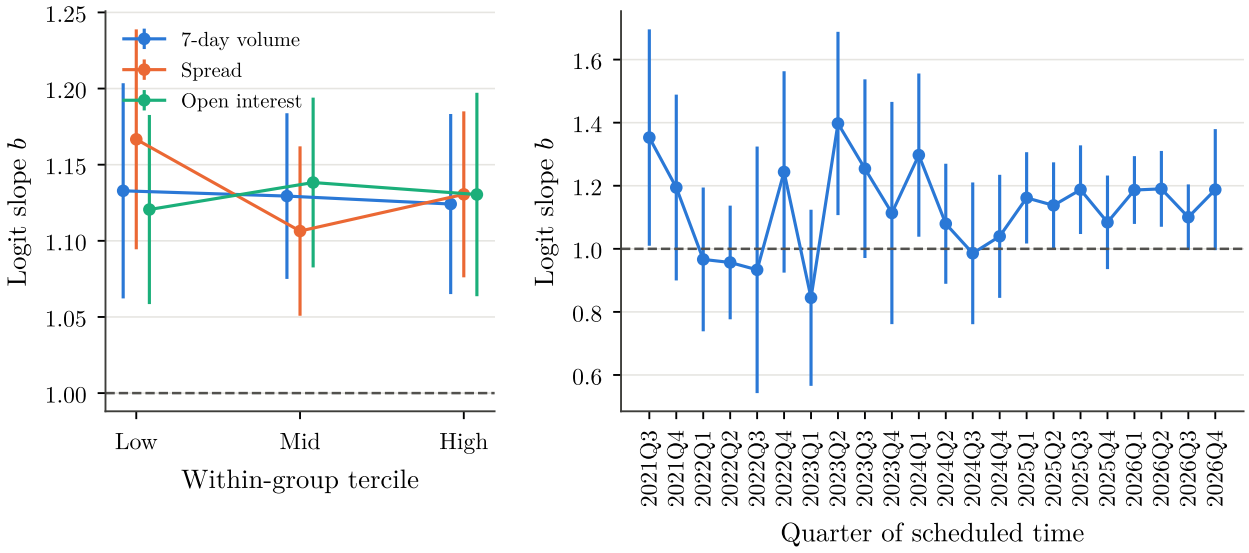


Figure 5: Left: calibration slope by within-group tercile of 7-day volume, quoted spread and open interest at the quote. Right: calibration slope by quarter of the scheduled time, all groups. Both market-weighted, pre-event leads pooled; 95% confidence intervals clustered by event.

6.1.5 Fees, the tick and the limits to arbitrage

Fees. Kalshi's taker fee is $0.07\,q(1-q)$ per contract, or $7(1-q)\%$ per dollar staked, so it is almost flat per dollar for longshots (Figure 6, left). Kalshi computes the fee per fill to \$0.000001 and rebates rounding over-payment [29], so the exact fee is the fee paid. Under a rule that rounded each order up to the cent, a single 1 ¢ contract would pay a 1 ¢ fee, 100% of the stake; the figure shows this case for comparison. Net of fees and the half-spread, longshot returns fall from -55.1% to -74.7% (Table 2). Fees and spread thus deepen the longshot buyer's loss without changing its sign.

Taking the other side. The arbitrageur's question is whether buying the favourite side of every longshot pays (Table 7, bottom). At the mid it returns 1.11% per dollar ($t = 9.8$). As a taker at the ask, net of fees, it returns -0.49% ($t = -1.7$): the spread and fee absorb the whole premium. Resting at the bid as a maker would return 2.64% ($t = 21.3$) if every order filled at the quote. That is an upper bound: fills on resting orders are adversely selected (Section 6.2).

The no-arbitrage band. Figure 6 (right) sets the mispricing per contract against the cost of trading against it.

- In the longshot buckets, price minus realised frequency is 0.49, 1.86 and 2.01 cents per contract for 1–2, 3–5 and 6–10 ¢.
- The taker's cost (half-spread plus fee on the opposite side) is 0.79, 2.48 and 2.56 cents.
- The mispricing lies inside the taker band in 12 of 12 buckets; in the three longshot buckets it takes up 62–78% of the band.

The longshot overpricing thus sits close to the edge of the band: most of the cost of removing it, and no more. This is the classic limit to arbitrage [46], here created by the spread and the fee schedule rather than by capital constraints. The capital term of the band in Section 4 adds little at these horizons: at a 5% annual cost of capital, holding a 95 ¢ favourite for 30 days costs about 0.4 ¢ per contract, and for one day about 0.01 ¢.

The tick floor. The floor binds often.

- 34% of quotes have a side priced at or below 2 ¢, and 16% sit at the floor with no bid and a 1 ¢ ask.
- Sides pinned there win 0.13% of the time against a 0.5 ¢ mid. A contract worth about a tenth of a cent cannot be quoted lower.
- Splitting the 1–2 ¢ bucket's loss of 0.49 cents per contract, 45% is mechanical, the loss of quotes at the floor.
- For a taker paying a 1–2 ¢ ask, the mechanical share is 68%: the 1 ¢ ask alone costs nearly a cent against a win rate of 0.29%.

The rest of the 1–2 ¢ loss comes from prices above the floor. Only 0.7% of markets quote on a 0.1 ¢ grid, too few for a clean comparison.

6.1.6 Robustness

The appendix collects the robustness checks (Tables 28 and 29).

- *Sample halves.* Splitting at the median observation date (2025-07-28) gives $\hat{\beta} = 1.10$ ($t = 3.5$) and 1.15 ($t = 5.3$), and longshot returns of −48.5% and −62.9%. The slope exceeds one in 7 of 7 groups in the first half and 6 of 6 in the second.
- *Without the tick-floor buckets.* Dropping quotes with a mid at or below 2 ¢ or at or above 98 ¢ leaves $\hat{\beta} = 1.16$ ($t = 7.9$). The slope is therefore not an artefact of the floor.
- *Other filters and weights.* Restricting to spreads of at most 5 ¢ ($\hat{\beta} = 1.12$) or quotes at most a day old (1.13), or reweighting by the inverse sampling probability (1.15, $t = 2.2$), changes little.
- *Clustering.* The test of $\beta = 1$ has $t = 7.7$ when clustered by market, 6.3 by series and 7.7 by calendar month.

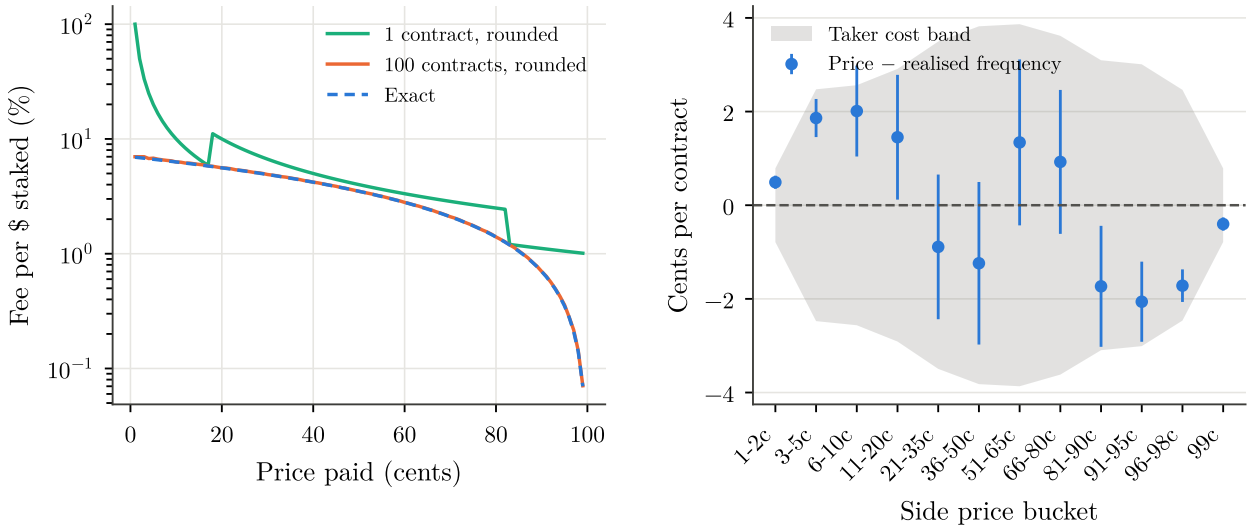


Figure 6: Limits to arbitrage. Left: Kalshi taker fee per dollar staked against the price paid, exact and rounded up to the cent for one order of 1 or 100 contracts (log scale). Right: mispricing per contract at the mid (side price minus realised frequency, with 95% confidence intervals clustered by event) against the cost of trading against it as a taker (half-spread plus the taker fee on the opposite side, shaded), by bucket of the side price.

Table 7: The tick floor and the other side of longshots

	Sides	Price (¢)	Win (%)	Loss (¢)	Mechanical (¢)	Share	Return (%)
All 1–2 ¢ sides (1 ¢ grid)	25,800	0.87	0.38	0.49	0.22	45%	−61.6 (−7.4)
at the floor (mid 0.5 ¢)	15,747	0.50	0.13	0.37	0.37	100%	−73.5 (−7.5)
above the floor	10,053	1.44	0.77	0.67	0.00	0%	−43.3 (−3.7)
0–2 ¢ sides, 0.1 ¢ grid	384	0.75	0.00	0.75	0.01	1%	−100.0
Taker at a 1–2 ¢ ask	20,948	1.25	0.29	0.95	0.65	68%	−81.6 (−17.1)

Buying the favourite side	Sides	Gross, mid (%)	Gross, ask (%)	Taker net (%)	Maker net (%)
All groups	40,552	+1.11 (9.8)	−0.27 (−0.9)	−0.49 (−1.7)	+2.64 (21.3)
Sports	9,108	+1.11 (4.8)	+0.06 (0.1)	−0.17 (−0.2)	+2.58 (9.8)
Weather	3,905	+1.51 (4.0)	+0.05 (0.1)	−0.19 (−0.3)	+3.41 (8.6)
Politics	11,768	+0.52 (2.2)	−1.38 (−2.5)	−1.59 (−2.9)	+1.95 (7.8)
Entertainment	8,939	+1.36 (6.9)	+0.16 (0.2)	−0.06 (−0.1)	+2.97 (13.1)
Mentions	1,918	+1.63 (3.4)	+0.77 (0.9)	+0.49 (0.6)	+2.92 (5.5)
Science & Tech	4,572	+1.16 (3.7)	−0.20 (−0.3)	−0.40 (−0.5)	+2.57 (7.3)
Other	342	+2.85 (5.7)	+2.66 (2.9)	+2.42 (2.6)	+4.40 (8.1)

Top: sides priced at most 2 ¢. Loss: price minus realised win rate, per contract. Mechanical: the loss of sides quoted at the floor (mid 0.5 ¢ with no bid, or a 1 ¢ ask for takers), $\max(\text{floor} - \text{win}, 0)$, times their weight share. Return: gross per dollar at the mid, or net at the ask for the taker row. Bottom: buying the favourite side ($q \geq 90$ ¢ at the mid) of every quote: at the mid; at the ask as a taker, gross and net of the taker fee; and resting at the bid as a maker, net of the maker fee where the series charges one. The maker column assumes a fill at the quote and ignores adverse selection, so it is an upper bound. Market-weighted; t-statistics clustered by event in parentheses.

6.1.7 Mechanical and behavioural components

The measures above separate the components that Section 4 distinguishes.

- *Mechanical.* The tick floor explains about half of the loss in the cheapest bucket (45% at the mid, 68% for a taker at the ask). Fees and spreads deepen buyers' losses and make the remaining mispricing uneconomic to trade against: the longshot overpricing lies inside the taker band, close to its edge. In this sense the bias survives fees for the buyer and does not survive them for the arbitrageur.
- *Not only mechanical.* The slope is 1.16 without the floor buckets. It does not shrink at short horizons, where capital costs vanish. It is as strong where volume and open interest are high as where they are low, and it is present in every group and in both halves of the sample. Mechanics set the band; something else keeps prices near its edge.
- *Not insiders.* In multi-outcome events, Shin's z is about 0.017 and largely a restatement of the spread, and the Shin adjustment leaves the field overpriced. This is contrary to the information hypothesis of Section 4.
- *Consistent with demand for lottery-like payoffs.* The bias is steepest in multi-outcome fields, in deadline contracts and in entertainment, where outcomes have no external anchor. It is weakest in politics and in pre-game sports with a known start, where polls and bookmaker lines provide one. This pattern is consistent with the preference and belief-dispersion explanations of Section 4, although the cross-group differences are modest. Sections 6.2 and 6.3 ask who is on each side of these trades.

6.2 Who pays: takers, trade size and timing

This subsection uses the main trade sample: 2,685,117 pre-resolution prints in 14,164 markets and 2,151 events. They make up 1,643,534 taker orders, 277 million contracts and $132 million of taker notional (Table 32). Two features of the data shape how we report the results.

- *Pooled means are dominated by sports and by a few upsets.* Sports carry 98.6% of the frame's volume, so design-weighted pooled means are close to sports means. Longshot payoffs are also extremely right-skewed. In the pooled taker longshot estimate, a single MLB home-run prop event contributes 40% of the design-weighted profit, and one Challenger tennis match another 29%. Pooled per-dollar returns on longshots are therefore imprecise. They can even change sign when one event is added or removed.
- *Each stratum is a self-weighting random sample.* We therefore rely on statistics that are robust to this: each stratum's own estimate, a sign test across the 18 independent strata, and the median stratum in the figures. We also rely on quantities that are not heavy-tailed, namely who trades at which price, spreads and fees.

Pooled estimates with event-clustered t-statistics are reported throughout as well.

6.2.1 Takers against makers

Across all prices, liquidity role does not decide who loses. Pooled over every print, takers earn -2.4% per dollar after fees ($t = -0.7$) and makers -0.4% ($t = -0.1$). The difference is -2.0 percentage points ($t = -0.3$; Table 8). Takers do worse than makers in only 9 of the 18 strata. Some groups show clear role differences, but they point both ways:

- in Mentions, takers lose 5.1% ($t = -2.3$) while makers earn 1.6%;
- in Weather, takers lose 3.0% ($t = -2.4$);
- in Science & Tech the sign reverses: takers earn 4.5% ($t = 5.8$) and makers lose 7.9% ($t = -6.5$).

So the maker advantage that Bürgi et al. [14] document on average is not uniform across categories.

At the longshot end, both roles lose, and takers lose more. Figure 7 and Table 9 show the per-stratum results.

- In the median stratum, takers who buy at 1–2 ¢ lose 89% of the stake. At 3–5 ¢ they lose 69%, and at 6–10 ¢ 52%.
- Makers who hold sides at the same prices lose 48%, 27% and 27%.
- Taker longshot positions lose in 14 of 18 strata (one-sided sign test $p = 0.015$). Maker longshot positions lose in 12 of 18 ($p = 0.12$). Takers do worse than makers on longshots in 13 of 18 ($p = 0.048$).
- The losses are large and precisely estimated in politics (-57%, $t = -3.0$), entertainment (-28%, $t = -3.1$), mentions (-63%, $t = -5.6$) and most sports leagues. In NBA, NCAAF, NHL and the other-sports stratum they reach -84 to -97%, with $|t| > 8$.
- The four exceptions are MLB, golf, tennis and Science & Tech. Each is driven by one or two events in which a heavily traded longshot won. Without its two most profitable events, the stratum's taker longshot return falls to -64% in golf and -85% in tennis, and to $+22\%$ in MLB and $+12\%$ in Science & Tech.
- Favourites earn roughly zero for both roles: between -1% and $+3\%$ per dollar in the median stratum at $q \geq 81$ ¢.

This is the pattern of Bürgi et al. [14]: the bias is present for both roles and stronger for takers. Table 34 gives the pooled estimates by bucket, and Figure 22 gives them by group.

Takers are the majority buyers of longshots. Takers hold 56–64% of the contracts bought at $q \leq 20$ ¢, but only 39–44% of those bought at $q \geq 81$ ¢ (Figure 7b). The typical longshot print is therefore a taker buying the longshot from a resting order that sells it, and the maker ends up holding the favourite.

Why takers lose more: the price of immediacy. Table 13 splits the taker's gross profit into the mispricing of the mid-quote and the effective half-spread.

- On longshots (mean price 5.5 ¢) the taker pays a half-spread of 0.51 ¢ per contract ($t = 14.4$) and a fee of 0.33 ¢.
- Together this is 14% of the stake: 14% in sports and 21% in non-sports.
- The same cost is 4% for mid-priced contracts and 1.5% for favourites.

The immediacy cost is precisely measured, and it falls on the taker alone. The remaining term, the gap between the outcome and the mid-quote, is the bias in the price itself. Takers and makers share it, and trade by trade it is too noisy to pin down in the pooled sample (+2.4 ¢, $t = 0.8$). Section 6.1 measures it from prices. Because the immediacy cost is a fixed amount per contract, it is largest per dollar on the cheapest contracts. Takers pay it there, which is exactly what a favourite–longshot pattern in taker returns requires.

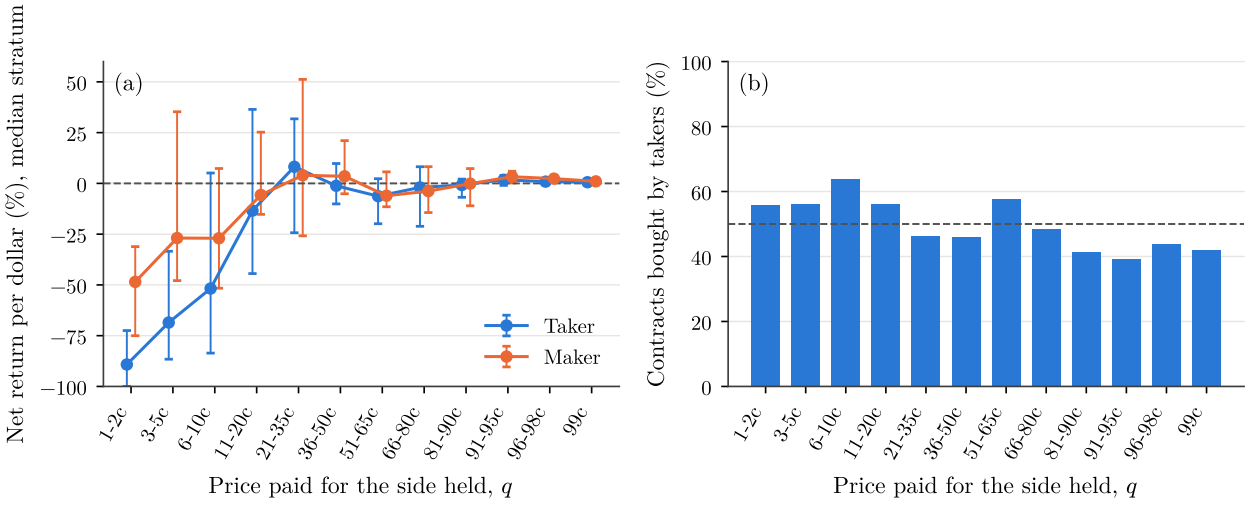


Figure 7: Takers against makers by the price of the side held. (a) Net return per dollar (dollar-weighted, after each role's fee) of taker and maker positions: the median across the 18 strata of each stratum's estimate, with bars spanning the interquartile range across strata (strata with at least five events in the cell). (b) Takers' share of the contracts held at each price, pooled and design-weighted. Main sample.

Table 8: Net return per dollar of takers and makers, by group

	All positions			Longshot positions ($q \leq 10$ ¢)			
Group	Taker	Maker	Difference	Taker	Maker	Taker share	Events
Sports	−2.4 (−0.7)	−0.4 (−0.1)	−2.0 (−0.3)	+52.0 (1.0)	+49.9 (0.8)	59	1,003
Weather	−3.0 (−2.4)	+1.2 (1.3)	−4.2 (−1.9)	−15.7 (−0.3)	+85.4 (0.7)	61	297
Politics	+1.1 (0.6)	−2.1 (−1.2)	+3.2 (0.9)	−57.2 (−3.0)	−37.9 (−2.2)	61	258
Entertainment	+0.3 (0.5)	−1.3 (−2.3)	+1.6 (1.4)	−27.9 (−3.1)	−28.4 (−2.4)	56	258
Mentions	−5.1 (−2.3)	+1.6 (1.7)	−6.8 (−2.1)	−62.9 (−5.6)	−36.9 (−2.2)	76	119
Science & Tech	+4.5 (5.8)	−7.9 (−6.5)	+12.4 (6.5)	+92.1 (1.5)	−8.2 (−0.2)	36	216
Non-sports	−0.9 (−1.0)	−0.4 (−0.5)	−0.5 (−0.3)	−31.0 (−1.5)	+11.9 (0.3)	60	1,148
All	−2.4 (−0.7)	−0.4 (−0.1)	−2.0 (−0.3)	+48.9 (1.0)	+48.4 (0.8)	59	2,151

Main sample, design-weighted. Net return per dollar is dollar-weighted (8) after each role's fee, in percent. t-statistics clustered by event are in parentheses. Difference: taker minus maker on the same prints, in percentage points. Longshot positions are positions in a side bought at $q \leq 10$ ¢: the taker's side when the taker bought the longshot, and the maker's side when the taker bought the favourite. Taker share: the takers' share of longshot contracts held. All pools the groups with design weights and is dominated by sports.

Table 9: Longshot positions by stratum: returns, buyers and spreads

		Longshot return (%/$)			Share at $q \leq 10\,¢$ (%)		Half-
Stratum	Events	Takers	Makers	Orders ≤ $100	Orders ≤ $10	Orders > $1k	spread (¢)
Combat	59	−65 (−2.4)	−75 (−3.1)	−63 (−2.1)	26	0.2	+0.42
Cricket	60	−57 (−1.8)	−36 (−0.9)	−59 (−2.3)	66	0.8	+0.55
Esports	85	−86 (−8.0)	+37 (0.4)	−80 (−5.7)	24	0.0	+0.96
Golf	30	+141 (2.1)	+73 (1.9)	+218 (2.3)	87	36.2	+0.47
MLB	87	+223 (3.9)	+96 (1.0)	+191 (3.5)	55	2.1	+0.61
NBA	49	−97 (−46.0)	−70 (−4.4)	−95 (−31.5)	43	0.0	+0.67
NCAAF	78	−86 (−11.6)	−35 (−1.1)	−78 (−7.0)	38	0.0	+0.29
NFL	60	−49 (−1.5)	+28 (0.8)	−63 (−3.1)	14	1.0	+0.48
NHL	60	−84 (−8.7)	−59 (−1.8)	−80 (−6.6)	54	0.0	+0.40
Other sports	123	−87 (−11.4)	−73 (−5.7)	−86 (−10.3)	77	0.0	+0.34
Soccer	113	−51 (−2.1)	−23 (−0.6)	−43 (−1.4)	39	0.5	+0.46
Tennis	139	+55 (0.5)	+130 (0.7)	+66 (0.6)	40	1.5	+0.53
WNBA	60	−66 (−2.8)	−5 (−0.2)	−63 (−2.5)	31	0.0	−0.11
Entertainment	258	−28 (−3.1)	−28 (−2.4)	−38 (−4.0)	79	0.3	+0.62
Mentions	119	−63 (−5.6)	−37 (−2.2)	−69 (−6.9)	79	0.2	+0.47
Politics	258	−57 (−3.0)	−38 (−2.2)	−39 (−1.6)	78	5.0	+0.65
Science & Tech	216	+92 (1.5)	−8 (−0.2)	+72 (1.0)	83	3.7	−1.43
Weather	297	−16 (−0.3)	+85 (0.7)	−14 (−0.3)	70	0.0	+0.71
Sign test, k/n (p)		14/18 (.015)	12/18 (.119)	14/18 (.015)	18/18 (<.001)		

Main sample, pre-resolution prints. Within a stratum every event has the same design weight, so each row is a self-weighting random sample. Longshot positions are sides held at $q \leq 10\,¢$; returns are dollar-weighted (8) after each role's fee, with event-clustered t-statistics in parentheses. Shares: the share of taker contracts at $q \leq 10\,¢$ within taker orders of each size. Half-spread: the effective half-spread s of taker prints at $q \leq 10\,¢$ with a tight mid-quote proxy. Last row: the number k of the n strata with a negative return (for the share columns, with a larger share in ≤ $10 orders than in > $1k orders) and the one-sided sign-test p-value.

6.2.2 Trade size: do small takers buy longshots?

Small takers buy longshots. Orders of at most \$10 are 57.8% of taker orders but only 5.0% of taker contracts. Orders above \$1k are 1.1% of orders and 42.3% of contracts (Table 10). The share of contracts bought at $q \leq 10\,¢$ falls steeply with order size:

- 38.9% in orders of at most \$10;
- 17.7% at \$10–100, 8.5% at \$100–1k and 1.2% above \$1k;
- the mean price paid rises from $22\,¢$ to $59\,¢$ across the same size buckets.

The gradient is the most robust result of this subsection.

- It holds in every one of the 18 strata (sign test $p < 0.001$; Table 9).
- It holds in both halves of the sample (41% against 1.1%, and 37% against 1.3%), under the strict pre-resolution rule (30% against 1.5%) and in the replication sample (62% against 5%; Table 15).
- It is steepest outside sports. In non-sports markets, 74% of the contracts in orders of at most \$10 are longshots, against 2.6% in orders above \$1k (sports: 36% against 1.2%; Table 36).

As a result, orders of at most \$100 place 17.5% of taker dollars but 59% of the taker dollars staked on longshots. Orders above \$1k place 51% of taker dollars but 8% of longshot dollars.

Is the loss concentrated in small trades? In quantity yes, in rate no.

- Per dollar, a longshot loses about the same at every size where it can be measured. The median-stratum taker return on longshots is −52% for orders of at most \$10, −55% at \$10–100 and −52% at \$100–1k (Figure 8). The full price profile is similar for each size bucket (Figure 9).
- Longshot orders above \$1k are too rare to estimate: only one stratum has five or more events with such orders.
- Small-order (at most \$100) longshot positions lose in 14 of 18 strata ($p = 0.015$).
- Favourites earn close to zero at every size.

The longshot loss is concentrated in small trades because small trades hold most of the longshots, not because small traders get worse prices for them. A small order pays the same half-spread per contract as a large one (Table 12).

6.2.3 YES and NO takers

Longshot takers buy YES. YES accounts for 79–85% of the taker contracts bought at $q \leq 20\,¢$, against 45–53% at $q \geq 91\,¢$ (Table 35). At equal side price, YES-takers do worse than NO-takers in 10 of the 12 buckets of the median stratum (Figure 10). Examples:

- at 6–10 ¢, −59% against −14%;
- at 11–20 ¢, −20% against +27%;
- at 81–90 ¢, −3% against +5%.

Table 10: Taker flow and returns by order size

	Share of taker		Share at $q \leq 10\,¢$			Taker net return per dollar		
Order size	orders	contracts	contracts	orders	Mean q (¢)	Longshots	Favourites	Longshots, per print
≤$10	57.8	5.0	38.9	10.5	22	+30.6 (0.8)	−2.6 (−1.4)	+11.1 (0.4)
$10–100	32.6	21.1	17.7	3.1	35	+52.3 (1.1)	−1.3 (−0.6)	+44.1 (1.2)
$100–1k	8.5	31.7	8.5	1.2	47	+26.1 (0.6)	−1.4 (−0.6)	+39.0 (0.8)
>$1k	1.1	42.3	1.2	0.4	59	+164.3 (1.2)	−3.6 (−0.9)	+67.4 (0.7)

Main sample, all groups, taker positions, design-weighted. Order size is the taker order's dollar notional (prints with the same market, timestamp and taker side). Shares in percent. Returns are dollar-weighted (8) after the taker fee, in percent, with event-clustered t-statistics in parentheses. Per print: each fill weighted equally. Longshots $q \leq 10\,¢$; favourites $q \geq 90\,¢$.

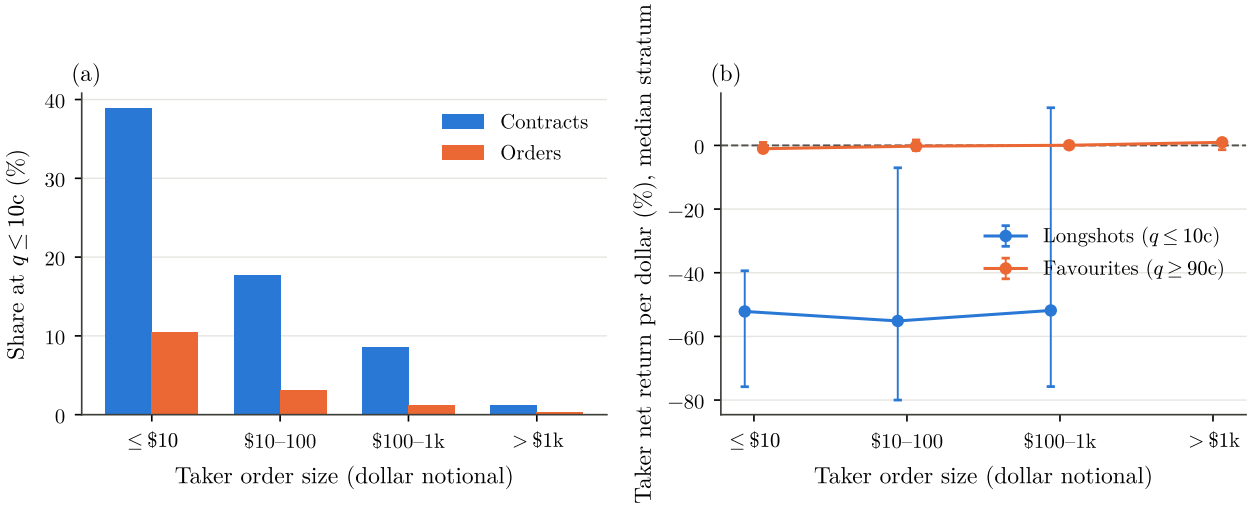


Figure 8: Order size and the longshot. (a) Share of taker contracts and taker orders at $q \leq 10\,¢$, by order size, pooled and design-weighted. (b) Taker net return per dollar on longshots ($q \leq 10\,¢$) and favourites ($q \geq 90\,¢$) by order size: median across strata with interquartile range, shown where at least six strata have five or more events in the cell (longshot orders above $1k qualify in only one stratum). Main sample.

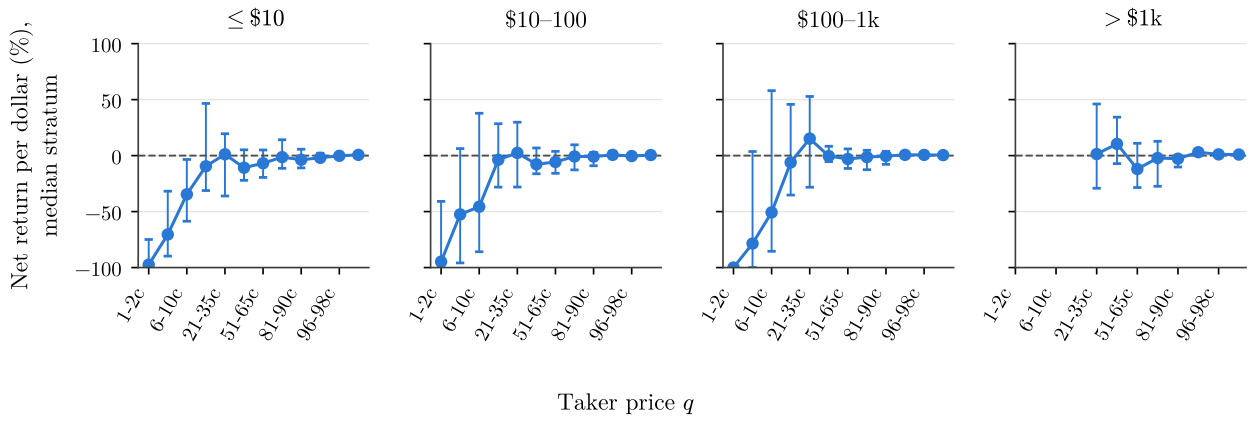


Figure 9: Taker net return per dollar by price bucket, one panel per order size: median across strata with interquartile range (cells with at least six qualifying strata). Main sample.

The pooled differences are significant at 66–80 ¢ (−20.1 percentage points, $t = -2.3$) and 81–90 ¢ (−25.8, $t = -3.2$). For favourites as a whole the pooled difference is −6.2 points ($t = -1.8$). For longshots it cannot be distinguished from zero ($t = 0.1$). There is a YES, or action, bias, but it is not concentrated in the longshot buckets. The link between the two biases is compositional: those who buy longshots overwhelmingly buy them as YES.

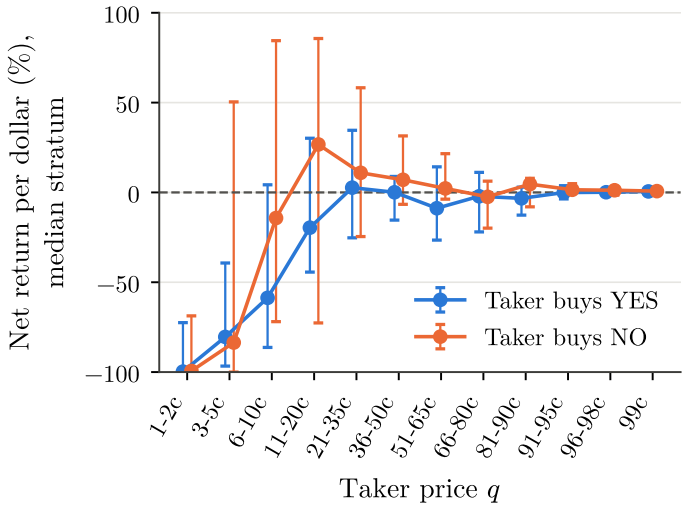


Figure 10: Net return per dollar of takers who bought YES and takers who bought NO, at the same side price: median across strata with interquartile range. Main sample.

6.2.4 Timing and attention

Attention does not bring in more small or longshot flow. Small orders dominate taker flow at every hour: about 89% of taker orders are of at most $100 throughout the day (Figure 11a; every cell is listed in Table 37). In the classic attention cells (evenings, weekends, primetime games and marquee events) neither the small-order share nor the longshot share moves significantly (Figure 11b). The busiest cells point the other way.

- In volume surges, in play, in a market's first 24 hours and in the last hour before expiry, the small-order share *falls*, by 2–4 percentage points (t between −2.9 and −5.8).
- During volume surges the longshot share of taker contracts falls by 13 points ($t = -2.6$).

Surges bring in relatively more large orders and fewer longshot purchases, not a retail rush into longshots.

Taker returns barely move with attention. Table 11 compares taker returns in each attention cell with the rest. Two contrasts are significant.

- *Weekends.* Taker returns are −11.0 points lower over all positions ($t = -2.3$) and −156 points lower on longshots ($t = -2.5$).
- *The last hour before scheduled expiry.* The bias steepens: longshots do 172 points worse ($t = -3.1$) and favourites 9.0 points better ($t = 2.2$).

Evening, primetime, marquee, in-play, new-listing and surge contrasts are all insignificant, with wide intervals. Of the 24 return contrasts in Table 11, four have $|t| > 2$, and they fall in only two cells. We therefore do not find that the bias in taker returns widens systematically when retail attention is high, in the sense of Barber and Odean [5] and Barber et al. [9]. The last-hour steepening is the exception: it moves both ends of the bias in the direction the bias predicts.

Table 11: Taker returns in high-attention trading

	All positions		Difference, cell minus rest (pp)			
Attention cell	In cell	Rest	All	Longshots	Favourites	Events
Evening, 18–24 ET	+0.9	−4.4	+5.3 (0.9)	−22.3 (−0.4)	+2.9 (0.6)	2151
Weekend (ET)	−10.2	+0.7	−11.0 (−2.3)	−156.3 (−2.5)	+7.5 (1.4)	2151
In play (sports)	−2.6	−3.4	+0.8 (0.1)	−57.1 (−0.5)	+2.6 (0.4)	777
Primetime game (sports)	−3.1	−2.7	−0.4 (−0.1)	−115.2 (−1.2)	+6.2 (0.9)	777
Marquee event (top decile)	−2.3	−3.3	+1.0 (0.2)	+73.1 (1.1)	−4.0 (−0.9)	2151
First 24 h after listing	−5.8	−1.4	−4.5 (−0.8)	+64.4 (0.6)	−10.4 (−1.1)	2151
Volume surge ($\geq 3\times$)	−1.5	−12.9	+11.3 (1.3)	−43.5 (−0.5)	+0.1 (0.0)	1515
Last hour before expiry	−5.2	−2.0	−3.3 (−0.7)	−172.4 (−3.1)	+9.0 (2.2)	2151

Main sample, all groups, taker positions, design-weighted. Net return per dollar is dollar-weighted (8) after the taker fee, in percent. Differences are in percentage points, with event-clustered t-statistics in parentheses. In-play and primetime contrasts are within sports games with a known scheduled start. A volume surge means the market's volume in the preceding hour is at least three times its own past hourly average. Marquee: the event's total volume is in the top decile of its stratum (ex post).

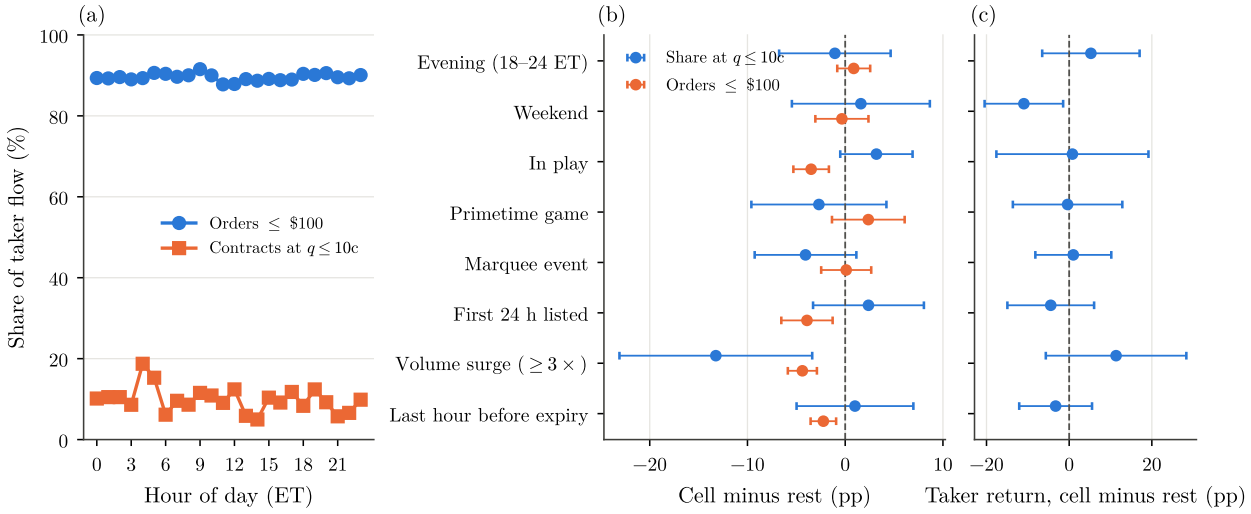


Figure 11: Timing and attention. (a) Share of taker orders of at most $100 and share of taker contracts at $q \leq 10\,¢$, by hour of day (ET). (b) Each attention cell minus the rest: the share of taker contracts at $q \leq 10\,¢$ and the share of taker orders of at most $100, in percentage points. (c) Each attention cell minus the rest: taker net return per dollar over all positions. 95% confidence intervals clustered by event. Main sample, design-weighted.

6.2.5 Order flow, price impact and information

Price impact rises with size. These results use the prints with a tight mid-quote proxy before and after the order: 36% of contracts, concentrated in the more liquid markets.

- Every size bucket pays a similar effective half-spread, 0.56–0.68 ¢ per contract ($t > 16$; Table 12).
- One minute after the order, the mid has moved in the taker's direction by 0.04 ¢ for orders of at most \$10 and by 0.41 ¢ for orders above \$1k (t from 2.5 to 4.9).
- One hour later, the median-stratum impact is 0.20, 0.66, 1.78 and 0.39 ¢ across the four size buckets.

Market-hour order flow. Regression (11) uses the 15,740 market-hours, in 651 events, with a tight proxy at both ends of the hour (Table 14, Figure 12).

- *Same hour.* An hour of entirely one-sided taker flow moves the mid by 0.28 ¢ if it is small ($t = 2.6$), 2.74 ¢ if medium ($t = 7.4$) and 3.26 ¢ if large ($t = 5.3$).
- *Next hour.* Small and medium flow keep moving the price in the same direction (0.35 ¢, $t = 2.8$; 0.58 ¢, $t = 2.6$). The impact of large flow neither continues nor reverts (−0.08 ¢, $t = -0.2$), while the mid itself partly reverses (coefficient on the same-hour change −0.12, $t = -2.2$).
- *To settlement.* No size class predicts the outcome. The estimates are 3.3 ¢ for small flow ($t = 1.3$), −0.4 ¢ for medium ($t = -0.3$) and −11.3 ¢ for large ($t = -1.6$).

Small flow is neither informed nor noise that reverts. Small taker flow moves prices little and does not predict outcomes, but its impact does not reverse within the hour either. This differs from equity markets, where the net buying of individual investors predicts returns over the following weeks [11, 30]. Large flow moves prices most, as the depth of a Kyle [32] market implies, but the outcome does not confirm its direction. If anything, the point estimate goes against it, in sports and non-sports alike. The per-order markouts to settlement are noisy and centred near zero (median stratum between −0.7 and +2.7 ¢). Combined with the immediacy cost above, this explains the taker's loss on longshots as a liquidity cost, not as adverse selection against informed makers. Takers lose because they pay the spread and the fee on cheap contracts. They do not lose because the side they buy predictably moves against them.

Table 12: Price impact and markouts of taker orders by size (cents per contract)

Order size	Half-spread s	Impact I_{1m}	Impact I_{1h}	Markout M_{1h}	Markout M_∞
≤\$10	+0.56 (21.2)	+0.04 (2.5)	+3.43 (2.6)	+2.87 (2.1)	+21.69 (1.7)
\$10–100	+0.58 (19.6)	+0.10 (3.5)	+2.91 (2.3)	+2.33 (1.8)	+13.28 (1.3)
\$100–1k	+0.68 (18.3)	+0.19 (4.9)	+1.03 (1.6)	+0.35 (0.5)	+2.18 (0.5)
>\$1k	+0.68 (16.8)	+0.41 (4.1)	−1.40 (−1.4)	−2.08 (−2.1)	−6.86 (−0.8)

Main sample, all groups, taker prints with a mid-quote proxy no wider than 5 ¢ just before the order and a proxy one minute and one hour after it. Contract-weighted and design-weighted, in cents per contract, from the taker's side (9). M_∞ is the gross profit to settlement. t-statistics clustered by event are in parentheses.

6.2.6 Robustness

Table 15 repeats the headline numbers in subsamples. The results that hold everywhere are the size gradient in longshot buying and the absence of a taker–maker gap averaged over all prices.

Table 13: Spread or price? Decomposing the taker's gross loss (cents per contract)

Sample	Taker side	Mean q (¢)	Gross profit $d(Y-p)$	Mid to outcome $d(Y-m_{t-})$	Half-spread s	Fee f^T	Immediacy cost (%/$)
Sports	Longshots ($q \leq 10$ ¢)	5.5	+2.00 (0.6)	+2.51 (0.8)	+0.50 (14.0)	0.33	14.2
	Middle	49.4	−0.15 (−0.1)	+0.64 (0.3)	+0.78 (13.0)	1.34	4.2
	Favourites ($q \geq 90$ ¢)	94.9	−2.52 (−0.6)	−1.51 (−0.4)	+1.02 (11.0)	0.32	1.4
Non-sports	Longshots ($q \leq 10$ ¢)	3.8	−0.17 (−0.1)	+0.45 (0.3)	+0.62 (11.7)	0.25	21.2
	Middle	47.8	+1.61 (0.9)	+3.94 (2.8)	+2.34 (6.0)	1.31	7.4
	Favourites ($q \geq 90$ ¢)	96.9	−2.05 (−0.6)	+0.56 (0.2)	+2.61 (5.3)	0.21	2.9
All	Longshots ($q \leq 10$ ¢)	5.5	+1.94 (0.6)	+2.44 (0.8)	+0.51 (14.4)	0.33	14.4
	Middle	49.4	−0.14 (−0.1)	+0.65 (0.3)	+0.79 (12.8)	1.34	4.2
	Favourites ($q \geq 90$ ¢)	95.0	−2.51 (−0.6)	−1.44 (−0.4)	+1.07 (10.8)	0.32	1.5

Main sample, taker prints with a mid-quote proxy no wider than 5 ¢ just before the order. Contract-weighted and design-weighted, in cents per contract, with event-clustered t-statistics in parentheses. By construction the gross profit equals the mid-to-outcome term minus the half-spread. The first term is the bias in the quoted price that any buyer of the side faces, and the second is the cost of immediacy. Fee: the exact taker fee per contract. Immediacy cost: $(s + f^T)/(q + f^T)$, dollar-weighted, in percent of the stake.

Table 14: Signed taker imbalance by size class and subsequent mid-quote changes

		Imbalance OIB_c			Same-hour	Market-	
Sample	Change	Small	Medium	Large	change	hours	Events
All	Same hour	+0.28 (2.6)	+2.74 (7.4)	+3.26 (5.3)	–	15,740	651
	Next hour	+0.35 (2.8)	+0.58 (2.6)	−0.08 (−0.2)	−0.12 (−2.2)	9,611	381
	To settlement	+3.28 (1.3)	−0.45 (−0.3)	−11.26 (−1.6)	+0.01 (0.2)	15,740	651
Sports	Same hour	−0.63 (−1.3)	+1.84 (3.6)	+2.87 (3.6)	–	3,281	308
	Next hour	+0.66 (1.1)	+0.33 (0.7)	−0.06 (−0.1)	−0.43 (−2.6)	2,084	173
	To settlement	+8.24 (1.2)	+1.20 (0.2)	−6.50 (−0.8)	+0.02 (0.3)	3,281	308
Non-sports	Same hour	+0.45 (4.0)	+3.11 (5.9)	+3.64 (3.8)	–	12,459	343
	Next hour	+0.25 (2.1)	+0.44 (1.8)	+0.05 (0.2)	−0.02 (−0.4)	7,527	208
	To settlement	+2.61 (1.0)	−0.76 (−0.5)	−13.50 (−1.4)	+0.00 (0.1)	12,459	343

OLS of regression (11) on market-hours with taker volume and a tight (≤ 5 ¢) mid-quote proxy at both ends of the hour (and of the next hour, for the next-hour change). The dependent variable is in cents. The imbalances (10) are signed contracts of each size class (small $\leq$ \$100, medium \$100–1k, large $>$ \$1k) over the hour's volume. Same-hour change: the control $m_h^{\text{end}} - m_h^{\text{start}}$. t-statistics (CR1, clustered by event) are in parentheses. Market-hours are unweighted.

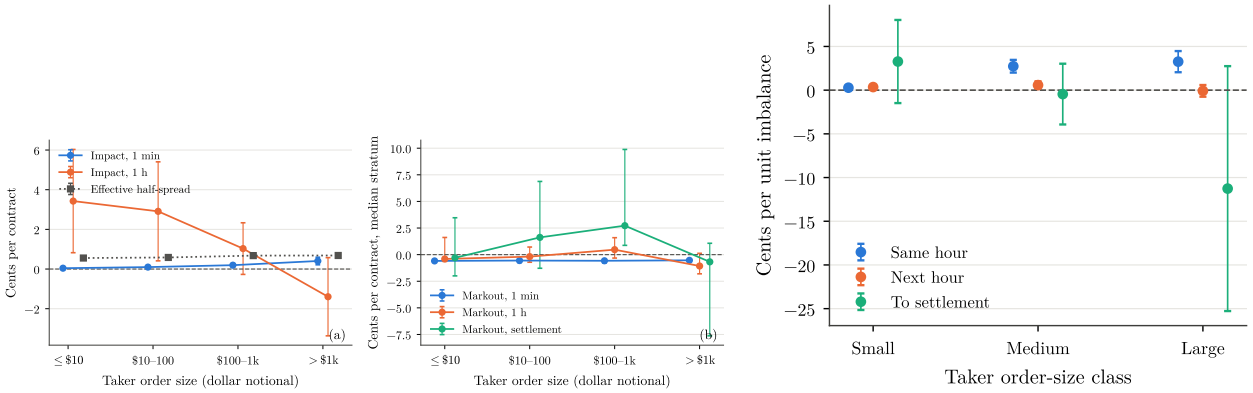


Figure 12: Price impact and information. Left: (a) effective half-spread and price impact at one minute and one hour (pooled, design-weighted, 95% intervals clustered by event), and (b) taker markouts at one minute, one hour and settlement (median across strata with interquartile range), by taker order size, in cents per contract. Right: coefficients of regression (11) on the signed imbalance of each size class, for the same-hour change, the next-hour change and the drift to settlement, with 95% intervals clustered by event.

- *Size gradient.* The longshot share of orders of at most \$10 exceeds that of orders above \$1k by 28 to 57 points in both halves, under the strict rule and in the replication sample.
- *No overall taker–maker gap.* Pooled taker returns lie between −0.9 and −3.5% and maker returns between −1.4 and +0.5%. Neither is significant in any variant. Clustering by market instead of event leaves the difference insignificant ($t = -0.1$).
- *Pooled longshot returns are unstable.* The pooled taker return on longshots is −35% in the first half and +125% in the second. The second half contains the MLB home-run prop that dominates the pooled sports estimate.
- *Small-order longshots.* Without design weights, longshot positions in orders of at most \$10 lose 28% ($t = -2.2$). In the replication sample, which is a much larger set of prints, they lose 37% ($t = -1.9$).
- *Excluded prices.* Dropping 1–2 ¢ and 99 ¢ prints changes nothing material.

Summary. The trade-level evidence partly supports the preference and retail-demand hypothesis (Hypothesis 2).

- Small takers are the longshot buyers. The size gradient holds in every stratum, and takers buy most longshots, mainly as YES.
- Takers lose more than makers on longshots. They pay a spread and a fee that together cost about 14% of the stake at the average longshot price of 5.5 ¢.
- But the per-dollar loss on a longshot is about the same for small and larger orders, and attention does not systematically widen it.

The information hypothesis (Hypothesis 3) finds no support at the trade level. Neither small nor large taker flow predicts outcomes beyond the price. The resting side of a longshot trade earns the spread; it is not exploiting the taker's ignorance of the outcome. Who pays the bias is therefore mostly a question of who demands immediacy on cheap contracts, and on Kalshi that is small takers.

Table 15: Headline results across subsamples and weights

	All positions		Taker longshots ($q \leq 10\,¢$)			LS share (%)
Sample	Taker	Maker	All orders	$\leq$ \$10 orders	$>$ \$1k orders	$\leq$ \$10 / $>$ \$1k
Main sample	−2.4 (−0.7)	−0.4 (−0.1)	+48.9 (1.0)	+30.6 (0.8)	+164.3 (1.2)	39 / 1
First half (by close)	−1.2 (−0.2)	−1.4 (−0.2)	−34.9 (−1.1)	−30.2 (−1.0)	−99.9 (−1853.9)	41 / 1
Second half	−3.5 (−1.3)	+0.5 (0.2)	+124.8 (1.7)	+75.2 (1.3)	+323.5 (1.6)	37 / 1
Strict pre-resolution	−3.1 (−0.3)	+0.1 (0.0)	+105.9 (1.2)	+109.4 (1.3)	+65.3 (0.4)	30 / 2
Excluding 1–2 ¢ and 99 ¢	−2.5 (−0.7)	−0.4 (−0.1)	+50.5 (1.0)	+39.9 (0.8)	+164.4 (1.2)	27 / 1
No design weights	−2.1 (−1.2)	−0.3 (−0.2)	−8.0 (−0.4)	−28.4 (−2.2)	−1.2 (−0.0)	57 / 2
Sports	−2.4 (−0.7)	−0.4 (−0.1)	+52.0 (1.0)	+36.1 (0.8)	+169.0 (1.2)	36 / 1
Non-sports	−0.9 (−1.0)	−0.4 (−0.5)	−31.0 (−1.5)	−35.2 (−1.3)	−85.1 (−4.6)	74 / 3
Replication sample	−3.0 (−0.7)	+0.2 (0.1)	+6.2 (0.2)	−36.6 (−1.9)	+40.3 (0.7)	62 / 5
Replication, strict	−3.2 (−0.8)	+0.4 (0.1)	+12.5 (0.4)	−24.2 (−1.0)	−5.5 (−0.1)	52 / 5

Net return per dollar, dollar-weighted (8) after each role's fee, in percent, with event-clustered t-statistics in parentheses. Halves split the main sample's markets at their median close time. Strict pre-resolution keeps only pre-game sports prints and other prints more than 24 hours before the close. The replication sample is the separate set of prints described in Section 3.2, which carries no design weights. LS share: the share of taker contracts at $q \leq 10\,¢$ in orders of at most \$10 and of more than \$1k.

6.3 Wallets, retail traders and venues

This subsection answers the four questions of the wallets workstream in turn: how Polymarket prices longshots, which wallets buy them, how concentrated and persistent wallet profits are, and how retail inflows and cross-venue prices behave.

6.3.1 Polymarket calibration and the bias by group

Table 16 and Figure 13 summarise calibration on Polymarket. Measured 24 hours before the information time, the pooled calibration slope is $\hat{\beta} = 0.99$ (SE 0.15; 2,121 markets), so $\beta = 1$ is not rejected ($t = -0.1$). The slope is 0.91 (SE 0.14) one hour before and 0.84 (SE 0.17) a week before. Weighting every sampled market equally gives 0.97 (SE 0.05). The slope is 1.28 in H1 and 0.97 in H2. By group, the 24-hour slopes are Sports 1.00, Politics 1.15, Weather 0.71, Entertainment 1.20, Mentions 0.84, Science & Tech 0.99, Other 0.82; 7 of the 7 are within two standard errors of one (Table 16).

Returns by price bucket are noisier. Buying every side priced at most 10 ¢ 24 hours ahead returns 84.5% ($t = 0.5$) per dollar market-weighted, −16.5% ($t = -0.2$) volume-weighted and 64.9% ($t = 0.8$) with equal weights. Buying every side priced at least 90 ¢ returns 1.9% ($t = 2.5$) and 2.3% ($t = 4.8$). Longshot returns are imprecise because a handful of 1–2 ¢ winners carry large design weights. The calibration slope pools all price levels and is the more reliable summary.

At the level of fills the bias is clear. Buying a longshot before the information time and holding it to settlement returns −53.8% ($t = -2.8$) per dollar net of fees; buying a favourite returns 2.2% ($t = 3.5$). The signs are the same in every group: longshot buys lose between 48% and 71% per dollar (significant at 5% in 5 of the 7 groups), and favourite buys earn between 1.1% and 3.5% (significant in 4; Table 16). The longshot loss is larger for takers (−71.2% ($t = -5.9$)) than for makers (−33.3% ($t = -1.2$)), and larger in H1 (−90.9% ($t = -18.7$)) than in H2 (−30.2% ($t = -1.0$)). Favourites earn 3.9% ($t = 6.9$) in H1 and 1.4% ($t = 1.7$) in H2. The two measures weight markets and moments differently. The fill-level return weights each dollar actually spent, including purchases in the last hours before the information time. The market-level calibration takes one price per market a day ahead, where the bias is weak. Section 6.1 reports the corresponding Kalshi estimates.

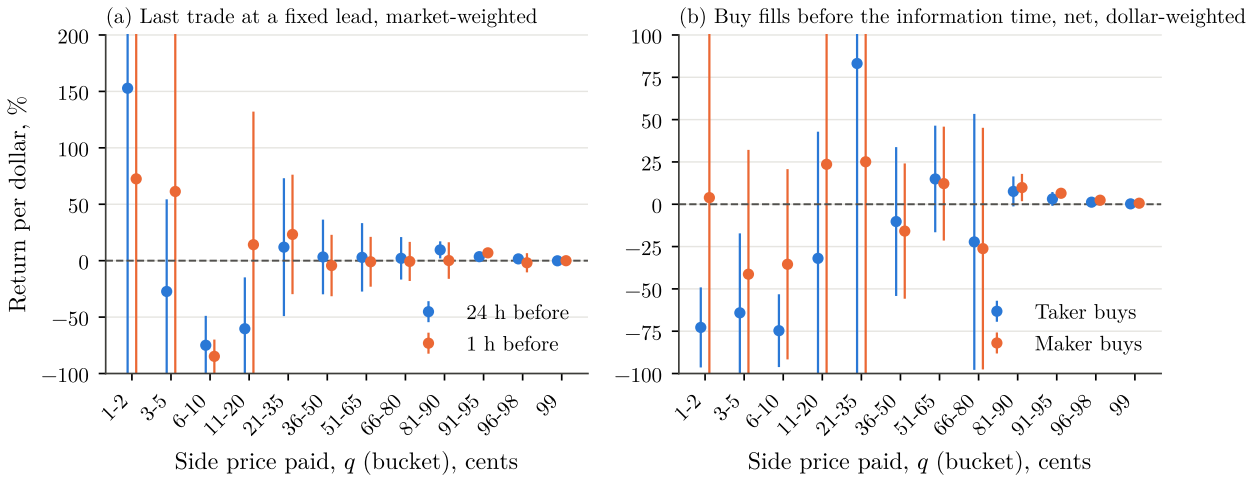


Figure 13: Return per dollar by side-price bucket on Polymarket. (a) Last trade 24 hours and 1 hour before the information time, both sides of each market, market-weighted with design weights, gross. (b) Buy fills before the information time held to settlement, net of taker fees, dollar-weighted, by liquidity role. Bars are 95% confidence intervals clustered by event, truncated at the axis limits.

6.3.2 Who buys longshots

Table 17 describes the five classes. Occasional small wallets ("retail") make 31.3% of buy fills and 12.4% of dollars bought, across 175,734 wallets. Market-maker-like wallets make 33.3% of fills, and whales 21.9% of dollars from only 1,677 wallets.

Longshots are a large share of what small wallets and market makers *buy*, but a small share of what any class *spends*:

- Longshots are 11.5% of retail buy fills, 15.7% of active-retail fills and 21.8% of market-maker fills, but only 1.5% of mid-size and 0.2% of whale fills (Figure 14).
- Because longshots are cheap, they are at most 1.3% of any class's dollars bought.
- Favourites take between 5.7% and 14.8% of each class's dollars: most for active retail and retail, least for whales.

Net flows show who takes the other side. Counting every buy and sell before the information time, retail wallets are the clear net buyers of longshots: 0.43% of the dollars they trade ($t = 3.9$), as takers (0.47%, $t = 2.7$) and with resting orders (0.39%, $t = 2.4$). Net flows sum to zero across classes, and retail makes 86% of all net longshot buying. The other classes are close to flat on net: active retail 0.04% ($t = 0.4$), mid-size wallets −0.06% ($t = -1.0$), whales −0.06% ($t = -1.6$), market makers −0.16% ($t = -1.3$). None of these is individually significant ($|t| \leq 1.6$). The longshot buying of small, occasional wallets is thus absorbed by many counterparties, not by one class of professional sellers. That small accounts tilt towards lottery-like payoffs matches the evidence on individual investors in equity markets [31].

The returns on these purchases differ by class (Figure 15). Per dollar, net longshot returns are −36.3% ($t = -1.4$) for retail, −59.4% ($t = -3.0$) for active retail and −42.3% ($t = -1.6$) for market makers; the few longshots bought by mid-size wallets and whales lose 90% and 100% of their cost. Across all price levels, class PnL per dollar traded is −2.1% ($t = -0.8$) for retail, 14.1% ($t = 2.5$) for active retail, −6.3% ($t = -1.2$) for mid-size wallets, 5.3% ($t = 1.1$) for whales and −10.8% ($t = -1.4$) for market makers (Table 17). Only the figure for active retail is significant at the 5%

Table 16: Polymarket calibration and longshot returns by group

		Last trade 24 h before t^{info}				Buy fills before t^{info}	
	Markets	$\hat{\beta}$ (SE) design	$\hat{\beta}$ (SE) sample	r, LS % (t)	r, FAV % (t)	r, LS % (t)	r, FAV % (t)
All markets	2,121	0.99 (0.15)	0.97 (0.05)	84.5 (0.5)	1.9 (2.5)	−53.8 (−2.8)	2.2 (3.5)
Sports	285	1.00 (0.24)	1.08 (0.18)	−93.6 (−20.3)	5.7 (3.6)	−60.1 (−2.2)	3.3 (2.0)
Politics	416	1.15 (0.18)	1.05 (0.13)	−80.3 (−8.3)	0.4 (0.5)	−48.0 (−1.5)	2.1 (2.0)
Weather	303	0.71 (0.25)	0.74 (0.14)	257.7 (0.7)	0.4 (0.5)	−70.6 (−6.2)	1.4 (7.1)
Entertainment	406	1.20 (0.19)	1.24 (0.15)	−84.4 (−8.3)	1.2 (2.4)	−59.2 (−3.9)	2.0 (3.2)
Mentions	227	0.84 (0.14)	0.77 (0.13)	−88.3 (−7.3)	6.7 (7.7)	−57.4 (−2.8)	3.5 (1.6)
Science & Tech	289	0.99 (0.16)	1.04 (0.13)	−26.1 (−0.5)	−1.2 (−0.7)	−61.0 (−2.8)	1.1 (1.8)
Other	195	0.82 (0.15)	0.89 (0.13)	18.3 (0.3)	−3.4 (−1.4)	−49.1 (−1.6)	1.4 (1.4)
H1	912	1.28 (0.16)	1.06 (0.09)	−36.1 (−1.5)	0.2 (0.2)	−90.9 (−18.7)	3.9 (6.9)
H2	1,209	0.97 (0.16)	0.91 (0.06)	93.3 (0.5)	2.0 (2.5)	−30.2 (−1.0)	1.4 (1.7)

Sample: stratified random sample of resolved in-scope Polymarket markets ($1k–$6M), January 2024 to October 2026. Left block: one observation per side of each market with a trade at least 24 h before its information time t^{info}; β from (2) on the outcome-0 side, with design weights or with every sampled market weighted equally ("sample"); LS: $q \leq 10\,¢$, FAV: $q \geq 90\,¢$; returns r in % per dollar, gross (Polymarket charged no fees before 2026), market-weighted (design weights). H1: markets resolving January 2024 to May 2025; H2: June 2025 to October 2026. Right block: buy fills (maker and taker) before t^{info}, held to settlement, % per dollar net of taker fees, dollar-weighted (notional × design weight). t-statistics in parentheses; standard errors clustered by event.

level. Table 40 in Appendix C shows that the ordering of longshot shares survives every change of cut-off.

Fill size gives the same gradient without any classification (Table 42). Longshots are 23.1% of buy fills of at most $10, 2.4% of $10–100 fills, 0.7% of $100–1k fills and 0.2% of fills above $1,000. Per dollar, however, the small longshot purchases lose least: −22.2% ($t = -0.6$) for fills of at most $10, against −48.4% ($t = -2.4$) at $10–100 and −77.6% ($t = -4.5$) at $100–1k. All-time leaderboard wallets put 6.8% of their buy fills into longshots, against 15.4% for other wallets.

Do longshot buyers hold to settlement? Table 43 in Appendix C follows each position opened by a buy. Retail sells at least half of 41.7% of its longshot positions before the close, active retail 6.9% and market makers 54.7%. For longshot entries, the realised PnL exceeds the hold-to-settlement PnL by 23.2% ($t = 2.2$) of cost: the shares sold were worth less at settlement than their sale price, as expected if longshots are overpriced when they are sold. Across all positions the difference is −1.4% ($t = -1.3$), and for favourite entries −1.2% ($t = -2.1$). Selling favourites early forgoes some of their small positive return.

6.3.3 Profit concentration and persistence

The accounting identity holds in the data. In 99.95% of the 2,197 fee-free markets the PnL of all wallets sums to zero within 10^{-6} of turnover, and the YES-equivalent positions sum to zero in 97.8%. In fee markets the residual is the fee charged in shares. PnL is therefore a transfer between wallets. Of 237,569 wallets, 50.7% end with positive PnL in the sampled markets. The gains are highly concentrated (Table 18, Figure 16):

- the top 1% of wallets take 85% of all gains and the top 0.1% take 51%;
- 220 wallets hold half of all gains;

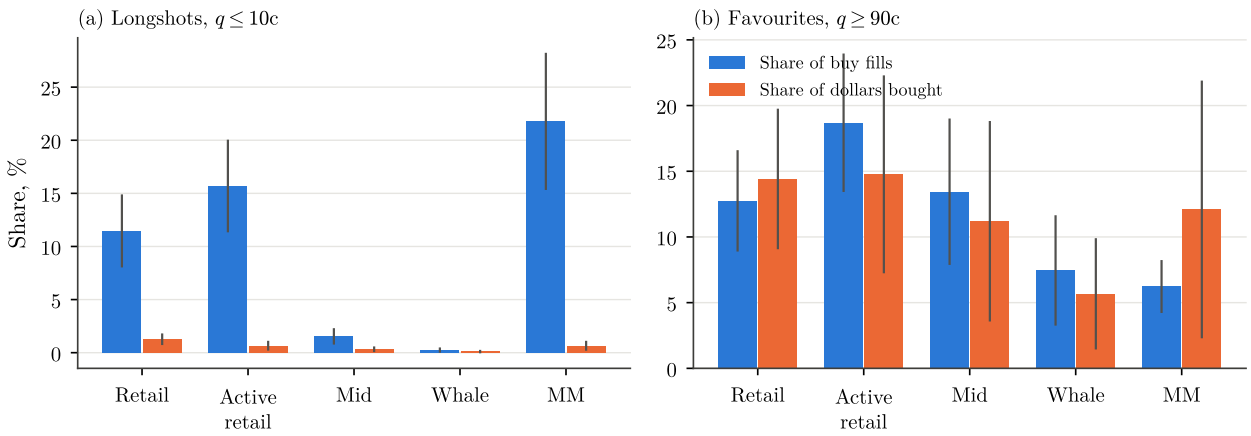


Figure 14: Longshot ($q \leq 10$c) and favourite ($q \geq 90$c) shares of buying by wallet class: share of design-weighted buy fills and of dollars bought, buy fills before the information time. Bars are 95% confidence intervals clustered by event.

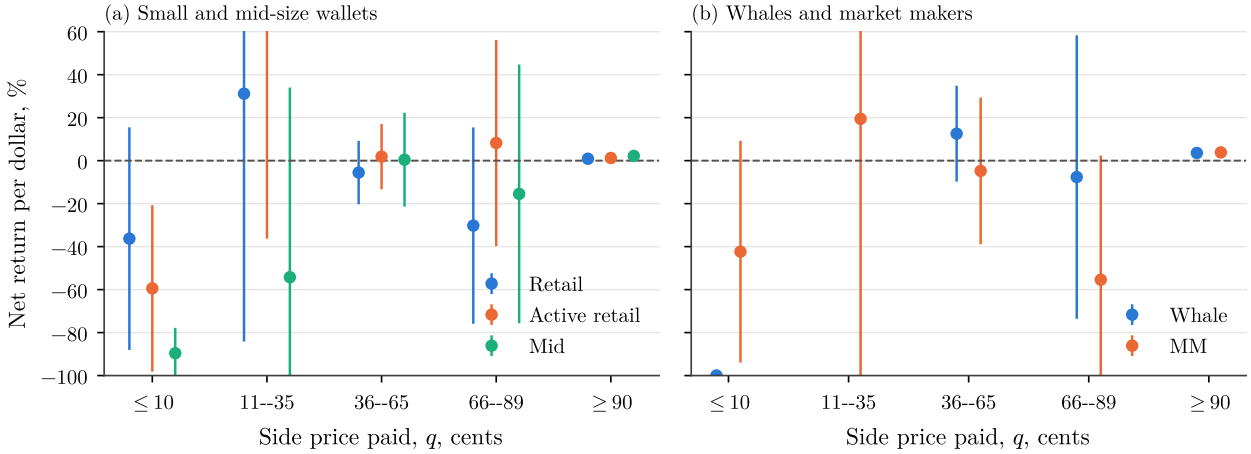


Figure 15: Net return per dollar of buy fills by wallet class and side price, held to settlement, dollar-weighted, buy fills before the information time. Bars are 95% confidence intervals clustered by event.

- the Gini coefficient of gains among winners is 0.97.

Losses are less concentrated (Gini 0.95; 326 wallets for half of all losses), and dollar volume is concentrated to a similar degree (Gini 0.96). The 96 all-time leaderboard wallets in the sample take 11.8% of gains on 10.4% of volume. Akey et al. [1] report a similar concentration of profits across all Polymarket users, and equity markets show the same pattern of losses spread over many individual traders [4, 6].

Concentration alone does not show skill: a market that pays out at long odds produces a few very large winners by chance. We therefore rank the 719 wallets active in both halves on their shrunk H1 return (13) and follow their H2 return (Figure 17):

- the rank correlation is 0.02 (95% CI -0.06 to 0.10), and 0.02 without shrinkage;
- the top H1 quintile earns -7.8% per dollar relative to the bottom quintile in H2 ($t = -1.2$);
- edge versus entry price, which removes favourite-versus-longshot style, has a rank correlation of -0.03.

Table 17: Who buys longshots: buying, returns and PnL by wallet class

		Share of buying		LS share		FAV share	LS r	Net LS flow	PnL per $
Class	Wallets	fills	$	fills	$	$	% (t)	% (t)	% (t)
Retail	175,734	31.3	12.4	11.5	1.3	14.4	−36.3 (−1.4)	0.43 (3.9)	−2.1 (−0.8)
Active retail	6,431	31.9	24.1	15.7	0.7	14.8	−59.4 (−3.0)	0.04 (0.4)	14.1 (2.5)
Mid	15,393	3.2	20.5	1.5	0.3	11.2	−89.6 (−14.9)	−0.06 (−1.0)	−6.3 (−1.2)
Whale	1,677	0.2	21.9	0.2	0.1	5.7	−100.0 (< −100)	−0.06 (−1.6)	5.3 (1.1)
MM	3,589	33.3	21.1	21.8	0.7	12.1	−42.3 (−1.6)	−0.16 (−1.3)	−10.8 (−1.4)

Classes are assigned to each fill from the wallet's history before it (Section 5.3). Wallets: distinct wallets with at least one buy in the class (a wallet can change class over time). Shares of buying and longshot ($q \leq 10\,¢$) / favourite ($q \geq 90\,¢$) shares use buy fills before t^{info}, in % of design-weighted fills or dollars. Longshot r: net return per dollar of those longshot buys held to settlement, dollar-weighted. Net LS flow: net longshot dollars bought per dollar traded (buys and sells, both roles; positive = net buyer of longshots). PnL per $: exact wallet-market PnL (12) net of fees per dollar traded, class at the wallet's first fill in the market. t-statistics clustered by event.

The luck placebo gives a benchmark. With outcomes redrawn from the last pre-information price, the same wallets and trades would produce a mean rank correlation of 0.15 (5th–95th percentile 0.09 to 0.20). That persistence comes from trading at prices better or worse than the last trade before the information time. The actual correlation on the placebo sample, −0.01, lies below all 200 placebo draws. So the price advantage measured against the last trade does not persist in realised returns. That is consistent with the last trade containing bid–ask bounce and with resting orders being filled when prices move against them. Likewise, 5.1% of wallets with at least 20 markets have a PnL t-statistic above 2, against 7.0% under the placebo. In this sample we find no persistent skill in wallet returns, whether measured by raw returns, shrunk returns or edge versus entry price. This contrasts with the persistence that Barber et al. [8] document among Taiwanese day traders. It also contrasts with the persistent skilled minority that Gómez-Cram et al. [21] report from Polymarket's full trade record, and agrees with the modest, possibly selection-driven persistence in Akey et al. [1]. Our panel is shorter and covers only the sampled markets of each wallet, so it has less power to detect skill. Within it, the concentration of gains is consistent with what a market with long-odds payoffs produces by chance.

6.3.4 Retail inflows, attention and herding

Account creation dates are available for 5,144 wallets of the address-selected subset. Figure 18(a) plots their creation dates by week. The ratio of creations in the 14 days around each event to the rate four to eight weeks before, with 95% intervals, is: US election 2024 2.01 [1.64, 2.47]; Super Bowl LIX 0.40 [0.32, 0.49]; Papal conclave 2025 1.03 [0.76, 1.40]; NYC mayor 2025 2.59 [1.99, 3.37]; Super Bowl LX 1.26 [0.96, 1.65]. Each event window contains between 64 and 180 creations. Creations rise significantly around US election 2024 and NYC mayor 2025. They fall around Super Bowl LIX. That fall reflects its baseline. The four to eight weeks before Super Bowl LIX, mid-December to mid-January, are the peak of the inflow that began in the run-up to the 2024 election (Figure 18(a)).

New accounts put a smaller share of their buy fills into longshots. Longshots are 14.9% of buy fills by accounts under a week old and 8.3% at 7–30 days, against 19.9% at 30–180 days and 22.7% for older accounts (Table 42). In a fill-level linear probability model, the difference between accounts under 30 days old and older accounts is −11.4 percentage points ($t = -3.3$, clustered by event). On a dollar basis the shares are 0.3% and 0.8%. Figure 18(b) shows the monthly shares. Attention-driven buying by new retail accounts, documented for equities by Barber and Odean

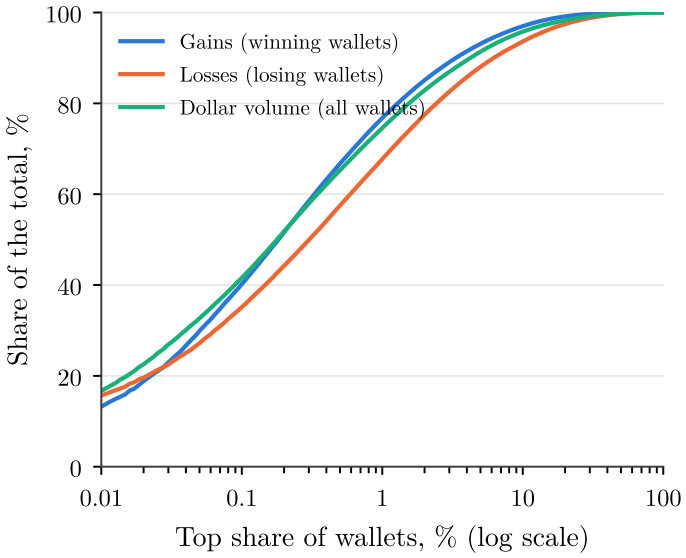


Figure 16: Concentration of wallet outcomes: share of all gains (winning wallets), all losses (losing wallets) and all dollar volume held by the top $x\%$ of wallets (equal shares would give $x\%$). Exact trade-based PnL over the sampled markets, net of fees.

[5] and Barber et al. [9], would predict a higher share for new accounts. The longshots that new accounts do buy, however, lose almost all of their cost: -97.2% ($t = -40.2$) per dollar for accounts under a week old and -88.3% ($t = -12.1$) at 7–30 days, against 18.2% ($t = 0.2$) at 30–180 days and -31.3% ($t = -0.5$) beyond 180 days.

Small retail buys cluster on one side. In one-hour windows with at least five small retail taker buys, the Lakonishok et al. [33] measure (14) is 0.095 ($t = 20.9$; 23,823 windows). Independent orders would give zero. The clustering is not specific to retail: taker buys by market-maker-like wallets cluster to the same degree (0.096, $t = 9.4$; 536 windows). Same-side clustering within the hour is therefore a feature of order flow around news, not of retail behaviour alone.

Retail herding moves prices only briefly. A window in which all small retail buys are on the YES side is followed by a YES-price change of 0.16 ¢ ($t = 3.4$) over the next hour, 0.09 ¢ ($t = 0.7$) over six hours and -0.18 ¢ ($t = -0.5$) over 24 hours, controlling for the previous hour's change. The previous hour's change itself partly reverses (coefficient -0.15, $t = -5.8$, at one hour). The same imbalance does not predict the outcome relative to the price: the coefficient is -1.53 ¢ ($t = -1.2$). Small-wallet herding thus neither carries information about outcomes nor leaves a reversal larger than the tick.

6.3.5 The bias across venues

On 523 matched team-sides in 441 games, Kalshi and Polymarket agree almost exactly before the start (Table 19, Figure 19). One hour before the start the prices correlate at 0.9993, and the mean absolute gap between the Polymarket price and the Kalshi midpoint is 0.40 ¢. A day before it is 0.59 ¢.

Game-winner markets rarely price a side at 10 ¢ or less an hour before the start: only 6 team-sides do, so the comparison cannot say whether one venue prices longshots higher. Between 11 ¢ and 80 ¢ (495 team-sides), the mean gap in every bucket is within 0.05 ¢ of zero. Above 80 ¢ (22 team-sides), Polymarket prices the favourite slightly below Kalshi's midpoint: by 0.59 ¢ at 90 ¢ or more ($t = -3.8$). Calibration slopes on the same team-sides are 0.71 for Kalshi and 0.71 for Polymarket ($t = -2.0$

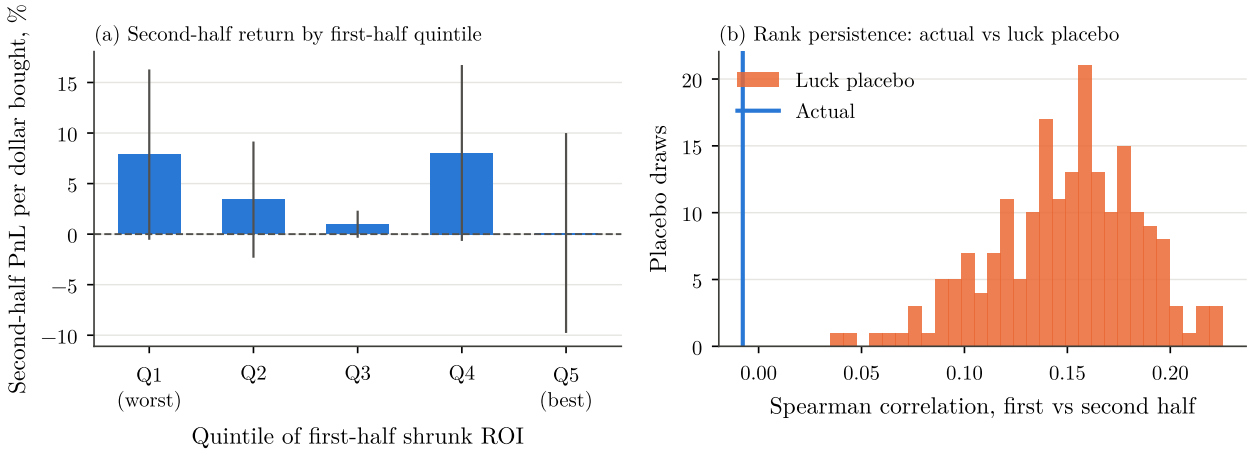


Figure 17: Persistence of wallet returns. (a) H2 PnL per dollar bought by quintile of shrunk H1 return, wallets active in both halves, with 95% bootstrap intervals. (b) Rank correlation between halves under 200 draws of the luck placebo (outcomes redrawn from the last pre-information price) and the actual value on the same wallets.

and $t = -2.0$ against one). The two are equal because the prices almost coincide (Table 19). Gaps are short-lived:

- the hour-to-hour autocorrelation of a pair's gap is 0.79, a half-life of about 3.0 hours;
- the gap 24 hours out has a correlation of 0.09 with the gap an hour before the start.

Only 2.3% of the one-hour gaps exceed the sum of the two venues' taker fees, and 0.2% exceed fees plus Kalshi's half-spread. Within Polymarket, Saguillo et al. [42] document arbitrage between related markets; across these two venues, price gaps on the same games are almost always inside trading costs. We use no Polymarket US prices (Section 3.3). By its published schedule, a small Polymarket US account pays the highest taker fee of the three venues (Table 38), so a gap involving that venue would have to be wider still to clear fees.

Summary. On identified Polymarket wallets, the pattern predicted by Hypothesis 2 appears in who trades longshots, not in who profits:

- longshots are a large share of the buying of small wallets and market makers, and almost none of the buying of whales and mid-size wallets;
- small, occasional wallets are significant net buyers of longshots, and their buying is absorbed by all other classes, none of which is individually a significant net seller;
- new accounts put a smaller share of their buying into longshots than established ones, but lose almost all of what they put in;
- longshot buys lose about half their cost on average and favourite buys earn a few per cent;
- wallet profits are extremely concentrated, but they do not persist out of sample;
- Kalshi and Polymarket price the same games almost identically.

Table 18: Profit concentration and persistence of wallet performance

Concentration of PnL across wallets	
Wallets in the sample	237,569
Share with positive PnL (%)	50.7
Share of all gains: top 0.1% / 1% / 10% of wallets (%)	51 / 85 / 99
Wallets holding half of all gains / half of all losses	220 / 326
Gini: gains (winners) / losses (losers) / dollar volume	0.97 / 0.95 / 0.96
Leaderboard wallets: number / share of gains / share of volume (%)	96 / 11.8 / 10.4
Persistence, H1 → H2	
Wallets active in both halves	719
Spearman, raw H1 ROI vs H2 ROI [95% CI]	0.02 [−0.06, 0.10]
Spearman, shrunk H1 ROI vs H2 ROI [95% CI]	0.02 [−0.06, 0.10]
Spearman, edge vs entry price, H1 vs H2	−0.03 (694 wallets)
H2 ROI, top minus bottom quintile (%, t)	−7.8 (−1.2)
Luck placebo: Spearman mean [5th, 95th pct.]	0.15 [0.09, 0.20]
Actual Spearman on the placebo sample (percentile)	−0.01 (0)
Wallets with ≥20 markets: share with $t > 2$, actual / placebo (%)	5.1 / 7.0

PnL is exact trade-based PnL (12) summed over the sampled markets, net of taker fees. Leaderboard: wallets on Polymarket's all-time overall, politics or sports top-100 (measured after the fact). Persistence: wallets with ≥ 5 markets and ≥ $50 bought in each half; shrunk ROI (13); edge = buy return minus its price bucket's mean in that half. The placebo redraws outcomes from the last pre-information price (200 draws). CIs from 500 wallet bootstraps.

6.4 What explains the bias

We ask how much of the longshot overpricing each explanation of Section 4 accounts for, using the methods of Section 5.4. The sample is the final panel of Section 6.1: 61,992 pre-event quotes on 19,007 markets in 4,770 events, with 40,552 longshot side observations. Quantities are in cents per contract at the calibration level, because pooled returns per dollar on longshots rest on a few large wins (Section 6.2). Supporting tables and figures are in Appendix D.

6.4.1 Mechanical frictions

Table 20 and Figure 20 decompose the longshot gap as in (15).

- *Size of the gap.* Longshot sides cost 2.54 ¢ at the mid and win 1.47% of the time, a gap of 1.07 ¢ per contract ($t = 10.0$).
- *Cost of selling as a taker.* It is 1.52 ¢:
  - 0.18 of half-spread at the tick floor;
  - 1.23 of other half-spread;
  - 0.07 of fees;
  - 0.04 of capital at 5% a year.
- *Residual.* It is $\bar{R} = -0.45$ ¢ ($t = -4.3$). Market-weighted, the mispricing lies inside the taker band in every price bucket, as in Table 7. In no group is the residual significantly positive.
- *Sides with a bid.* Where a taker sale is possible at all, the residual is −0.52 ¢ ($t = -1.9$). This is the per-contract counterpart of taking the other side of longshots in Table 7.

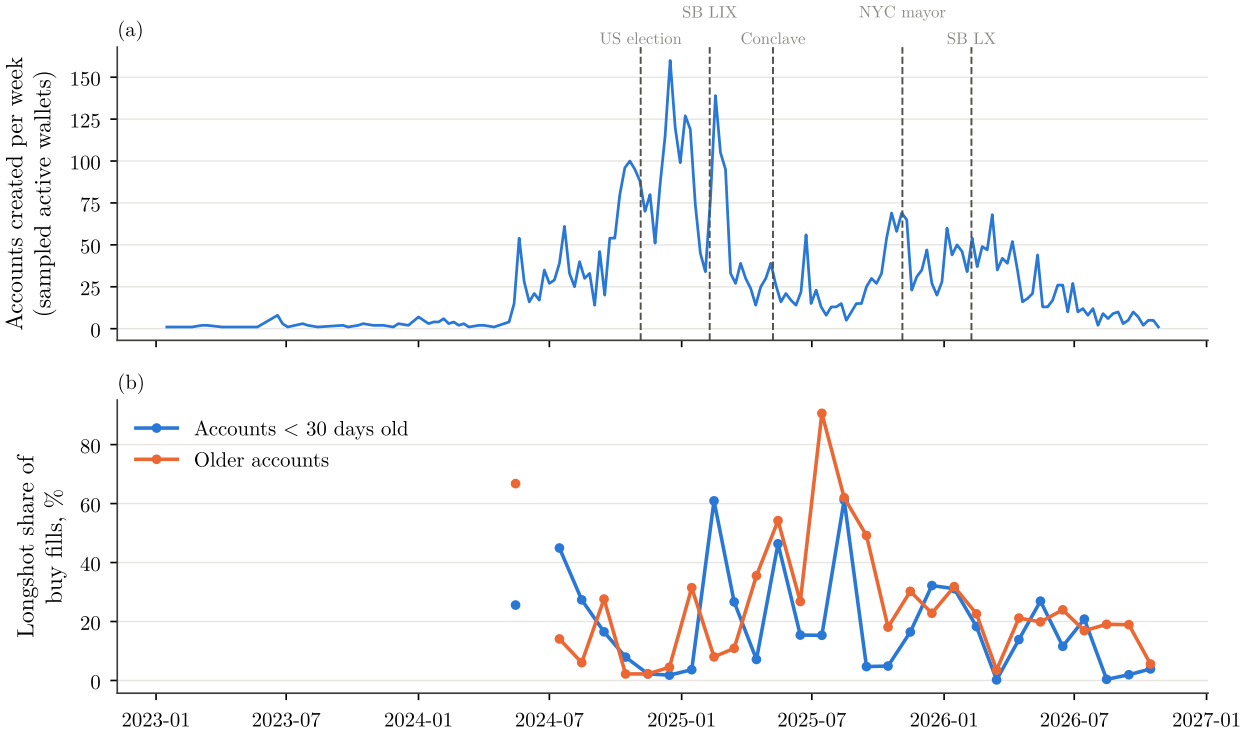


Figure 18: Retail inflow and longshot buying. (a) Weekly account creations of the address-selected subset of sampled wallets, with five salient events marked. (b) Longshot share of design-weighted buy fills by accounts under 30 days old and by older accounts, by month (months with at least 30 fills), buy fills before the information time.

- *Short leads.* At leads of a day or less, the gap sits at the edge of the band (residuals of −0.14, −0.09 and 0.03 ¢).

Frictions thus explain why takers do not trade the bias away, but not where it comes from.

1. *The floor explains the cheapest bucket only.* The floor's half-spread is 61% of the 1–2 ¢ gap. Table 7 finds less because it counts the realised loss of floor quotes. Across all longshots, the floor is 17% of the gap. On the 0.1 ¢ grid, whose floor is ten times lower (384 sides, 31 events), the 1–2 ¢ gap is 0.75 ¢, against 0.49 ¢ on the 1 ¢ grid, with a residual of 0.17 ¢ ($t = 2.3$).

2. *The bias does not grow with the horizon.* Capital costs at most 0.40 ¢ at 30 days. Within price buckets, the gap *falls* by −0.14 ¢ per unit of log lead ($t = -2.7$), and the residual by −0.18 ($t = -3.4$). The raw gap is larger at 30 days (1.55 against 1.13 ¢) because longshots quoted that far ahead are dearer (3.38 against 2.05 ¢ at the mid). Hypothesis 1 and Page and Clemen [38] predict growth with the horizon.

3. *The cost of capital is immaterial.* The residual is −0.42 ¢ at $\rho = 0$ and −0.49 ¢ at $\rho = 10\%$.

4. *Where the volume is, the bias exceeds the band.* Volume-weighted, the gap is 1.56 ¢ but the band only 0.77 ¢, because active markets have tight spreads. The residual is 0.79 ¢ ($t = 2.7$).

5. *Makers could earn the gap.* Selling at the ask would earn $\bar{R}^{mk} = 2.44$ ¢ ($t = 21.3$). This is an upper bound that ignores adverse selection, consistent with Section 6.2.

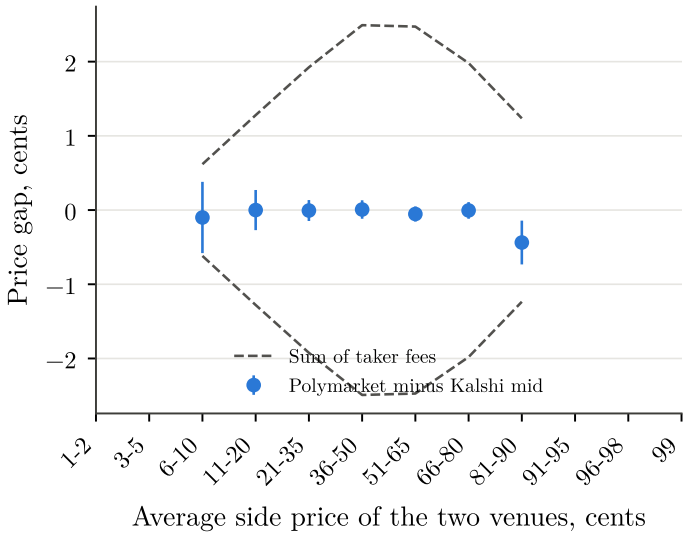


Figure 19: Polymarket price minus Kalshi bid–ask midpoint for the same team, one hour before the scheduled start, by the two venues' average side price, with 95% intervals clustered by game. Dashed: mean sum of the two venues' taker fees at those prices.

Table 19: Kalshi and Polymarket prices on the same games

Lead	Team		Corr.	Mean $\lvert g\rvert$	g, $\bar{q} \le 10\,¢$		g, $\bar{q} \ge 90\,¢$		$\lvert g\rvert >$ fees (%)		β	
(h)	sides	Games	of prices	(¢)	¢	(t)	¢	(t)	fees	+spread	Kalshi	PM
1	523	441	0.9993	0.40	−0.10	(−0.5)	−0.59	(−3.8)	2.3	0.2	0.71	0.71
3	523	441	0.9993	0.40	−0.01	(−0.0)	−0.04	(−0.1)	2.7	0.2	0.70	0.70
6	523	441	0.9993	0.42	−0.02	(−0.1)	−0.12	(−0.4)	2.5	0.0	0.70	0.69
12	523	441	0.9967	0.54	+0.31	(0.9)	+0.19	(0.6)	3.8	1.1	0.69	0.69
24	514	432	0.9976	0.59	−0.19	(−0.4)	+0.23	(0.6)	4.9	0.6	0.69	0.69

Kalshi NBA, NHL, NFL, college-football and WNBA game-winner markets (August–October 2026) matched to the same team's Polymarket outcome, observed on the same hourly timestamp L hours before the scheduled start. $g = p^{\mathrm{PM}} - p^{\mathrm{K}}$ with p^{K} the Kalshi bid–ask midpoint; $\bar{q}$ is the two venues' average price of that side. Fees: Kalshi taker $0.07\,p(1-p)$ plus each Polymarket market's taker rate; "+spread" adds Kalshi's half-spread. β: calibration slope (2) on the same team-sides. t-statistics clustered by game.

6.4.2 Probability weighting

In Table 21, the LLO curvature is $\gamma = 1/\beta$ for the slope β of Table 2, estimated on the identical sample.

- *Pooled.* $\gamma = 0.89$ (standard error 0.016; $t_{\gamma<1} = 7.4$), and the Prelec curvature is also 0.89. Both overweight small probabilities, with near-neutral elevation ($\delta = -0.06$ and 1.01).
- *Groups.* γ is below one in 6 of 6 groups (0.80–0.93). A common γ is not rejected market-weighted ($p = 0.127$). It is rejected volume-weighted ($p = 0.026$), where the distortion is stronger ($\gamma = 0.68$).
- *Leads and halves.* Across leads γ ranges from 0.81 to 0.89 ($p = 0.033$). It is 0.91 and 0.87 in the two halves.
- *Out of sample.* Fitted on one half of the months and scored on the other, pooled LLO improves

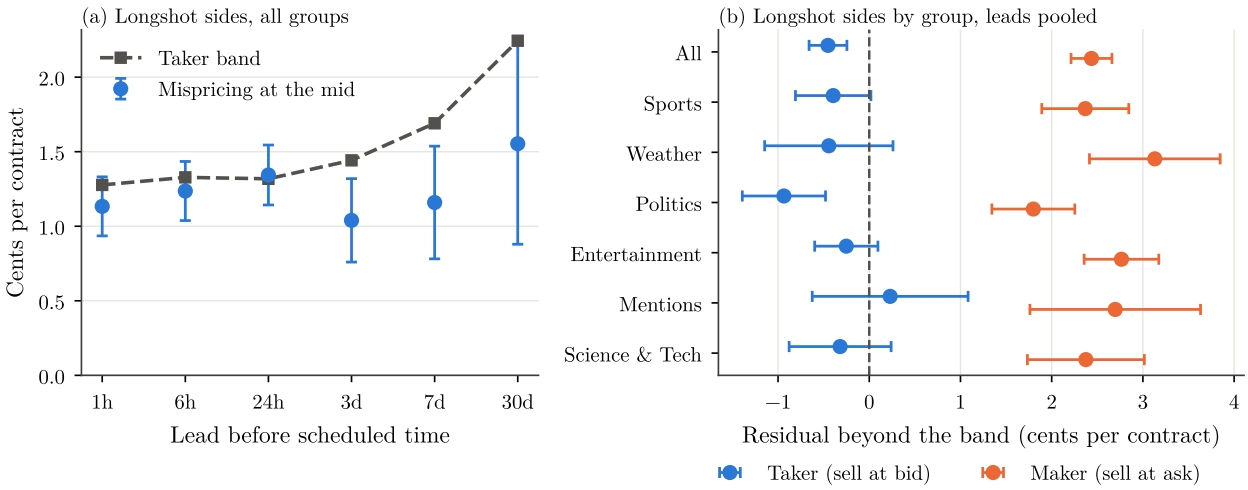


Figure 20: The longshot mispricing and the no-arbitrage band. Left: price at the mid minus realised win rate of longshot sides ($q \leq 10\,¢$) by lead, against the taker band (tick floor, half-spread, fee and capital at $\rho = 5\%$). Right: the residual beyond the band, the mean profit per contract of selling longshots at the bid (taker) or at the ask (maker, an upper bound), by group with leads pooled. Market-weighted; 95% confidence intervals clustered by event.

Table 20: Longshot mispricing and the no-arbitrage band

				Gross	Band components				Residual	Maker	In band
Cell	Sides	Events	$\bar{q}$	$\bar{q} - \bar{w}$	Floor	Half-spr.	Fee	Capital	$\bar{R}$	$\bar{R}^{mk}$	(%)
All longshots	40,552	3,091	2.54	1.07 (10.0)	0.18	1.23	0.07	0.04	−0.45 (−4.3)	2.44 (21.3)	100
By lead, all groups											
1h	9,847	2,690	2.05	1.13 (11.2)	0.23	0.99	0.05	0.00	−0.14 (−1.5)	2.35 (21.7)	100
6h	9,721	2,633	2.17	1.24 (12.2)	0.21	1.06	0.06	0.00	−0.09 (−0.9)	2.49 (23.1)	100
24h	8,025	1,962	2.10	1.34 (13.1)	0.20	1.04	0.06	0.01	0.03 (0.3)	2.57 (22.9)	98
3d	7,056	1,742	2.44	1.04 (7.3)	0.17	1.16	0.07	0.04	−0.40 (−2.9)	2.32 (15.1)	100
7d	4,149	1,180	2.56	1.16 (6.0)	0.15	1.38	0.07	0.09	−0.53 (−2.8)	2.59 (12.5)	100
30d	1,754	670	3.38	1.55 (4.5)	0.09	1.64	0.11	0.40	−0.69 (−2.0)	2.89 (8.0)	100
By group, leads pooled											
Sports	9,108	502	2.29	1.07 (4.9)	0.22	1.14	0.06	0.04	−0.40 (−1.9)	2.37 (9.7)	100
Weather	3,905	721	3.46	1.42 (4.0)	0.11	1.62	0.12	0.02	−0.44 (−1.2)	3.13 (8.6)	100
Politics	11,768	835	2.51	0.51 (2.2)	0.15	1.17	0.08	0.04	−0.94 (−4.0)	1.80 (7.8)	100
Entertainment	8,939	494	2.22	1.31 (7.1)	0.19	1.30	0.05	0.03	−0.25 (−1.4)	2.76 (13.2)	100
Mentions	1,918	91	3.22	1.55 (3.5)	0.20	0.96	0.14	0.02	0.23 (0.5)	2.70 (5.7)	85
Science & Tech	4,572	401	2.30	1.12 (3.8)	0.18	1.14	0.07	0.05	−0.32 (−1.1)	2.37 (7.3)	100

Side observations with $q \leq 10\,¢$ at the mid, pre-event quotes. All entries in cents per contract. Gross: price at the mid minus the realised win rate. The band components (Equation (15)) are the half-spread of sides pinned at the tick floor (Floor), the half-spread of other sides, the taker fee and the capital cost at $\rho = 5\%$ a year over the lead. $\bar{R}$: residual beyond the taker band, the mean net profit per contract of selling the side at its bid; $\bar{R}^{mk}$: selling at its ask as a maker (an upper bound: no adverse selection). In band: share of the gross mispricing the taker band covers, min(band, gross)/gross. Market-weighted (each market once per cell; within a lead, equal weights). Cells by price bucket and structure are in Tables 44 and 45; the band per bucket agrees with Table 7. t-statistics clustered by event in parentheses.

the log score of face-value prices by only 0.46 ($t = 0.9$) and 0.04 milli-nats per quote, and Prelec by 0.75 and 0.19. The log loss itself is 0.309 nats. A constant YES shift lowers the score (-1.32, $t = -2.9$). Group-specific LLO is worse than pooled LLO by -3.10 ($t = -4.2$) and -1.83 ($t = -3.1$).

One mild inverse-S function thus describes all categories. It is precisely estimated but small next to the information in prices.

Table 21: Probability weighting: fitted functions, equality tests and forecasts

			LLO			Prelec		$\Delta \log L$	
Group	Quotes	Events	γ	δ	$t_{\gamma<1}$	γ	δ	LLO	Prelec
Panel A. Fitted weighting functions									
All groups	61,992	4,770	0.89 (0.016)	−0.06 (0.029)	7.4	0.89 (0.016)	1.01 (0.016)	1.4	1.6
Sports	18,303	1,521	0.91 (0.031)	−0.04 (0.047)	2.9	0.90 (0.027)	1.00 (0.027)	0.8	1.2
Weather	6,796	1,065	0.87 (0.040)	−0.10 (0.047)	3.3	0.89 (0.039)	1.02 (0.022)	1.6	1.6
Politics	15,774	968	0.93 (0.036)	−0.07 (0.065)	1.9	0.93 (0.035)	1.03 (0.032)	0.4	0.5
Entertainment	10,424	530	0.80 (0.034)	0.04 (0.076)	6.0	0.85 (0.035)	0.97 (0.041)	5.1	5.0
Mentions	4,743	103	0.89 (0.049)	−0.03 (0.106)	2.3	0.88 (0.052)	0.99 (0.060)	1.5	2.0
Science & Tech	5,472	510	0.85 (0.049)	−0.17 (0.141)	3.0	0.84 (0.052)	1.08 (0.083)	3.2	3.6
Panel B. Equal slopes (γ) across cells: Wald tests									
Groups			$\chi^2_5 = 8.6$, $p = 0.127$, 4,697 events						
Groups, γ and δ			$\chi^2_{10} = 16.6$, $p = 0.084$, 4,697 events						
Structures			$\chi^2_3 = 7.5$, $p = 0.057$, 4,770 events						
Leads			$\chi^2_5 = 12.1$, $p = 0.033$, 4,770 events						
Groups, volume-weighted			$\chi^2_5 = 12.8$, $p = 0.026$, 4,357 events						
Panel C. Out-of-sample log score against calibrated prices (milli-nats per quote)									
Constant shift			first → second half: −1.32 (−2.9); second → first: −0.73 (−4.4)						
LLO, pooled			first → second half: 0.46 (0.9); second → first: 0.04 (0.1)						
Prelec, pooled			first → second half: 0.75 (1.7); second → first: 0.19 (0.3)						
LLO, by group			first → second half: −2.65 (−2.3); second → first: −1.79 (−1.6)						

YES sides of pre-event quotes, market-weighted. LLO: $\text{logit}\, q = \delta + \gamma \,\text{logit}\, \pi$, estimated as (2) with $\gamma = 1/\beta$ and $\delta = -\alpha/\beta$. Prelec: $q = \exp(-\delta(-\ln \pi)^\gamma)$. Maximum likelihood on outcomes; standard errors in parentheses clustered by event (delta method for γ, δ). $\Delta \log L$: in-sample gain in mean log likelihood over calibrated prices, milli-nats per market. Panel B: cluster-robust Wald tests from one fully interacted logit. Panel C: models fitted on one half of the months and scored on the other; differences against $\mathbb{P}(y = 1) = q$ with event-clustered t in parentheses.

6.4.3 Insiders

Table 22 extends Table 6 to two-leg events and partition ladders, by group and lead. Of 5,101 event-leads, 1,107 miss a quote on some leg and 39 lack exactly one winner. That leaves 3,955 on 1,421 events. The asks sum to less than one in 39 of them and the mids in 1,069. Mentions events are not partitions.

- *The overround restates the spread.* z from the asks averages 0.030, and from the mids 0.004. It rises by 0.0168 per cent of half-spread ($t = 51.0$; $R^2 = 0.68$ with leg count). Given the spread, it is higher in the information groups by 0.004 ($t = 5.5$), small against its mean of 0.030.

- *The shading the outcomes require.* Fitted to outcomes, $\hat{z} = 0.080$ [0.020, 0.130], more than twice the margin's z. It is 0.085 in the information groups and 0.070 in Sports and Weather (difference 0.015, [−0.110, 0.090]). Politics needs the least of any group (0.025).

- *Forecasts* (Table 48). Longshot legs are overpriced by 0.93 ¢ at the mid ($t = 4.9$), 0.83 after normalisation and 0.94 after Shin's correction from the asks. Shin beats the normalised asks

by 3.47 milli-nats per leg ($t = 8.5$), as Štrumbelj [48] found for bookmakers, but it does so by undoing the spread. A z fitted on the first half shrinks the second-half gap from 1.08 to 0.05 ¢ ($t = 0.2$), acting as a calibration curve like LLO.

- *Overpricing and z.* Longshot overpricing rises with an event's z from the asks (22.0 cents per unit, $t = 2.1$). Since that z mostly measures the spread, the band predicts the same.

Table 22: Shin's insider share in events whose legs partition the outcomes

Sample	Event-leads	Events	Legs (med.)	$\sum a - 1$ (%)	$\sum p - 1$ (%)	z from asks	z from mids	z fitted to outcomes [95% CI]
All partition events	3,955	1,421	2	7.8 (33.0)	1.3 (12.8)	0.030 (0.0005)	0.004	0.080 [0.020, 0.130]
Two-leg events	2,009	740	2	3.0 (40.6)	−0.2 (−4.0)	0.032 (0.0008)	0.003	0.025 [0.005, 0.095]
Multi-outcome events	1,289	381	4	10.1 (17.1)	2.2 (7.6)	0.026 (0.0008)	0.005	0.095 [0.025, 0.245]
Partition ladders	657	300	9	16.7 (38.3)	3.9 (16.3)	0.030 (0.0008)	0.008	0.105 [0.005, 0.225]
Sports	2,373	916	2	4.7 (20.7)	0.3 (2.6)	0.030 (0.0007)	0.003	0.060 [0.020, 0.125]
Weather	237	139	6	15.5 (25.5)	4.1 (11.4)	0.034 (0.0014)	0.010	0.110 [0.020, 0.420]
Politics	1,011	280	6	12.6 (23.3)	2.6 (12.3)	0.028 (0.0008)	0.005	0.025 [0.005, 0.165]
Entertainment	120	42	8	17.9 (8.2)	6.3 (5.8)	0.036 (0.0034)	0.010	0.070 [0.005, 0.230]
Science & Tech	214	44	7	7.6 (5.4)	1.8 (2.3)	0.018 (0.0019)	0.005	0.195 [0.090, 0.600]
Information minus public						−0.003		0.015 [−0.110, 0.090]

Mutually exclusive events (Kalshi flag) and strike ladders that partition the outcome space, at leads where every leg has a usable quote and exactly one leg settles YES. a, p: YES ask and mid. z from asks or mids solves (4) ($z = 0$ when the prices sum to at most one). Fitted z: the value on a grid over $[0, 0.6]$ that maximises the log score of the winning leg, with the scale chosen so that probabilities sum to one; 95% event-bootstrap interval. Each event weighs one across its leads; Table 6 weights event-leads equally, hence the small differences in the multi-outcome row. Information groups: Politics, Entertainment, Mentions, Science & Tech; public: Sports, Weather. Standard errors (of z) or t-statistics (of the overrounds) clustered by event in parentheses.

6.4.4 Belief dispersion

The slope interactions of (18) are in Table 49 and Figure 25.

- *Number of outcomes.* More outcomes steepen the slope by 0.041 per standard deviation ($t = 1.7$; within groups 0.045, $t = 1.8$). This is the sign Hypothesis 4 predicts, but not significant.
- *Turnover.* Turnover flattens the slope (−0.041, $t = -1.5$). The interaction is positive in only 1 of 6 groups.
- *Forecast anchor.* The anchor interaction is −0.049 ($t = -1.1$), and it flips sign between halves (0.096 and −0.204).
- *Horizon and spread.* Longer horizons flatten the slope (−0.048, $t = -2.8$). Wider spreads leave the slope unchanged but raise the longshot gap in cents (0.57, $t = 3.4$), as the band predicts.

6.4.5 Synthesis

Table 23 summarises the evidence by group; Table 51 lists every prediction tested.

Table 52 applies each explanation's map from prices to probabilities to the second-half longshot gap of 1.12 ¢.

- The taker band covers 100% of the gap.

Table 23: The evidence by group

	Hypothesis 1: band		Hypothesis 2: weighting	Hypothesis 3: insiders	Hypothesis 4: dispersion
Group	Residual $\bar{R}$ (¢)	In band (%)	γ ($t_{\gamma<1}$)	fitted z [95% CI]	Turnover
All groups	+ −0.45 (−4.3)	100	+ 0.89 (7.4)	0.080 [0.02, 0.13]	0 −0.043 (−1.5)
Sports	+ −0.40 (−1.9)	100	+ 0.91 (2.9)	− 0.060 [0.02, 0.12]	0 0.025 (0.6)
Weather	+ −0.44 (−1.2)	100	+ 0.87 (3.3)	− 0.110 [0.02, 0.42]	0 −0.054 (−0.4)
Politics	+ −0.94 (−4.0)	100	0 0.93 (1.9)	+ 0.025 [0.01, 0.17]	− −0.127 (−2.7)
Entertainment	+ −0.25 (−1.4)	100	+ 0.80 (6.0)	+ 0.070 [0.01, 0.23]	0 −0.101 (−1.8)
Mentions	+ 0.23 (0.5)	85	+ 0.89 (2.3)	–	0 −0.107 (−1.7)
Science & Tech	+ −0.32 (−1.1)	100	+ 0.85 (3.0)	+ 0.195 [0.09, 0.60]	0 −0.088 (−1.3)

Mark: + consistent with the hypothesis's prediction for the group, − inconsistent, 0 inconclusive. Hypothesis 1: + when the longshot residual beyond the taker band is not significantly positive. Hypothesis 2: + when $\gamma < 1$ significantly. Hypothesis 3: in Politics, Entertainment, Mentions and Science & Tech + when the interval for the fitted z excludes zero; in Sports and Weather, where outcomes are public, + when it includes zero and − when it does not. Hypothesis 4: sign and significance of the turnover interaction (within-group slopes for all groups). Sources and clustering as in Tables 20 to 22 and 49.

- LLO fitted on the first half removes 61% (residual 0.43 ¢, $t = 2.7$). Prelec removes 68%, and group-specific LLO 82%.
- In partition events, normalisation removes 6%, Shin with the margin's z 2%, and Shin with a fitted z 95%.

The shares are not additive. The band measures what frictions *allow*; the fitted maps measure what a calibration curve *describes*.

Prices and outcomes cannot separate four things:

- risk-love from probability weighting, which needs bet sizes or exotic bets [47];
- weighting from heterogeneous beliefs, since both yield a monotone calibration curve that LLO fits by construction [17, 35];
- insiders from the spread, which itself prices informed trading [18];
- the true cost of capital.

We therefore report no variance decomposition.

Summary of verdicts.

1. *Mechanical (Hypothesis 1): partly supported (2 of 5 predictions hold).* Spreads, fees and the tick floor explain why the bias survives: market-weighted, it is no larger than the cost of selling it as a taker in any group. They do not explain where it comes from. It falls rather than grows with the horizon, a finer tick does not remove it, and volume-weighted it exceeds the band.
2. *Preference (Hypothesis 2): partly supported (4 of 5).* Prices behave as if traders weight probabilities with one mild inverse-S function, $\gamma \approx 0.89$, below one in every category, though it adds little out of sample. Small takers and retail wallets buy the longshots and lose (Sections 6.2 and 6.3). Together, this makes preference the best-supported explanation.

3. *Information (Hypothesis 3): partly supported (2 of 4).* The predictions that hold are small and confounded with the spread. The margin's shading removes none of the longshot overpricing. The shading the outcomes require is not larger where private information is plausible, and it is smallest in Politics.

4. *Belief dispersion (Hypothesis 4): not supported (0 of 4).* The bias is not significantly steeper with more outcomes or turnover, and anchors do not reliably flatten it.

7 Discussion

7.1 What the three data sets agree on

The three samples measure the bias in different ways. The quote panel uses mid-quotes at fixed leads. The Kalshi trade sample uses the prices at which orders executed. The Polymarket sample uses fills by identified wallets. They agree on four points.

- *Longshots lose and favourites roughly break even.* On Kalshi, longshot sides return −55% per dollar at the mid. On Polymarket, longshot buys return −54% per dollar. In Kalshi's trade sample, taker longshot positions lose in 14 of 18 strata. Favourites earn between zero and a few per cent on both venues.

- *The bias is small per contract.* In probability units the mispricing is a fraction of a cent to a few cents (Section 6.1.5). Per dollar it is large only because the price is small. Sides quoted at 3–5 ¢ return −54% per dollar at the mid, so they win about half as often as their price implies, yet the absolute error is only about two cents.

- *Small traders hold the longshots.* On both venues, longshots are a large share of small orders or small wallets and almost none of large orders or whales.

- *Nobody is reliably on the other side.* Kalshi makers earn little more than takers overall, and on Polymarket no class of wallets is a significant net seller of longshots. The wallets that profit do not persist.

They differ in one respect. Polymarket's market-level calibration slope a day ahead, 0.99 (SE 0.15), is imprecise and cannot be distinguished from Kalshi's 1.13, but the equal-weight estimate, 0.97 (SE 0.05), is lower. Polymarket trades on a sub-cent grid near the extremes (89% of its fills below 4 ¢ or above 96 ¢ are at sub-cent prices) and charged no fees for most of the sample, so both mechanical frictions of Section 4 were weaker there. The samples, price measures and periods also differ, so we do not read the gap as a venue effect. On the 441 games observed on both venues, prices are almost identical.

7.2 Interpreting the bias

Mechanics set the band, not the position within it. The one-cent tick and the fee schedule explain why the bias cannot be traded away. A contract worth a tenth of a cent cannot be quoted below a cent, and a taker who sells the longshot pays a spread and a fee that absorb the whole premium. This is a limit to arbitrage in the sense of Shleifer and Vishny [46], created here by market design rather than by capital constraints. The band permits mispricing; it does not require it. Without the tick-floor buckets the slope is unchanged, and at a one-hour horizon, where the capital term of Proposition 1 vanishes, the bias is at least as strong as at thirty days. Something

keeps longshot prices near the upper edge of the band. That something is demand. The structural analysis sharpens this reading (Section 6.4).

- Market-weighted, the band covers the whole longshot gap in every group.
- Volume-weighted, the gap exceeds the band by 0.79 ¢ per contract ($t = 2.7$). In the busiest markets, where spreads are tight, prices go beyond what frictions allow.
- On the few markets quoted on a 0.1 ¢ grid, 1–2 ¢ sides remain overpriced beyond their band.

Who supplies that demand. The trade-level evidence identifies the buyers.

- Longshots are bought disproportionately by small orders and small wallets, as Hypothesis 2 predicts.
- The buying is not concentrated in moments of retail attention. On Kalshi, volume surges bring *fewer* longshots and fewer small orders, and only weekends and the last hour before expiry show a steeper bias.
- Small traders do not lose more per dollar than large ones on the longshots they buy. They bear most of the bias because they hold most longshots.

This is the pattern of lottery demand by individual investors in equity and option markets [13, 31]: the demand is for the payoff profile, not the result of worse execution.

Information plays a small role. Insider trading would make longshot flow informative and longshot prices protective. Neither holds. No order-size class predicts outcomes on Kalshi, small-order flow neither predicts nor reverts, and Polymarket's herding by small wallets moves prices for an hour without predicting outcomes. In multi-outcome events Shin's insider share is small, and the longshot field stays overpriced after the Shin adjustment. The insider share implied by the quoted margin is mostly a restatement of the spread. The share that best fits outcomes is no higher where private information is plausible, and it is lowest in politics.

Preferences and beliefs. Two explanations remain: preferences, either risk-love or probability weighting, and heterogeneous beliefs. Prices and outcomes alone cannot separate them. Both produce a monotone calibration curve, which a weighting function fits by construction [17, 35]. Separating risk-love from weighting would need bet sizes or exotic bets [47]. Three findings tilt the balance towards preferences.

- *One weighting function fits everywhere.* The fitted curvature is below one in every category (0.80 to 0.93), and a common value is not rejected under market weights. The pooled function predicts out of sample as well as group-specific ones. Its gain in log score over face-value prices is negligible, because most quotes lie in the middle of the price range, where prices are calibrated.
- *Dispersion proxies fail.* The bias is not significantly steeper in events with more outcomes or more turnover. An external forecast does not reliably flatten it.
- *The buyers are small.* Longshots are bought by small orders and small wallets, the traders that the evidence on lottery stocks and options associates with skewness preference [12, 13, 31].

Belief dispersion among these traders is not ruled out. A minority of noise traders with dispersed beliefs produces the same pattern [17], and our proxies for dispersion are coarse. What the evidence does rule out is a bias set by informed traders or by frictions alone.

7.3 Implications

- *Reading prices as probabilities.* Event-contract prices are informative but biased in the tails. Users who read them as forecasts should shrink longshot prices towards zero and favourite prices towards one, or apply the calibration function of Section 6.1. The correction matters most for multi-outcome fields and deadline contracts.
- *Market design.* The tick floor alone accounts for about half of the loss in the cheapest bucket. A finer grid near the extremes would let prices of very unlikely outcomes fall towards their probabilities. The fee per dollar is largest on longshots, so fees shape who can trade against the bias but do not explain why it exists.
- *Retail participation.* The losses on longshots fall mainly on small accounts, and wallet profits are concentrated in a few accounts that do not persist. This mirrors the distribution of outcomes in retail options trading [13].

7.4 Limitations

- *Retail is proxied.* On Kalshi we observe order size, not identity, and makers are anonymous. Polymarket wallet classes use each wallet's history within the sampled markets only.
- *Heavy tails.* Per-dollar longshot returns are dominated by rare wins. Pooled trade-level longshot means are imprecise and can change sign between samples, so we rely on per-stratum estimates, sign tests, calibration slopes and per-contract measures.
- *Samples and periods.* The trade sample covers two and a half months of 2026. The quote panel applies the November 2026 fee schedule to its whole history, so net returns before 2025 are indicative. The Polymarket sample excludes markets with more than $6 million of volume, which carry 42% of its frame's volume.
- *Maker returns.* Returns to resting orders assume fills at the quote and are an upper bound, because resting orders are adversely selected.
- *Multiple testing.* We report many cells. We therefore emphasise patterns that hold across categories, halves of the sample and data sets, rather than single cells.
- *Polymarket US.* The data host of Polymarket's US exchange disallows our agent in its `robots.txt`, so we use only its published fee schedule.

8 Conclusion

Event-contract exchanges reproduce the oldest anomaly in betting markets. On Kalshi, longshots lose more than half of every dollar staked on them and favourites roughly break even. The pattern holds across sports, weather, politics, entertainment, mentions and science, at every horizon from a month to an hour, and in liquid and illiquid markets alike. It has not shrunk as the venue has grown by three orders of magnitude.

Who buys the longshot answers much of why the bias persists.

- Small orders and small wallets buy the longshots. They do not trade at worse prices than large traders, but they hold most of the longshots and so bear most of the loss.

- Takers pay the spread and the fee on top, which together come to about a seventh of a longshot's stake.
- Liquidity suppliers take the other side at prices that just cover these costs, so the bias survives arbitrage without being a reliable source of profit for anyone. Wallet profits are extremely concentrated but do not persist.

Frictions explain why the bias is not traded away, but not why it arises. Neither insiders nor dispersion in beliefs explains the pattern well. A single mild probability-weighting function fits every category.

Two directions for future work follow. First, data that link identified accounts to their bet sizes over time would separate probability weighting from risk-love and from belief dispersion, which prices and outcomes alone cannot do. Second, the tick floor and fee schedule are design choices. Natural experiments in either, such as a move to a finer grid near the extremes, would show how much of the longshot premium market design can remove.

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A Additional results: calibration and the cross-section

A.1 Coverage

A.2 Supplementary tables

Table 25 gives returns by price bucket for every group. Tables 26 and 27 give the liquidity and calendar-time cuts behind Figure 5. Tables 28 and 29 collect the robustness checks of Section 6.1. Table 30 and Figure 21 add detail on multi-outcome events.

Table 24: Sampling frame, sample and coverage by group

	Frame		Analysed (pre-event quotes)				Analysed markets by year of T_m					
Group	Markets	Eligible	Markets	Events	First	Last	2021	2022	2023	2024	2025	2026
Sports	2,577,710	1,598,723	6,284	1,521	2024-12	2026-10	–	–	–	4	3,258	3,022
Weather	303,119	109,053	2,901	1,065	2021-07	2026-10	124	566	546	581	531	553
Politics	26,399	23,818	3,906	968	2021-07	2026-10	100	495	859	859	844	749
Entertainment	123,123	101,257	2,665	530	2021-08	2026-10	61	27	114	850	853	760
Mentions	61,798	40,894	1,760	103	2025-02	2026-10	–	–	–	–	827	933
Science & Tech	7,732	6,722	1,375	510	2021-07	2026-10	174	156	24	94	343	584
Other	207	126	116	73	2021-07	2026-01	38	66	–	–	10	2
All	3,100,088	1,880,593	19,007	4,770	2021-07	2026-10	497	1,310	1,543	2,388	6,666	6,603

Frame: settled single-leg non-financial markets, July 2021 to October 2026 (sports from July 2024). Eligible: scheduled time known and at most 31 October 2026, open at least one hour before it, event volume of at least 10 contracts. Analysed: sampled markets with at least one usable pre-event quote (spread ≤ 10 ¢). First and last: months of the scheduled time.

Table 25: Gross return per dollar at the mid by price bucket and group (%)

Bucket	All	Sports	Weather	Politics	Entert.	Mentions	Sci. & Tech	In-play
1-2c	−62.0	−62.1	−7.9	−57.7	−73.9	−100.0	−73.2	−91.1
	(−7.5)	(−3.3)	(−0.2)	(−4.5)	(−5.9)		(−5.3)	(−10.7)
3-5c	−54.1	−62.7	−46.7	−21.2	−81.1	−69.3	−55.7	−54.7
	(−9.1)	(−5.6)	(−3.0)	(−1.3)	(−11.0)	(−4.9)	(−2.8)	(−3.0)
6-10c	−29.2	−26.5	−46.8	−12.9	−26.6	−34.6	−35.2	−47.8
	(−4.5)	(−1.7)	(−3.8)	(−1.0)	(−1.5)	(−1.8)	(−1.7)	(−2.1)
11-20c	−10.6	−11.0	−11.4	+0.3	−22.1	−16.9	−14.6	−11.8
	(−2.4)	(−1.2)	(−1.2)	(0.0)	(−1.6)	(−1.4)	(−0.7)	(−0.5)
21-35c	+3.4	+6.0	+0.6	+3.9	−3.7	−0.8	+10.2	+2.0
	(1.2)	(1.2)	(0.1)	(0.7)	(−0.3)	(−0.1)	(0.6)	(0.2)
36-50c	+3.0	+3.4	+6.6	+0.0	−10.1	+5.3	−5.9	+15.8
	(1.4)	(1.1)	(1.5)	(0.0)	(−1.3)	(0.9)	(−0.6)	(1.9)
51-65c	−2.3	−2.7	−5.4	+0.6	+6.4	−3.7	+5.9	−9.8
	(−1.5)	(−1.1)	(−1.7)	(0.2)	(1.1)	(−0.9)	(0.9)	(−1.6)
66-80c	−1.3	−2.2	−0.4	−1.9	+2.0	+0.1	−1.4	+0.7
	(−1.2)	(−1.1)	(−0.2)	(−1.0)	(0.5)	(0.0)	(−0.2)	(0.1)
81-90c	+2.0	+2.0	+2.6	+0.1	+3.7	+3.2	+1.2	+3.3
	(2.6)	(1.2)	(1.6)	(0.1)	(1.7)	(1.6)	(0.4)	(0.9)
91-95c	+2.2	+2.4	+3.3	+0.7	+2.2	+2.5	+3.0	+2.4
	(4.7)	(2.2)	(3.6)	(0.7)	(1.9)	(1.4)	(2.0)	(1.2)
96-98c	+1.8	+2.0	+1.4	+0.9	+2.6	+2.5	+1.9	+1.9
	(9.7)	(5.7)	(2.4)	(1.9)	(13.4)	(5.6)	(3.6)	(1.9)
99c	+0.4	+0.4	+0.2	+0.3	+0.5	+0.7	+0.4	+0.7
	(5.6)	(2.7)	(0.5)	(2.0)	(4.6)	(28.4)	(1.6)	(15.6)

Mean of $(w - q)/q$ at the mid over YES and NO sides in each bucket of the side price, market-weighted, pre-event leads pooled (sports in-play: leads after the scheduled start). t-statistics clustered by event in parentheses. Cells with fewer than 30 sides are blank.

Table 26: The bias by liquidity tercile

Sort	Tercile	Markets	$\hat{\beta}$ ($t_{\beta=1}$)	LS gross (%)	FAV gross (%)	LS net (%)
7-day volume	Low	8,922	1.133 (3.7)	−53.3 (−5.0)	+1.3 (7.3)	−75.7 (−13.9)
	Mid	9,335	1.129 (4.7)	−51.0 (−6.5)	+1.2 (5.6)	−71.1 (−15.1)
	High	7,360	1.124 (4.1)	−60.9 (−7.9)	+0.9 (4.8)	−76.6 (−18.3)
Quoted spread	Low	8,767	1.167 (4.5)	−62.9 (−7.2)	+0.5 (4.5)	−77.8 (−17.3)
	Mid	9,729	1.106 (3.7)	−40.4 (−4.2)	+1.0 (4.9)	−65.3 (−12.7)
	High	10,590	1.131 (4.7)	−57.9 (−11.8)	+2.2 (8.8)	−80.0 (−27.8)
Open interest	Low	8,997	1.121 (3.8)	−48.4 (−5.4)	+1.3 (6.9)	−71.4 (−14.4)
	Mid	8,898	1.138 (4.9)	−54.7 (−6.9)	+1.2 (6.1)	−74.3 (−16.6)
	High	6,206	1.130 (3.8)	−63.0 (−6.7)	+0.8 (4.4)	−78.7 (−16.4)

Terciles are formed within group from quotes at the observation time: contracts traded in the previous 7 days, the quoted spread, and open interest. Market-weighted; pre-event leads pooled. Gross: return per dollar at the mid. Net: at the ask, net of the exact taker fee. t-statistics clustered by event in parentheses.

Table 27: The bias by quarter of the scheduled time

Quarter	Markets	Events	$\hat{\beta}$ ($t_{\beta=1}$)	LS gross (%)	LS net (%)
2021Q3	191	181	1.353 (2.0)	−80.4 (−5.2)	−82.2 (−5.8)
2021Q4	306	236	1.194 (1.3)	−15.4 (−0.3)	−36.9 (−1.0)
2022Q1	248	203	0.966 (−0.3)	+57.7 (0.7)	−14.1 (−0.3)
2022Q2	359	179	0.957 (−0.5)	+9.1 (0.2)	−33.3 (−1.1)
2022Q3	281	102	0.933 (−0.3)	+41.3 (0.4)	−14.1 (−0.2)
2022Q4	422	74	1.244 (1.5)	−35.8 (−1.0)	−59.3 (−2.8)
2023Q1	404	74	0.845 (−1.1)	+113.6 (1.2)	+13.1 (0.3)
2023Q2	380	63	1.397 (2.7)	−77.8 (−7.9)	−83.6 (−10.9)
2023Q3	347	61	1.254 (1.8)	−69.2 (−4.2)	−85.3 (−9.9)
2023Q4	412	87	1.114 (0.6)	−53.4 (−2.2)	−69.8 (−4.2)
2024Q1	554	110	1.297 (2.3)	−51.8 (−1.9)	−76.8 (−5.4)
2024Q2	622	100	1.079 (0.8)	+19.4 (0.3)	−40.0 (−1.4)
2024Q3	577	95	0.986 (−0.1)	−61.5 (−4.8)	−77.8 (−9.5)
2024Q4	635	150	1.040 (0.4)	−66.6 (−4.7)	−81.6 (−9.6)
2025Q1	1,288	278	1.162 (2.2)	−83.5 (−16.4)	−89.6 (−24.2)
2025Q2	1,765	352	1.138 (2.0)	−84.2 (−14.9)	−91.1 (−26.1)
2025Q3	1,828	441	1.188 (2.6)	−69.4 (−5.5)	−84.8 (−12.8)
2025Q4	1,785	456	1.084 (1.1)	−28.0 (−0.8)	−63.2 (−3.4)
2026Q1	1,835	486	1.186 (3.4)	−76.1 (−11.7)	−84.3 (−18.4)
2026Q2	1,960	479	1.190 (3.1)	−63.7 (−4.6)	−79.1 (−9.8)
2026Q3	2,137	501	1.100 (1.9)	−66.3 (−7.0)	−79.2 (−13.3)
2026Q4	671	156	1.188 (1.9)	−90.4 (−13.4)	−95.2 (−23.2)

All groups pooled, market-weighted, pre-event leads pooled. Quarters with fewer than 150 quotes or 20 events are omitted. t-statistics clustered by event in parentheses.

Table 28: Robustness of the pooled measures

Sample	Markets	$\hat{\beta}$ ($t_{\beta=1}$)	LS gross (%)	FAV gross (%)	LS net (%)
Full sample	19,007	1.129 (6.5)	−55.1 (−10.0)	+1.1 (9.8)	−74.7 (−25.3)
First half	9,395	1.099 (3.5)	−48.5 (−6.4)	+1.1 (7.2)	−70.6 (−17.0)
Second half	9,724	1.148 (5.3)	−62.9 (−8.1)	+1.1 (6.7)	−79.3 (−19.2)
Excluding 1–2 ¢ and 99 ¢	15,059	1.163 (7.9)	−56.2 (−15.7)	+2.5 (13.3)	−72.1 (−27.5)
Spread $\leq$ 5 ¢	16,656	1.116 (5.5)	−52.5 (−8.4)	+0.8 (7.2)	−72.6 (−22.2)
Quote age $\leq$ 24 h	18,511	1.134 (6.7)	−54.4 (−9.6)	+1.2 (10.1)	−74.4 (−24.3)
Design-weighted	19,007	1.153 (2.2)	−56.3 (−5.2)	+1.4 (2.9)	−72.8 (−9.7)
Scheduled time: close	5,883	1.162 (3.9)	−69.0 (−8.1)	+1.1 (8.1)	−83.5 (−19.4)
Scheduled time: date	9,683	1.111 (4.3)	−33.0 (−3.2)	+1.2 (5.4)	−61.3 (−10.5)
Scheduled time: deadline	2,509	1.224 (3.9)	−68.3 (−10.7)	+0.9 (4.0)	−81.1 (−20.0)
Scheduled time: start	932	1.080 (0.8)	−52.5 (−2.7)	+1.0 (0.6)	−63.5 (−4.1)

Market-weighted, pre-event leads pooled, all groups. Halves split at the median observation time. Excluding extreme buckets drops quotes with a mid at or below 2 ¢ or at or above 98 ¢. Design-weighted: each event weighted by the inverse of its sampling probability within its cell. Scheduled time: subsamples by the source of T_m (Section 3.1). t-statistics clustered by event in parentheses.

Table 29: Standard errors under alternative clustering

Cluster	Clusters	$\hat{\beta}$	se	($t_{\beta=1}$)	LS gross (%)	se (%)	(t)
Event	4,770	1.129	0.020	(6.5)	−55.1	5.5	(−10.0)
Market	19,007	1.129	0.017	(7.7)	−55.1	5.0	(−11.1)
Series	1,670	1.129	0.020	(6.3)	−55.1	6.4	(−8.6)
Month	64	1.129	0.017	(7.7)	−55.1	6.2	(−8.9)

Pooled pre-event sample, market-weighted. The clusters column counts clusters in the logit regression.

Table 30: Overround and Shin insider share by group

Group	Legs	Events	Event-leads	Mean z	Median z	$\sum a - 1$ (%)	$\sum p - 1$ (%)
Politics	2	74	376	0.031	0.015	+2.4	+0.0
Sports	2	660	1,609	0.030	0.020	+2.8	−0.1
Entertainment	>=3	35	94	0.032	0.029	+19.6	+6.9
Politics	>=3	46	217	0.029	0.021	+11.2	+3.1
Science & Tech	>=3	44	214	0.018	0.015	+7.2	+1.6
Sports	>=3	256	764	0.024	0.016	+8.7	+1.6

Mutually exclusive events with every leg quoted at the lead and exactly one leg settling YES. z is computed from the asks (Section 5.1). Cells with at least 20 event-leads and 10 events.

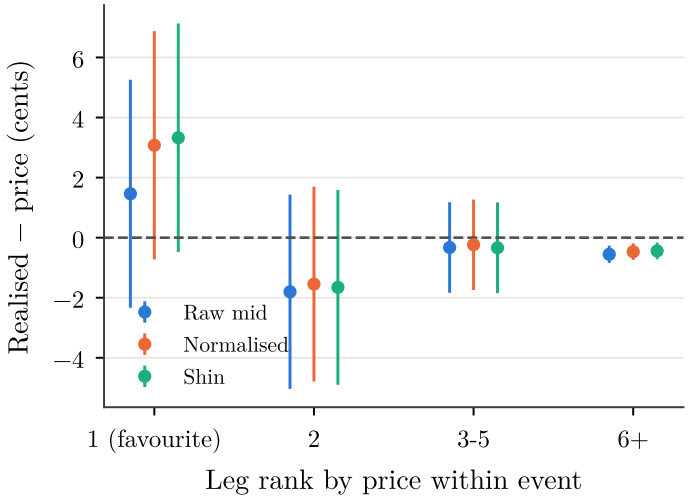


Figure 21: Multi-outcome events (mutually exclusive, at least three legs): realised frequency minus price, in cents, by the leg's price rank within its event, for the raw mid, the mid normalised to sum to one, and the Shin probability $\hat{p}_k(z)$. 95% confidence intervals clustered by event.

B Additional results: trade-level flow

B.1 Sample design and coverage

Table 31 lists the strata of the trade sample. The design weight of an event is its stratum's frame-to-sample event ratio, times $L_e/20$ when the event has $L_e > 20$ traded legs. Sports leagues are identified from the series ticker. "Other sports" collects the remaining leagues, including motor racing, minor soccer and basketball leagues, table tennis and the Ballon d'Or.

Table 31: Sampling frame and stratified sample of the trade-level analysis

	Frame		Sample		
Stratum	Events	Markets	Events	Markets	Weight per event
Combat	1,295	4,688	60	208	21.6
Cricket	1,154	3,463	60	213	19.2
Esports	32,301	67,181	150	306	215.3
Golf	779	12,229	40	231	19.5
MLB	22,948	231,556	100	624	229.5
NBA	3,935	61,082	50	609	78.7
NCAAF	11,871	126,743	80	891	148.4
NFL	7,995	99,542	60	491	133.2
NHL	4,621	44,803	60	375	77.0
Other sports	43,694	147,595	150	475	291.3
Soccer	32,087	134,290	120	473	267.4
Tennis	30,489	65,940	150	318	203.3
WNBA	2,646	19,812	60	488	44.1
Entertainment	1,160	12,095	260	2,492	4.5
Mentions	349	7,230	120	2,048	2.9
Politics	801	4,589	260	1,515	3.1
Science & Tech	426	3,200	220	1,615	1.9
Weather	12,303	82,218	300	1,957	41.0

Frame: settled, non-financial Kalshi markets that closed 15 August–31 October 2026 and traded at least once. Events are drawn uniformly at random within each stratum (seed 20261111). Weight per event: frame events divided by sample events.

Table 32: Coverage of the main trade sample by group

Group	Markets	Events	Prints	Taker orders	Contracts (m)	Taker $ (m)	Dropped (%)	Complete (%)
Sports	5,026	1,003	1,529,207	994,330	191.84	92.30	40.9	100
Weather	1,720	297	203,699	96,491	4.03	1.79	17.7	100
Politics	1,489	258	339,908	213,216	37.93	17.81	10.2	100
Entertainment	2,435	258	278,713	161,512	22.48	10.10	9.4	100
Mentions	1,921	119	245,400	126,128	9.21	3.07	43.5	100
Science & Tech	1,573	216	88,190	51,857	11.19	6.59	3.3	100

Counts after the pre-resolution filter. Taker $ is the taker side's notional, $\sum nq$. Dropped: the share of all fetched prints removed by the filter. Complete: the median across markets of contracts printed over the market's reported volume.

B.2 Choosing the pre-resolution cut

Table 33 motivates the cut-offs. In sports the share of contracts traded at $\leq 3\,¢$ or $\geq 97\,¢$ is 3–10% more than half an hour before the close. It then rises to about 20% at 10–30 minutes, about 35% at 5–10 minutes and 65–75% in the last five minutes. These are games that are already decided and sweeps after the final whistle. In non-sports markets that closed early, the share jumps from 22–35% to 72–89% in the last hour, while the event that closed the market unfolds. The four-hour cut is deliberately wider than that, because the moment the event occurred is not observed. In markets that closed on schedule, extreme prices are common at every horizon. They come from the far legs of multi-outcome events, which are genuine longshots, so they are kept.

Table 33: Share of contracts traded at extreme prices by minutes before the close (%)

	Sports		Non-sports		Share of
Minutes before close	On schedule	Early	On schedule	Early	volume
0–5	75	65	61	89	5.2
5–10	35	34	68	86	4.8
10–30	22	18	54	80	15.6
30–60	9	10	62	72	15.0
60–240	4	3	67	22	32.5
> 240	5	8	45	35	26.8

All fetched prints of the main sample, before any filter. Extreme: YES price $\leq 3\,¢$ or $\geq 97\,¢$; contract-weighted, unweighted by design. Closed early: closed more than 30 minutes before the scheduled expiration. Share of volume: the window's share of all contracts.

B.3 Returns by price bucket

Table 34 and Figure 22 give pooled, design-weighted returns by price bucket for both roles, overall and by group. Table 35 splits takers by the side they bought. The pooled longshot cells are dominated by a few upsets (Section 6.2.1); the per-stratum medians in the main text are the robust summary.

B.4 Trade size by sector

Table 36 repeats Table 10 separately for sports and non-sports markets. The longshot share of small orders is twice as high outside sports.

B.5 Timing cells

Table 37 lists taker flow and returns in every timing and attention cell used in Section 6.2.4.

C Additional results: wallets and venues

C.1 Data access and robots.txt record

Following the project's data policy, we use a host only if its own `robots.txt` allows our access. We checked each host on 11 November 2026.

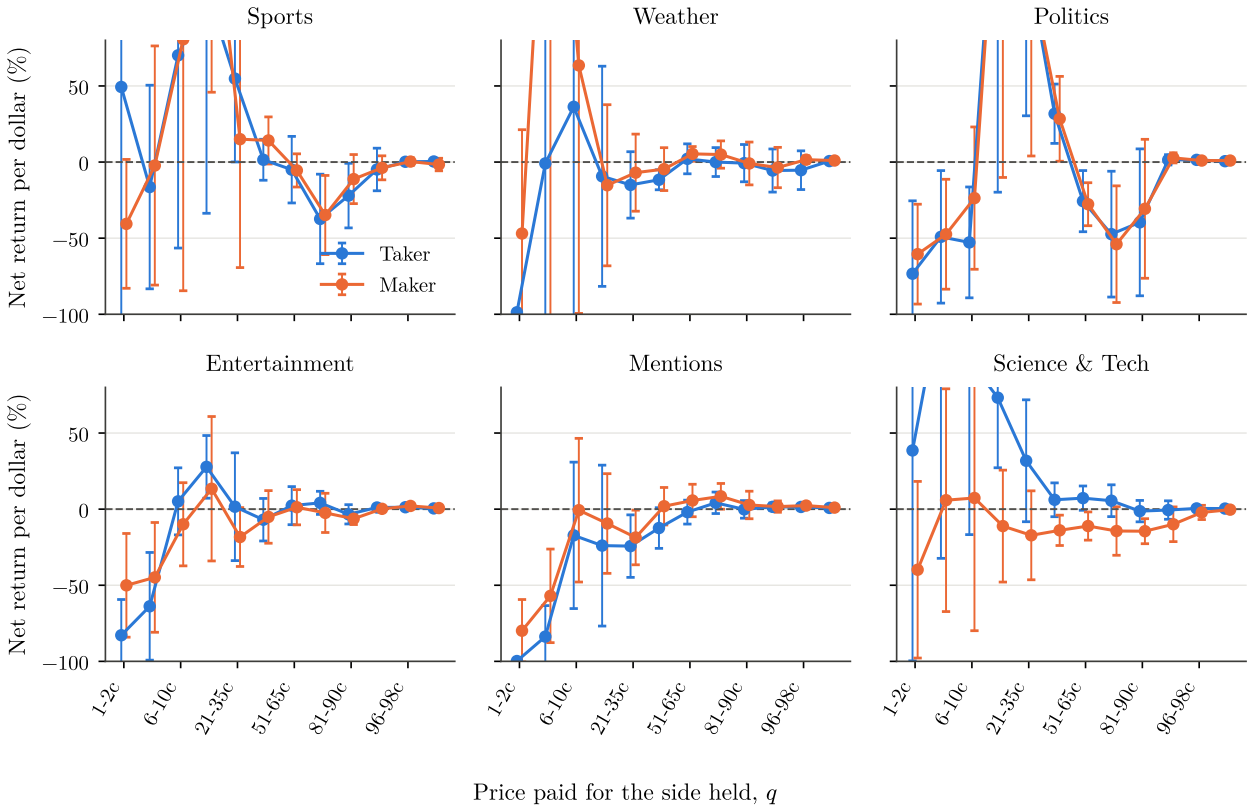


Figure 22: Takers against makers by price bucket and group. Net return per dollar (dollar-weighted, after each role's fee) of taker and maker positions, by the price of the side held, with 95% confidence intervals clustered by event. Main sample, design-weighted.

Table 34: Taker and maker returns by the price of the side held

	Taker	Taker		Maker	
Side price q	share (%)	Gross (¢)	Net (%/$)	Gross (¢)	Net (%/$)
1–2 ¢	56	+0.48 (0.4)	+29.5 (0.3)	−0.55 (−2.1)	−41.9 (−2.2)
3–5 ¢	56	−0.46 (−0.3)	−17.2 (−0.5)	+0.06 (0.0)	+0.8 (0.0)
6–10 ¢	64	+6.18 (1.2)	+68.3 (1.1)	+6.40 (1.0)	+78.7 (1.0)
11–20 ¢	56	+19.67 (1.5)	+113.3 (1.5)	+31.02 (2.4)	+183.1 (2.6)
21–35 ¢	46	+17.22 (2.2)	+54.7 (2.0)	+4.77 (0.4)	+15.1 (0.4)
36–50 ¢	46	+2.29 (0.7)	+1.5 (0.2)	+6.54 (1.9)	+14.1 (1.8)
51–65 ¢	58	−1.40 (−0.2)	−5.0 (−0.5)	−2.93 (−0.9)	−5.5 (−1.0)
66–80 ¢	49	−26.70 (−2.3)	−37.2 (−2.5)	−25.62 (−2.5)	−34.7 (−2.6)
81–90 ¢	41	−17.95 (−2.0)	−21.9 (−2.0)	−9.49 (−1.4)	−11.2 (−1.4)
91–95 ¢	39	−4.02 (−0.6)	−4.8 (−0.7)	−3.40 (−0.9)	−3.7 (−0.9)
96–98 ¢	44	+0.30 (0.3)	+0.1 (0.1)	+0.41 (0.4)	+0.4 (0.3)
99 ¢	42	+0.47 (2.5)	+0.4 (2.1)	−1.17 (−0.7)	−1.2 (−0.7)

Main sample, all groups, design-weighted. Each print contributes a taker position at q and a maker position at $1 - q$, and both are sorted by the price of the side held. Taker share: the takers' share of the contracts held at that price. Gross: contract-weighted (7) with $f = 0$, in cents. Net: dollar-weighted (8) after fees, in percent. t-statistics clustered by event are in parentheses.

Table 35: YES-takers against NO-takers at equal side price

Taker price q	YES share (%)	YES-taker, %/$	NO-taker, %/$	Difference
1–2 ¢	79	+57.3 (0.5)	−90.8 (−21.0)	+148.1 (1.2)
3–5 ¢	85	−14.2 (−0.4)	−33.3 (−0.9)	+19.0 (0.4)
6–10 ¢	85	+67.0 (1.0)	+75.3 (0.9)	−8.3 (−0.1)
11–20 ¢	85	+124.8 (1.5)	+48.4 (1.0)	+76.4 (0.8)
21–35 ¢	78	+57.1 (1.8)	+46.5 (2.5)	+10.6 (0.4)
36–50 ¢	73	−6.3 (−0.6)	+23.2 (1.4)	−29.5 (−1.3)
51–65 ¢	74	−10.4 (−0.6)	+11.2 (1.2)	−21.6 (−1.0)
66–80 ¢	71	−42.9 (−2.7)	−22.9 (−1.8)	−20.1 (−2.3)
81–90 ¢	62	−31.7 (−2.5)	−5.9 (−1.1)	−25.8 (−3.2)
91–95 ¢	48	−9.1 (−0.9)	−0.7 (−0.2)	−8.4 (−1.4)
96–98 ¢	45	+0.1 (0.1)	+0.1 (0.1)	−0.0 (−0.0)
99 ¢	53	+0.4 (1.7)	+0.4 (1.3)	+0.0 (0.1)

Main sample, all groups, taker positions only, design-weighted. Net return per dollar is dollar-weighted (8), in percent. YES share: the share of taker contracts at that price bought as YES. Difference: YES minus NO, in percentage points, with an event-clustered t that accounts for shared events.

Table 36: Taker flow and returns by order size, sports and non-sports

		Share of taker		Share at	Mean	Taker net return per dollar	
Sector	Order size	orders	contracts	$q \leq 10$ ¢	q (¢)	Longshots	Favourites
Sports	≤$10	51.7	4.7	36.3	23	+36.1 (0.8)	−2.7 (−1.2)
	$10–100	37.7	20.9	17.0	35	+55.3 (1.1)	−1.0 (−0.4)
	$100–1k	9.5	31.8	8.3	47	+27.3 (0.6)	−1.4 (−0.5)
	>$1k	1.2	42.7	1.2	58	+169.0 (1.2)	−3.8 (−1.0)
Non-sports	≤$10	67.1	20.7	74.0	13	−35.2 (−1.3)	−2.4 (−1.1)
	$10–100	24.8	32.3	46.4	30	−23.1 (−1.0)	−3.4 (−1.1)
	$100–1k	7.0	27.7	21.6	56	−29.7 (−1.3)	−0.9 (−0.6)
	>$1k	1.1	19.4	2.6	81	−85.1 (−4.6)	+0.5 (0.6)

As Table 10, separately for sports and non-sports markets. Shares are in percent within the sector. Returns are dollar-weighted (8) after the taker fee, with event-clustered t-statistics in parentheses.

Table 37: Taker flow and returns by timing and attention cell

Cell	Volume share (%)	Taker return per dollar (%)		Share at $q \leq 10\,¢$ (%)	Orders $\leq \$100$ (%)
		Net	Gross		
Hour of day (ET)					
00–06	8.1	−10.3 (−1.7)	−8.2 (−1.4)	11.2	89.5
06–12	11.8	−11.0 (−2.3)	−8.9 (−1.8)	9.5	89.5
12–18	41.2	−1.2 (−0.2)	+1.3 (0.2)	8.8	88.8
18–24	38.8	+0.9 (0.3)	+3.4 (1.1)	8.2	89.9
Day					
Weekday	72.6	+0.7 (0.2)	+3.2 (0.9)	8.4	89.5
Weekend	27.4	−10.2 (−2.3)	−8.1 (−1.8)	10.0	89.2
Sports game phase					
Pre-game	33.4	−3.4 (−0.4)	−0.9 (−0.1)	5.3	91.6
In play	66.6	−2.6 (−1.1)	−0.3 (−0.1)	8.5	88.1
Sports game start					
Before 19:00 ET	55.0	−2.7 (−0.4)	−0.4 (−0.1)	8.7	88.3
19:00–23:59 ET	45.0	−3.1 (−1.2)	−0.7 (−0.3)	6.0	90.6
Time to scheduled expiry					
< 1 h	13.4	−5.2 (−2.4)	−3.2 (−1.4)	9.7	87.4
1–6 h	70.2	−0.4 (−0.1)	+2.1 (0.6)	7.5	88.1
6–24 h	10.7	−1.5 (−0.1)	+1.1 (0.1)	9.2	92.8
1–7 days	4.5	−26.8 (−2.2)	−25.2 (−2.0)	17.3	96.1
> 7 days	1.3	−1.1 (−0.4)	+0.9 (0.3)	43.9	90.1
Market age					
First 24 h	23.0	−5.8 (−1.6)	−3.7 (−1.0)	10.7	86.3
Later	77.0	−1.4 (−0.3)	+1.1 (0.3)	8.3	90.2
Past-hour volume vs own average					
$< 0.5\times$	2.7	−13.8 (−2.0)	−11.7 (−1.7)	27.2	92.0
0.5–$3\times$	4.1	−12.4 (−1.1)	−10.5 (−0.9)	16.0	93.9
3–$10\times$	7.2	−12.2 (−1.4)	−10.2 (−1.1)	11.5	93.3
$\geq 10\times$	86.0	−0.7 (−0.2)	+1.8 (0.5)	6.9	88.4
Event volume					
Below top decile	14.1	−3.3 (−1.2)	−1.1 (−0.4)	12.4	89.3
Top decile (marquee)	85.9	−2.3 (−0.6)	+0.2 (0.0)	8.3	89.4

Main sample, all groups, taker positions, design-weighted. Volume share: the cell's share of taker contracts within the panel. Returns are dollar-weighted (8), with event-clustered t-statistics in parentheses. Share at $q \leq 10\,¢$: the share of taker contracts bought at a longshot price. Orders $\leq \$100$: the share of taker orders with a notional of at most \$100. Game phase and start time cover sports games with a known scheduled start. The past-hour volume ratio needs six hours of trading history.

Table 38: Taker fee per contract by venue (cents)

	Kalshi	Polymarket International			Polymarket US	
q (¢)		Sports	Politics	Geopolitics	< $10k	≥ $250k
2	0.14	0.10	0.08	0	0.19	0.14
5	0.33	0.24	0.19	0	0.45	0.33
10	0.63	0.45	0.36	0	0.85	0.63
25	1.31	0.94	0.75	0	1.78	1.30
50	1.75	1.25	1.00	0	2.38	1.74

Exact fee $\theta\, q(1-q)$ per contract before rounding. Kalshi: $\theta = 0.07$; it computes fees per fill to $0.000001, rebates rounding over-payment, and charges makers $0.0175\, q(1-q)$ on some series. Polymarket International (`docs.polymarket.com/trading/fees`): makers pay nothing; fees are in force from March–April 2026. Polymarket US (`docs.polymarket.us/fees`): the two columns are its lowest and highest volume tiers, with banker's rounding to the cent per order. Schedules checked on 11 November 2026.

- `data-api.polymarket.com` (trades, account statistics, price histories) and `gamma-api.polymarket.com` (markets, events, tags, resolutions) serve no `robots.txt`: the request returns HTTP 410 or 404, which places no restriction. Both are used, at no more than three requests per second.
- `polymarket.com` allows all agents (`User-agent: * / Disallow:`) and points AI agents to an `llms.txt`. We fetched nothing from it.
- `docs.polymarket.com` and `docs.polymarket.us` allow all agents (`Content-Signal: ai-input=yes`). We read their fee pages.
- `gateway.polymarket.us`, the Polymarket US data host, disallows `anthropic-ai`, `ClaudeBot` and `Claude-Web`. We therefore use no Polymarket US market data. The venue enters the comparison only through its published fee schedule.

No order was placed on any venue, and no geoblocking workaround was used. Polymarket's international exchange is used purely as a public data source.

C.2 Fee schedules

Table 38 lists the taker fee per contract on each venue at several side prices. All three venues charge $\theta\, q(1-q)$ per contract, so the fee is largest in dollars at 50 ¢ and largest relative to the price for longshots: at $q = 5$ ¢ a positive taker fee of 0.19–0.45 ¢ is 4–9% of the price.

- Polymarket International charges no maker fees. Its taker rate depends on the category (0.05 for sports, culture, weather and other; 0.04 for politics, mentions and tech; zero for geopolitics). Fees were switched on for almost every market in March–April 2026. Before that, almost all trading was fee-free.
- Polymarket US sets θ by the account's trailing 30-day taker volume, from 0.095 below $10,000 to 0.0695 above $250,000 (effective 28 October 2026). Small accounts therefore pay the highest rate on any of the three venues.

Table 39: Polymarket frame and trade sample

	Frame		Trade sample	
	Markets	Volume ($M)	Markets	Fills
Sports	461,685	51,677	481	295,004
Weather	115,966	1,099	362	360,575
Politics	29,382	19,858	480	508,491
Entertainment	14,526	2,060	466	478,494
Mentions	10,948	315	398	224,015
Science & Tech	3,524	431	315	515,981
Other	489	400	263	191,915
$1k-10k	383,120		1,126	185,467
$10k-100k	183,614		1,222	787,395
$100k-1M	59,808		386	1,232,656
$1M-6M	8,638		31	368,957
$>$6M	1,340		–	–

Frame: resolved Polymarket markets with $\geq$ $1,000 volume in events ending January 2024 to October 2026, excluding financial tags. Sample: markets whose complete trade history was downloaded (maker and taker rows). Volume tiers by market volume; > $6M markets are not sampled.

C.3 Sample, coverage and data checks

Table 39 lists the frame and the trade sample by group and volume tier. The sample design balances the groups. Sports make up 72.5% of the frame's markets, but each group contributes several hundred sampled markets, so group-level estimates are comparably precise and pooled estimates are re-weighted with the design weights. Two checks support the trade data:

- *Exposure balance.* In 97.8% of the 2,197 fee-free markets the YES-equivalent exposures of all wallets sum to zero, and in 99.95% their PnL sums to zero within 10^{-6} of turnover, as (12) requires when every fill is observed. In the 568 fee-charging markets the figure is 85.9%, because the taker's buy size there is net of the fee taken in shares. The total PnL of all wallets is within 1% of turnover in 100% of them.
- *Resolution agreement.* For the cross-venue sample, Kalshi and Polymarket settle every matched team-side the same way (100%).

Polymarket's taker fees cover 14% of frame markets resolving in March 2026, 85% in April and 100% from May; earlier markets are fee-free. 0 sampled markets exceeded the 150-page cap and are excluded from the trade analyses.

C.4 Robustness of the class results

Table 40 moves each class cut-off in turn. Table 41 splits the class results by half of the sample. Table 42 replaces the classes with fill size, account age and leaderboard status. Table 43 reports the hold-or-exit decomposition by class.

Table 40: Robustness of the class results to the cut-offs

	Retail		Active retail		Mid		Whale		MM	
Variant	LS$	r_{LS}	LS$	r_{LS}	LS$	r_{LS}	LS$	r_{LS}	LS$	r_{LS}
Baseline	1.3	−37	0.7	−59	0.3	−89	0.1	−100	0.7	−42
Small ≤ $50	1.6	−26	0.8	−57	0.3	−89	0.1	−100	0.7	−42
Small ≤ $250	1.1	−45	0.5	−60	0.4	−97	0.1	−100	0.7	−42
Whale ≥ $500	1.3	−37	0.7	−59	0.4	−89	0.1	−99	0.7	−42
Whale ≥ $5,000	1.3	−37	0.7	−59	0.2	−91	0.1	−100	0.7	−42
Active: ≥ 10 fills	1.6	−36	0.7	−56	0.3	−89	0.1	−100	0.7	−42
Active: ≥ 50 fills	1.1	−42	0.6	−59	0.3	−89	0.1	−100	0.7	−42
MM loose (10, 0.5, 0.3)	1.3	−37	0.5	−58	0.3	−90	0.1	−100	0.7	−48
MM strict (50, 0.9, 0.7)	1.3	−37	0.7	−60	0.3	−89	0.1	−100	0.7	−2

LS$: longshot share of dollars bought (%); r_{LS}: net return per dollar of longshot buys (%), both dollar-weighted, buy fills before t^{info}. Each row moves one cut-off of the baseline (small $100, whale $1,000, active 20 earlier fills, MM: 20 fills, maker share 0.7, two-sided share 0.5).

Table 41: Class results by half of the sample

	H1: January 2024 – May 2025				H2: June 2025 – October 2026			
	LS fills	r_{LS}	Net LS flow		LS fills	r_{LS}	Net LS flow	
Retail	15.2	−89	1.67	(2.4)	11.1	−18	0.29	(3.3)
Active retail	17.2	−90	0.80	(0.8)	15.6	−39	−0.03	(−0.5)
Mid	5.3	−96	−0.10	(−0.2)	1.2	−84	−0.06	(−1.1)
Whale	2.1	−100	−0.30	(−0.9)	0.1	−100	−0.04	(−1.6)
MM	38.0	−87	−1.17	(−1.5)	20.0	−9	−0.04	(−0.8)

LS fills: longshot share of design-weighted buy fills (%); r_{LS}: net return per dollar of longshot buys (%, dollar-weighted); net LS flow: net longshot dollars bought per dollar traded (%, t in parentheses). Buy and sell fills before t^{info}; t-statistics clustered by event.

D Additional results: structural decomposition

This appendix supports Section 6.4. The CSV files under `data/research/flb_structural/` also hold every bucket × lead × group cell of the band, the fits for each group and lead, and the z of every event-lead.

- *Band.* Table 44 splits the band decomposition by price bucket and group. Table 45 varies:
  - the cost of capital;
  - the weights;
  - the sample half;
  - the quote age and the spread;
  - the tick grid and the contract structure.

  It also drops the five events with the largest absolute contribution to the residual.
- *Weighting.* Table 46 and Figure 23 give the weighting functions by lead, structure and group.
- *Shin.* Tables 47 and 48 and Figure 24 give z by lead, group and structure, and compare forecasts.

Table 42: Longshot buying by fill size, account age and leaderboard status

			Longshot share (%)			FAV	Longshot r	
	Wallets	Fills	fills	\$	(t)	\$ (%)	%	(t)
Fill size								
≤\$10	133,995	828,526	23.1	7.6	(6.0)	14.8	−22.2	(−0.6)
\$10–100	67,035	383,136	2.4	1.9	(4.6)	22.9	−48.4	(−2.4)
\$100–1k	20,785	97,225	0.7	0.6	(2.4)	15.4	−77.6	(−4.5)
>\$1k	3,278	10,420	0.2	0.1	(1.6)	7.9	−100.0	(< −100)
Account age at the fill								
<7 days	905	1,768	14.9	0.4	(1.1)	3.5	−97.2	(−40.2)
7–30 days	1,031	3,540	8.3	0.2	(1.4)	9.1	−88.3	(−12.1)
30–180 days	1,765	9,318	19.9	0.8	(2.7)	9.8	18.2	(0.2)
>180 days	1,016	8,608	22.7	0.8	(2.5)	14.3	−31.3	(−0.5)
All-time leaderboard (ex post)								
Leaderboard	86	33,380	6.8	0.1	(2.3)	8.2	−85.0	(−11.3)
Other	188,115	1,285,927	15.4	0.6	(3.2)	11.9	−52.6	(−2.7)

Buy fills before t^{info}. Account age uses the creation date of the address-selected wallet subset with account statistics. Shares and returns as in Table 17; t-statistics clustered by event.

Table 43: Holding to settlement versus exiting, by class and entry price

	All positions			Longshot entries ($q \leq 10\,¢$)				FAV entries	
	Pos.	Exited %	Exit VA % (t)	Pos.	Exited %	Hold r % (t)	Exit VA % (t)	Exited %	Exit VA % (t)
Retail	262,487	21.4	−3.3 (−1.1)	43,801	41.7	−66.9 (−4.0)	28.1 (1.8)	28.2	−0.5 (−2.1)
Active retail	46,392	6.3	0.0 (0.1)	9,693	6.9	−91.8 (−21.6)	4.4 (1.7)	10.3	−0.3 (−2.4)
Mid	19,761	12.8	−2.1 (−1.2)	880	16.9	−97.1 (−64.0)	19.1 (2.6)	27.4	−0.6 (−3.0)
Whale	1,885	21.3	−4.9 (−1.9)	33	0.5	−99.0 (−81.4)	−0.1 (−0.1)	55.5	−8.8 (−1.4)
MM	43,662	18.6	2.4 (1.9)	13,153	54.7	−77.4 (−6.8)	51.5 (3.0)	23.6	−1.1 (−2.5)
All	374,187	16.1	−1.4 (−1.3)	67,560	28.9	−81.6 (−10.6)	23.2 (2.2)	24.4	−1.2 (−2.1)

Positions: wallet × market × outcome, opened by a buy before t^{info}. Exited: share (%, cost- and design-weighted) with at least half of the shares sold before the close. Exit VA: realised minus hold-to-settlement PnL per dollar of cost (%). Hold r: return per dollar if every share were held. Gross of fees; t-statistics clustered by event.

- *Belief dispersion.* Tables 49 and 50 and Figure 25 report the slope interactions, pooled, by group and by half.
- *Synthesis.* Table 51 lists every prediction tested and Table 52 the counterfactual shares.

Table 44: The band by group and longshot bucket

				Gross	Band components				Residual	Maker	In band
Cell	Sides	Events	$\bar{q}$	$\bar{q} - \bar{w}$	Floor	Half-spr.	Fee	Capital	$\bar{R}$	$\bar{R}^{mk}$	(%)
All groups											
1–2 ¢	26,184	2,118	0.87	0.49 (6.9)	0.30	0.48	0.01	0.03	−0.33 (−4.6)	1.24 (16.9)	100
3–5 ¢	9,521	2,007	3.53	1.86 (9.0)	0.00	2.40	0.08	0.04	−0.65 (−3.2)	4.21 (19.4)	100
6–10 ¢	4,847	1,707	7.53	2.01 (4.1)	0.00	2.22	0.34	0.04	−0.60 (−1.2)	4.18 (8.4)	100
Sports											
1–2 ¢	6,211	303	0.79	0.46 (3.1)	0.35	0.39	0.00	0.05	−0.32 (−2.2)	1.14 (7.4)	100
3–5 ¢	1,914	280	3.56	2.25 (5.8)	0.00	2.67	0.06	0.04	−0.52 (−1.4)	4.85 (11.3)	100
6–10 ¢	983	276	7.63	2.02 (1.7)	0.00	2.21	0.32	0.03	−0.54 (−0.5)	4.16 (3.4)	100
Weather											
1–2 ¢	2,126	317	0.97	0.24 (0.7)	0.27	0.56	0.01	0.03	−0.63 (−1.9)	1.04 (3.2)	100
3–5 ¢	1,122	455	3.68	1.57 (2.6)	0.00	2.38	0.09	0.02	−0.92 (−1.5)	3.94 (6.4)	100
6–10 ¢	657	384	7.52	3.29 (3.4)	0.00	2.33	0.34	0.02	0.59 (0.6)	5.60 (5.8)	82
Politics											
1–2 ¢	7,720	728	0.93	0.41 (2.7)	0.25	0.56	0.01	0.03	−0.44 (−2.9)	1.19 (7.9)	100
3–5 ¢	2,597	563	3.49	0.66 (1.2)	0.00	2.14	0.09	0.05	−1.62 (−3.0)	2.76 (5.1)	100
6–10 ¢	1,451	482	7.54	0.70 (0.7)	0.00	2.17	0.35	0.06	−1.89 (−1.9)	2.81 (2.8)	100
Entertainment											
1–2 ¢	6,011	377	0.90	0.66 (6.1)	0.29	0.54	0.00	0.02	−0.20 (−1.9)	1.47 (12.3)	100
3–5 ¢	2,169	353	3.48	2.77 (10.2)	0.00	2.76	0.05	0.04	−0.09 (−0.3)	5.49 (18.6)	100
6–10 ¢	759	266	7.39	1.74 (1.3)	0.00	2.33	0.33	0.07	−0.99 (−0.7)	4.01 (2.9)	100
Mentions											
1–2 ¢	982	58	0.72	0.72 (24.0)	0.39	0.18	0.01	0.02	0.12 (5.9)	1.27 (25.0)	83
3–5 ¢	423	64	3.60	2.57 (5.7)	0.00	1.53	0.14	0.02	0.88 (2.0)	4.08 (7.5)	66
6–10 ¢	513	82	7.65	2.38 (1.6)	0.00	2.02	0.37	0.02	−0.03 (0.0)	4.38 (2.9)	100
Science & Tech											
1–2 ¢	2,967	311	0.87	0.48 (2.4)	0.30	0.47	0.01	0.03	−0.33 (−1.6)	1.21 (5.8)	100
3–5 ¢	1,184	267	3.38	1.93 (3.2)	0.00	2.13	0.09	0.06	−0.35 (−0.6)	3.97 (6.4)	100
6–10 ¢	421	189	7.37	2.48 (1.7)	0.00	2.24	0.34	0.09	−0.19 (−0.1)	4.60 (3.1)	100

As Table 20, by group and price bucket, leads pooled, $\rho = 5\%$, market-weighted. t-statistics clustered by event in parentheses.

Table 45: The band: cost of capital, weights, halves and the tick grid

Cell	Sides	Events	$\bar{q}$	Gross $\bar{q}-\bar{w}$	Band components Floor	Half-spr.	Fee	Capital	Residual $\bar{R}$	Maker $\bar{R}^{mk}$	In band (%)
Main	40,552	3,091	2.54	1.07 (10.0)	0.18	1.23	0.07	0.04	−0.45 (−4.3)	2.44 (21.3)	100
$\rho = 0$	40,552	3,091	2.54	1.07 (10.0)	0.18	1.23	0.07	0.00	−0.42 (−3.9)	2.47 (21.6)	100
$\rho = 10\%$	40,552	3,091	2.54	1.07 (10.0)	0.18	1.23	0.07	0.07	−0.49 (−4.6)	2.40 (20.9)	100
Volume-weighted	40,552	3,091	2.18	1.56 (5.1)	0.27	0.39	0.10	0.02	0.79 (2.7)	2.19 (7.3)	50
First half (to 2025-07)	22,146	1,762	2.67	0.81 (5.1)	0.17	1.30	0.08	0.03	−0.76 (−4.8)	2.24 (13.3)	100
Second half	18,406	1,341	2.69	0.97 (5.4)	0.18	1.19	0.08	0.04	−0.52 (−2.9)	2.29 (12.2)	100
Drop 5 influential events	40,507	3,086	2.54	1.16 (11.7)	0.18	1.23	0.07	0.04	−0.36 (−3.7)	2.53 (23.6)	100
Quote age $\leq$ 24 h	33,432	3,055	2.72	1.11 (9.2)	0.16	1.29	0.08	0.04	−0.46 (−3.9)	2.52 (19.7)	100
Spread $\leq 5\,¢$	35,751	2,868	2.11	0.80 (7.4)	0.21	0.80	0.07	0.04	−0.31 (−2.9)	1.77 (16.0)	100
Pinned at the floor	15,801	1,445	0.50	0.37 (7.5)	0.50	0.00	0.00	0.03	−0.16 (−3.3)	0.83 (16.9)	100
Above the floor	24,751	2,889	3.67	1.46 (9.1)	0.00	1.91	0.11	0.04	−0.61 (−3.8)	3.33 (20.1)	100
With a bid ($q^b > 0$)	12,404	2,494	4.98	1.42 (5.3)	0.00	1.68	0.22	0.04	−0.52 (−1.9)	3.05 (11.3)	100
Binary	8,484	1,324	3.26	1.58 (6.5)	0.12	1.47	0.11	0.03	−0.14 (−0.6)	3.14 (12.4)	100
Multi-outcome	10,045	291	1.69	1.04 (7.3)	0.27	0.89	0.03	0.04	−0.20 (−1.5)	2.16 (12.1)	100
Ladder leg	13,819	884	2.87	0.84 (3.9)	0.13	1.42	0.09	0.03	−0.83 (−3.9)	2.35 (10.6)	100
Deadline	8,204	608	2.10	0.90 (4.1)	0.22	1.01	0.06	0.07	−0.46 (−2.1)	2.06 (8.9)	100
1 ¢ grid, 1–2 ¢	25,800	2,093	0.87	0.49 (6.8)	0.30	0.48	0.01	0.03	−0.34 (−4.7)	1.24 (16.7)	100
0.1 ¢ grid, 1–2 ¢	384	31	0.75	0.75 (10.2)	0.01	0.50	0.02	0.05	0.17 (2.3)	1.20 (8.7)	77
1 ¢ grid, all LS	39,960	3,061	2.55	1.07 (9.9)	0.18	1.23	0.07	0.04	−0.45 (−4.3)	2.44 (21.2)	100
0.1 ¢ grid, all LS	592	36	1.94	0.92 (1.3)	0.00	1.01	0.06	0.07	−0.24 (−0.3)	1.87 (2.5)	100

Longshot sides ($q \leq 10\,¢$ at the mid), leads pooled; columns as in Table 20. Halves split at the month that divides the quotes most evenly. The influential events are the five with the largest absolute contribution to the residual. Grid: markets quoted on Kalshi's 0.1 ¢ price grid against the standard 1 ¢ grid; the floor is one tick on each grid. t-statistics clustered by event in parentheses.

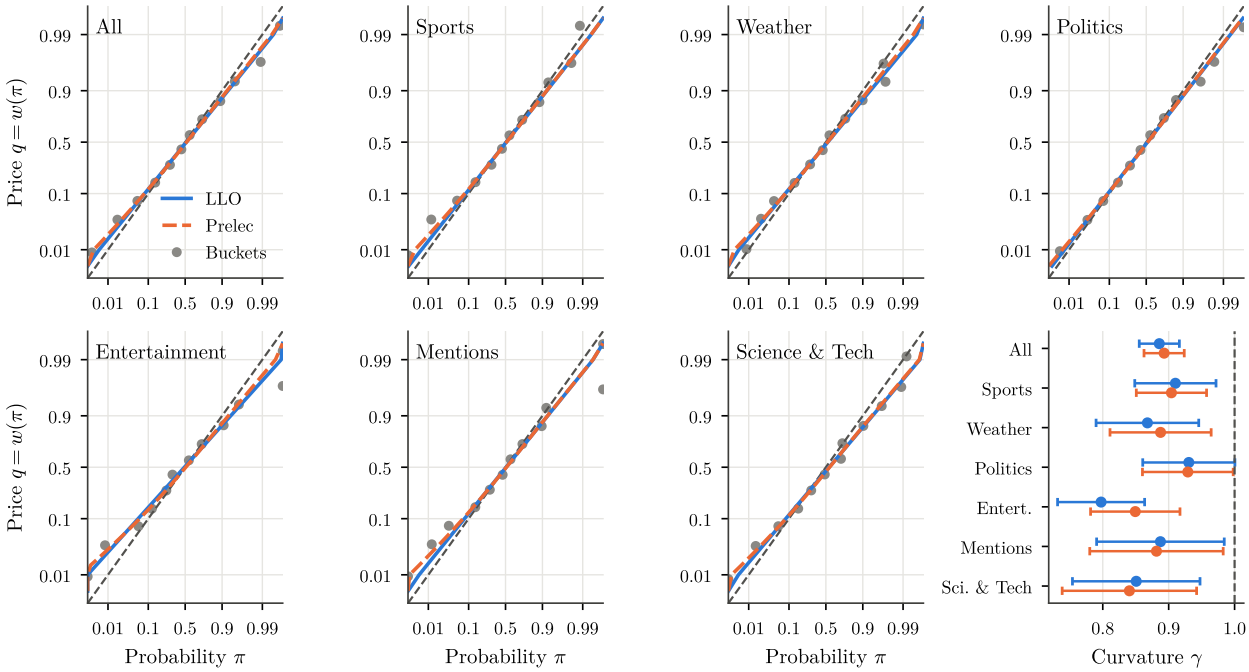


Figure 23: Fitted probability-weighting functions $q = w(\pi)$ on log-odds axes: linear in log odds (LLO) and Prelec, pooled and by group, with the calibration points (realised frequency, mean YES price) of the price buckets. Buckets with no or only winning outcomes sit at the edges. The dashed diagonal is calibration. Last panel: curvature γ by group with 95% confidence intervals clustered by event (LLO upper, Prelec lower marker). YES sides of pre-event quotes, market-weighted.

Table 46: Weighting-function parameters by lead and structure

			LLO				Prelec	
Cell	Quotes	Events	α	β ($t_{\beta=1}$)	γ	δ	γ	δ
By lead								
1h	15,784	4,191	0.06	1.18 (7.1)	0.85 (0.018)	−0.05 (0.033)	0.87 (0.018)	1.00 (0.018)
6h	15,827	4,163	0.07	1.20 (7.7)	0.84 (0.018)	−0.06 (0.032)	0.85 (0.017)	1.00 (0.017)
24h	11,688	2,739	0.13	1.23 (7.8)	0.81 (0.020)	−0.11 (0.040)	0.83 (0.020)	1.03 (0.022)
3d	10,358	2,380	0.03	1.17 (5.9)	0.86 (0.021)	−0.02 (0.045)	0.89 (0.022)	0.99 (0.024)
7d	5,553	1,478	−0.12	1.16 (3.8)	0.86 (0.031)	0.10 (0.069)	0.91 (0.033)	0.92 (0.036)
30d	2,782	885	−0.32	1.12 (2.3)	0.89 (0.043)	0.28 (0.087)	0.98 (0.048)	0.83 (0.041)
By structure								
Binary	17,163	2,644	−0.04	1.12 (3.7)	0.89 (0.026)	0.04 (0.038)	0.91 (0.023)	0.96 (0.022)
Multi-outcome	12,822	474	0.14	1.28 (4.4)	0.78 (0.039)	−0.11 (0.037)	0.84 (0.032)	1.01 (0.013)
Ladder leg	22,392	1,030	0.14	1.12 (2.9)	0.89 (0.032)	−0.13 (0.062)	0.89 (0.033)	1.05 (0.033)
Deadline	9,615	644	0.47	1.22 (3.9)	0.82 (0.039)	−0.38 (0.108)	0.77 (0.041)	1.21 (0.067)
By group, volume-weighted								
All groups	46,505	4,430	−0.01	1.48 (3.6)	0.68 (0.061)	0.01 (0.130)	0.76 (0.064)	0.96 (0.082)
Sports	11,242	1,298	−0.29	1.37 (2.0)	0.73 (0.098)	0.21 (0.119)	0.86 (0.089)	0.83 (0.067)
Weather	5,616	1,007	−0.10	1.29 (2.0)	0.77 (0.089)	0.07 (0.133)	0.86 (0.079)	0.92 (0.068)
Politics	12,490	941	1.13	1.80 (6.6)	0.56 (0.038)	−0.63 (0.305)	0.53 (0.063)	1.39 (0.228)
Entertainment	7,609	506	0.76	1.34 (3.0)	0.74 (0.064)	−0.57 (0.249)	0.68 (0.060)	1.34 (0.164)
Mentions	4,517	102	0.07	1.35 (4.6)	0.74 (0.041)	−0.05 (0.116)	0.77 (0.044)	0.99 (0.064)
Science & Tech	4,678	503	−0.13	1.56 (4.0)	0.64 (0.058)	0.08 (0.170)	0.73 (0.058)	0.92 (0.092)
Other	353	73	0.34	1.43 (1.5)	0.70 (0.144)	−0.24 (0.361)	0.71 (0.134)	1.11 (0.230)

As Table 21. Within a lead each market counts once. Standard errors (or t) clustered by event in parentheses.

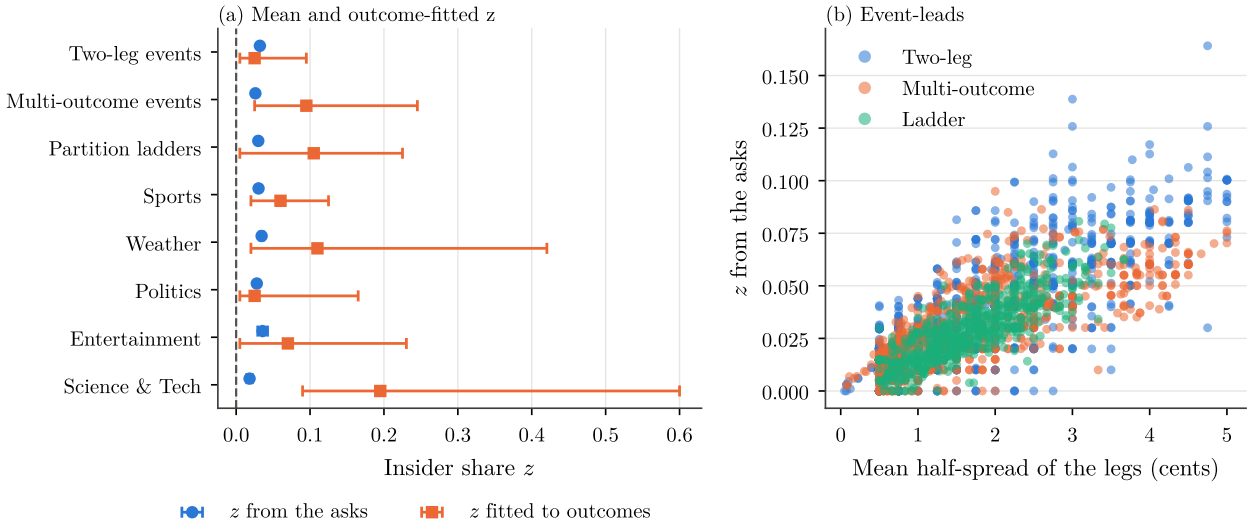


Figure 24: Shin's insider share in partition events. Left: mean z implied by the asks (95% confidence intervals clustered by event) and z fitted to outcomes (95% event-bootstrap intervals; the grid ends at 0.6), by structure and group. Right: z implied by the asks against the mean half-spread of the event's legs, one point per event and lead.

Table 47: Shin's insider share by lead, group and structure

Sample	Event-leads	Events	Legs (med.)	$\sum a - 1$ (%)	$\sum p - 1$ (%)	z from asks	z from mids	z fitted to outcomes [95% CI]
By lead								
1h	1,143	1,143	2	6.9 (24.9)	1.2 (9.1)	0.031 (0.0008)	0.004	0.060 [0.025, 0.135]
6h	1,187	1,187	2	7.4 (23.2)	1.2 (7.5)	0.030 (0.0007)	0.005	0.135 [0.075, 0.225]
24h	729	729	3	8.4 (20.7)	1.4 (8.2)	0.030 (0.0008)	0.004	0.115 [0.065, 0.185]
3d	559	559	3	10.6 (18.3)	1.9 (8.8)	0.027 (0.0009)	0.004	0.010 [0.000, 0.095]
7d	242	242	2	8.1 (9.4)	1.6 (3.9)	0.030 (0.0013)	0.005	0.010 [0.000, 0.080]
30d	95	95	2	10.1 (6.9)	1.8 (2.5)	0.038 (0.0032)	0.007	0.020 [0.000, 0.600]
By group and structure								
Sports, two-leg	1,609	660	2	3.1 (38.3)	−0.2 (−4.2)	0.032 (0.0008)	0.003	0.020 [0.000, 0.095]
Sports, multi	764	256	3	9.0 (12.4)	1.4 (4.1)	0.026 (0.0010)	0.003	0.155 [0.035, 0.530]
Weather, partition ladder	237	139	6	15.5 (25.5)	4.1 (11.4)	0.034 (0.0014)	0.010	0.110 [0.020, 0.420]
Politics, two-leg	376	74	2	2.5 (13.9)	0.0 (0.0)	0.031 (0.0020)	0.003	0.050 [0.015, 0.370]
Politics, multi	217	46	6	11.5 (11.8)	3.2 (7.5)	0.030 (0.0024)	0.008	0.020 [0.005, 0.331]
Politics, partition ladder	418	160	9	17.7 (29.4)	3.7 (11.8)	0.026 (0.0009)	0.006	0.100 [0.000, 0.265]
Entertainment, two-leg	24	6	2	4.3 (4.8)	0.5 (2.0)	0.057 (0.0132)	0.006	0.600 [0.600, 0.600]
Entertainment, multi	94	35	10	19.6 (8.4)	6.9 (6.0)	0.032 (0.0031)	0.010	0.035 [0.000, 0.220]
Science & Tech, multi	214	44	7	7.6 (5.4)	1.8 (2.3)	0.018 (0.0019)	0.005	0.195 [0.090, 0.600]

As Table 22. Regression of z from the asks on the mean half-spread, log legs and group dummies (Sports omitted), weighted, event-clustered t in parentheses: half-spread (¢) 0.0168 (50.5); log legs −0.0037 (−5.8); Weather 0.0029 (2.7); Politics 0.0039 (4.7); Entertainment 0.0114 (5.7); Science & Tech 0.0059 (4.2); $R^2 = 0.68$, 3,955 event-leads.

Table 48: Partition events: raw, normalised and Shin probabilities

	Longshot legs (¢)		Log score gain (milli-nats)		
Forecast	All months	Second half	All months	Second half	against
Raw mid	0.93 (4.9)	1.08 (4.3)	–	–	–
Normalised mids	0.83 (4.4)	1.01 (4.1)	−0.41 (−3.2)	−0.69 (−3.6)	raw mid
Normalised asks	–	–	–	–	–
Shin, from the asks	0.94 (5.0)	1.06 (4.3)	3.47 (8.5)	3.47 (6.2)	normalised asks
Shin, from the mids	–	–	0.86 (5.0)	0.89 (3.3)	normalised mids
Shin, z fitted on the first half	−0.21 (−1.1)	0.05 (0.2)	1.45 (1.7)	1.79 (1.5)	normalised mids

Legs of the partition events of Table 22. Longshot legs: mid at most 10 ¢ (8,906 legs in all months, 3,747 in the second half); mean forecast minus outcome in cents. Log score: mean gain per leg, over all legs, against the normalisation of the same prices. The fitted z is estimated on the first half of the months, so the second half is out of sample. Each event weighs one across its leads; t-statistics clustered by event in parentheses.

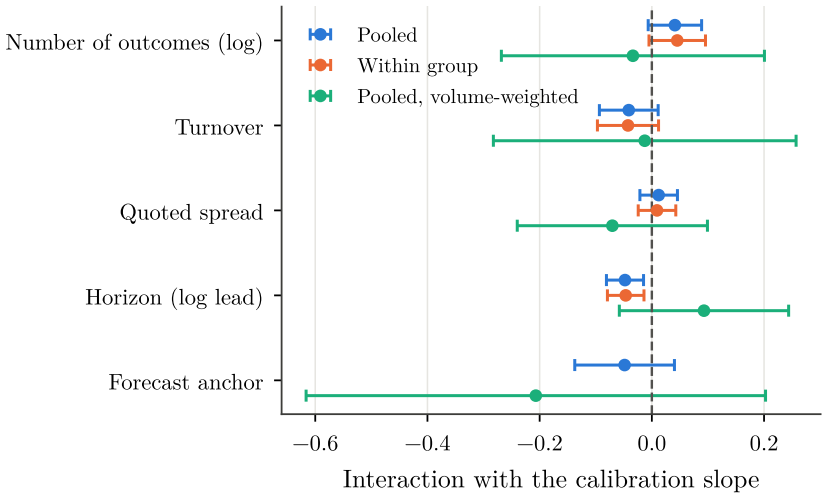


Figure 25: Interaction of each characteristic with the calibration slope, b_k in (18): the change in the logit slope per standard deviation (per unit for the 0/1 anchor). Positive values mean a stronger favourite–longshot bias. Pooled, within groups (group-specific slopes; the anchor is absorbed) and pooled volume-weighted. 95% confidence intervals clustered by event.

Table 49: Where is the bias steeper? Calibration-slope interactions

	Slope interaction (logit)				Longshot overpricing (¢)	
	Pooled	Within group	Volume-wtd.	Split anchor	Within group	Split anchor
Number of outcomes (log)	0.041 (1.7)	0.045 (1.8)	−0.034 (−0.3)	0.044 (1.8)	0.01 (0.1)	−0.03 (−0.4)
Turnover	−0.041 (−1.5)	−0.043 (−1.5)	−0.013 (−0.1)	−0.042 (−1.6)	−0.06 (−0.6)	−0.01 (−0.2)
Quoted spread	0.012 (0.7)	0.009 (0.5)	−0.070 (−0.8)	0.012 (0.7)	0.57 (3.4)	0.57 (3.4)
Horizon (log lead)	−0.048 (−2.8)	−0.047 (−2.8)	0.093 (1.2)	−0.046 (−2.7)	−0.28 (−3.0)	−0.29 (−3.0)
Forecast anchor (0/1)	−0.049 (−1.1)		−0.207 (−1.0)			
Anchor: Sports (0/1)				−0.065 (−1.3)		0.12 (0.4)
Anchor: Weather (0/1)				−0.007 (−0.1)		−0.06 (−0.2)
Events	4,770	4,770	4,430	4,770	3,091	3,091
Observations	61,992	61,992	46,505	61,992	40,552	40,552

Columns 1–4: coefficients b_k of (18), the change in the calibration slope per standard deviation of each characteristic (per unit for 0/1 anchors), YES sides of pre-event quotes; $b_k > 0$ means a stronger bias. Pooled: one slope plus interactions, group intercepts; within group: group-specific slopes, so the group-level anchor is absorbed. Columns 5–6: least squares of the price minus outcome of longshot sides (cents per contract) on the standardised characteristics with price-bucket (and in column 5 group) fixed effects. Market-weighted unless stated; t-statistics clustered by event in parentheses.

Table 50: Calibration-slope interactions by group and by half

Sample	Outcomes	Turnover	Spread	Horizon	Quotes	Events
Sports	–	0.025 (0.6)	0.019 (0.6)	−0.013 (−0.5)	18,303	1,521
Weather	–	−0.054 (−0.4)	0.043 (0.9)	0.087 (1.9)	6,796	1,065
Politics	–	−0.127 (−2.7)	0.036 (1.2)	−0.080 (−2.0)	15,774	968
Entertainment	–	−0.101 (−1.8)	0.034 (0.6)	−0.203 (−3.8)	10,424	530
Mentions	–	−0.107 (−1.7)	−0.064 (−1.1)	−0.059 (−1.2)	4,743	103
Science & Tech	–	−0.088 (−1.3)	−0.002 (0.0)	−0.099 (−1.4)	5,472	510
Pooled, first half (to 2025-07)	0.025 (0.9)	−0.067 (−2.0)	0.036 (1.5)	−0.041 (−1.7)	31,006	2,443
anchor	0.096 (1.5)					
Pooled, second half	0.049 (1.5)	−0.032 (−0.8)	−0.009 (−0.4)	−0.062 (−2.9)	30,986	2,339
anchor	−0.204 (−3.4)					

As Table 49, column 1, estimated separately by group (a characteristic enters when at least ten events lie on each side of its group median) and by half of the sample. Positive interactions across groups: number of outcomes 0 of 0, turnover 1 of 6, quoted spread 4 of 6, horizon 1 of 6. t-statistics clustered by event in parentheses.

Table 51: The four explanations scored against their predictions

Prediction	Statistic	Holds
H1: partly supported (2 of 5 predictions)		
Longshot mispricing inside the taker band ($\bar{R}$, ¢)	−0.45 (−4.3)	yes
Inside the taker band, volume-weighted ($\bar{R}$, ¢)	0.79 (2.7)	no
Inside the taker band in every group (Bonferroni)	6 of 6	yes
Gross mispricing rises with the lead (¢ per log hour)	−0.143 (−2.7)	no
Concentrated at the tick floor (floor share ≥ 0.5)	0.17	no
H2: partly supported (4 of 5 predictions)		
Pooled $\gamma < 1$ (γ, $t_{\gamma<1}$)	0.89 (7.4)	yes
$\gamma < 1$ in every group (point estimates)	6 of 6	yes
Equal γ across groups not rejected (Wald χ^2)	8.6 ($p = 0.127$)	yes
LLO beats calibrated prices out of sample, both directions	0.46, 0.04 (0.9, 0.1)	no
Pooled LLO predicts as well as group-specific LLO	−3.10, −1.83 (−4.2, −3.1)	yes
H3: partly supported (2 of 4 predictions)		
Shin from the asks removes the longshot-leg overpricing (¢)	0.94 (5.0)	no
z from the asks higher in information groups, given the spread	0.004 (5.5)	yes
Fitted z higher in information than in public groups	0.015 [−0.110, 0.090]	no
Longshot overpricing rises with z (¢ per unit)	22.0 (2.1)	yes
H4: not supported (0 of 4 predictions)		
Steeper slope with more outcomes, within groups	0.045 (1.8)	no
Steeper slope with more turnover, within groups	−0.043 (−1.5)	no
Flatter slope with a public forecast anchor	−0.049 (−1.1)	no
Turnover interaction positive in most groups	1 of 6	no

Each row is a prediction of Hypotheses 1–4 as made operational in Section 5.4, tested at the 5% level; the every-group band test uses a one-sided Bonferroni threshold across groups. Statistics: cents per contract with t (Hypothesis 1 band and horizon rows; the floor row is a share), γ with $t_{\gamma<1}$, χ^2 with p, out-of-sample log-score differences in milli-nats per quote for the two directions with t, cents per contract, z differences with 95% bootstrap intervals, and slope interactions with t. All t-statistics and intervals clustered by event. Verdict: supported when every prediction holds, partly supported when some do.

Table 52: How much of the longshot loss does each explanation map remove?

Map from prices to probabilities	Removed (¢)	Residual (¢)	Share removed (%)
Longshot sides, second half (18,406 sides, 1,341 events); gross mispricing 1.12 (6.9) ¢			
Taker band, $\rho = 5\%$ (band 1.48 ¢)	1.12	–	100
LLO, pooled, fitted on the first half	0.69	0.43 (2.7)	61
LLO by group, fitted on the first half	0.92	0.20 (1.2)	82
Prelec, pooled, fitted on the first half	0.76	0.36 (2.3)	68
Longshot legs of partition events, second half (3,747 legs, 269 events); gross mispricing 1.08 ¢			
Normalised mids	0.06	1.01 (4.1)	6
Shin from the asks	0.02	1.06 (4.3)	2
Shin with z fitted on the first half	1.02	0.05 (0.2)	95

Gross mispricing: price at the mid minus outcome, cents per contract. Band: the share of the gross mispricing that lies inside the taker band (Table 20). Each other row replaces the price by a fitted probability and reports the remaining price-minus-outcome gap (residual) and the share removed. The maps are alternative readings of the same calibration curve, not additive components. Market-weighted (partition events: each event once); t-statistics clustered by event in parentheses.